
HTML
CSS
JavaScript
Week 1
Introduction to the Web Technology
Bootstrap
Node.js

รายละเอียดวิชา

2-241-215 วิชาการเขียนโปรแกรมบนเว็บ 3 (2-2-5)

วิชาบังคับก่อน

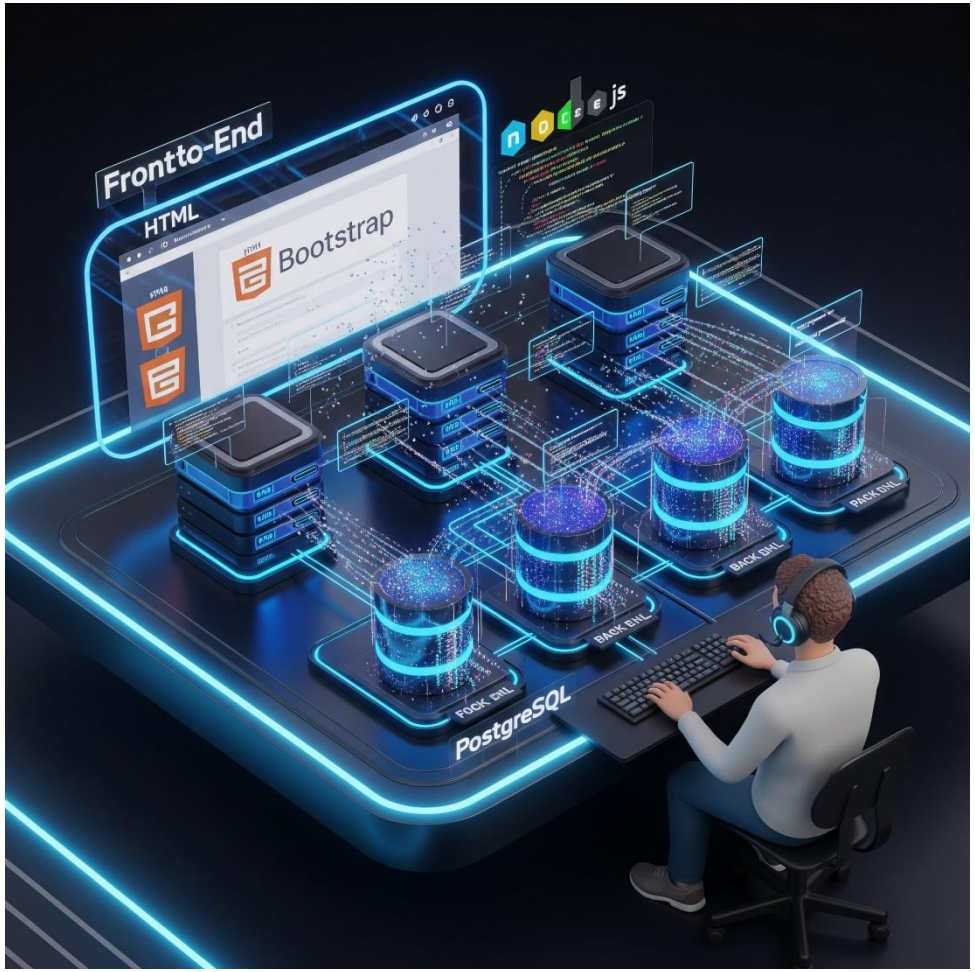
2-231-103 ระเบียบวิธีการเขียนโปรแกรม

2-243-212 ระบบจัดการฐานข้อมูล

เทคโนโลยีของเว็บ สื่อดิจิทัล สถาปัตยกรรมของการให้บริการโปรแกรมประยุกต์บนเว็บ การพัฒนาโปรแกรมประยุกต์บนเว็บ การพัฒนาโปรแกรมประยุกต์บนเว็บเพื่อติดต่อฐานข้อมูล แนวทางการพัฒนาเว็บให้ปลอดภัย เครื่องมือที่ใช้ในการพัฒนาเว็บเฟรมเวิร์คและเทคโนโลยีที่เกี่ยวข้อง

การแบ่งคะแนน

- จิตพิสัย (มาเรียน) 5
- บันทึกการเรียน 10
- คะแนนระหว่างภาค 25
  - งาน
    - รายสัปดาห์
    - โปรเจคพัฒนาเว็บไซต์
  - ทดสอบรายหน่วย
- สอบกลางภาค 30
- สอบปลายภาค 30



การประเมินผล

- 80 ขึ้นไป A
- 75-79 B+
- 70-74 B
- 65-69 C+
- 60-64 C
- 55-59 D+
- 50-54 D
- ต่ำกว่า 50 F



All Items

- Introduction to the Web Technology
- HTML
- Multi-Page websites
- CSS
- Bootstrap Framework
- Javascript
- The Document Object Model (DOM)
- jQuery
- Node.js
- Git, Github and Version Control
- Database (PostgreSQL)
- REST (Representational State Tranfer)
- Authentication & Security

เนื้อหา

หน่วย	ชื่อหน่วยเรียน	เนื้อหาหลัก
1	บทนำสู่เทคโนโลยีเว็บ	- ประวัติและวิวัฒนาการของเว็บ - HTML, CSS, JavaScript พื้นฐาน - โครงสร้าง Client-Server - Protocol: HTTP, HTTPS
2	โครงสร้างและองค์ประกอบของเว็บ	- HTML5 (Semantic Elements) - CSS Layout (Flexbox, Grid) - Responsive Design - Web Accessibility (WCAG)
3	การเขียนโปรแกรมฝั่งผู้ใช้ (Client-side)	- JavaScript เบื้องต้น- DOM Manipulation - Event Handling - AJAX และ Fetch API
4	การเขียนโปรแกรมฝั่งเซิร์ฟเวอร์ (Server-side)	- แนะนำ Node.js หรือ PHP - การทำ Routing และ Template - รับค่าจากฟอร์ม (POST/GET) - การจัดการ Session และ Cookie
5	การเชื่อมต่อฐานข้อมูล (Database Connectivity)	- Introduction to SQL - การเชื่อมต่อฐานข้อมูล (เช่น MySQL, SQLite) - CRUD Operations - ORM (พื้นฐาน)

เนื้อหา (ต่อ)

หน่วย	ชื่อหน่วยเรียน	เนื้อหาหลัก (**Key Content**)
6	สถาปัตยกรรมเว็บแอปพลิเคชัน	- MVC Concept - RESTful APIs - JSON/XML - แนะนำ Web Service
7	แนวทางการพัฒนาเว็บให้ปลอดภัย	- Security Concepts (XSS, SQL Injection, CSRF) - การยืนยันตัวตนและการเข้ารหัส - การจัดการสิทธิ์การเข้าถึง - HTTPS และ TLS
8	เครื่องมือและเฟรมเวิร์คสำหรับการพัฒนาเว็บ	- Version Control (Git + GitHub - VS Code/IDE Tools - Front-end Frameworks (เช่น Bootstrap) - Server Frameworks (Express.js, EJS เบื้องต้น)
9	โครงงานและการพัฒนาเว็บแอปจริง	- การออกแบบ UI/UX เบื้องต้น - การวางแผนและแบ่งงาน (Agile/Scrum เบื้องต้น) - การพัฒนาเว็บโปรเจกต์อย่างเป็นขั้นตอน
10	สรุปและประเมินผลการพัฒนาเว็บแอปพลิเคชัน	- นำเสนอผลงาน / Code Review - การทดสอบระบบ (Testing & Debugging) - การ Deployment (แนะนำ GitHub Pages, Vercel, หรือ Render)

Three-tier architecture

Presentation tier

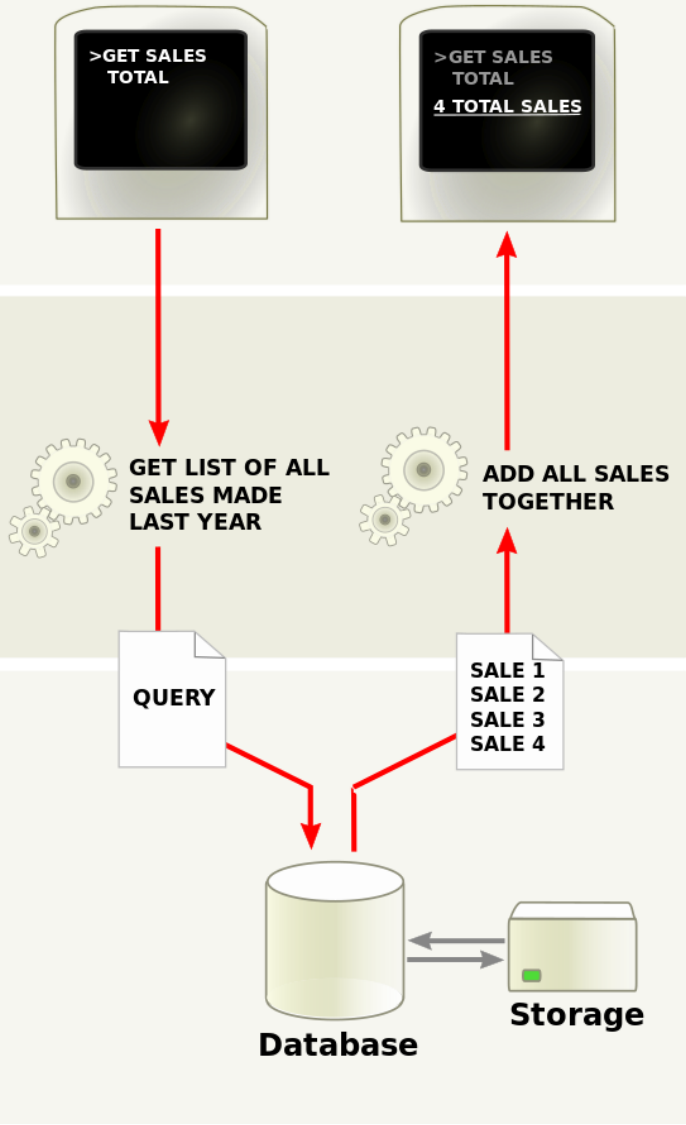
The top-most level of the application is the user interface. The main function of the interface is to translate tasks and results to something the user can understand.

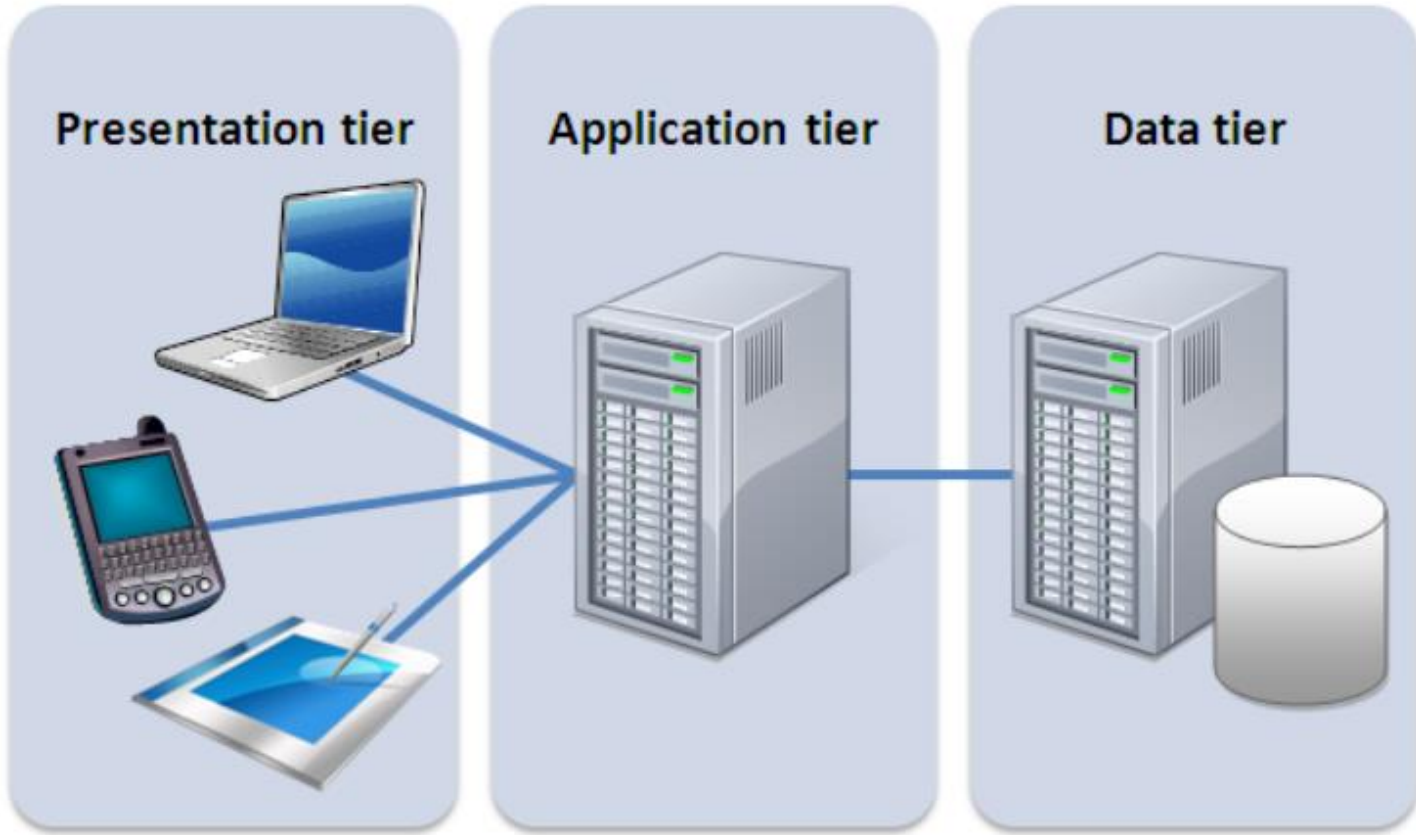
Logic tier

This layer coordinates the application, processes commands, makes logical decisions and evaluations, and performs calculations. It also moves and processes data between the two surrounding layers.

Data tier

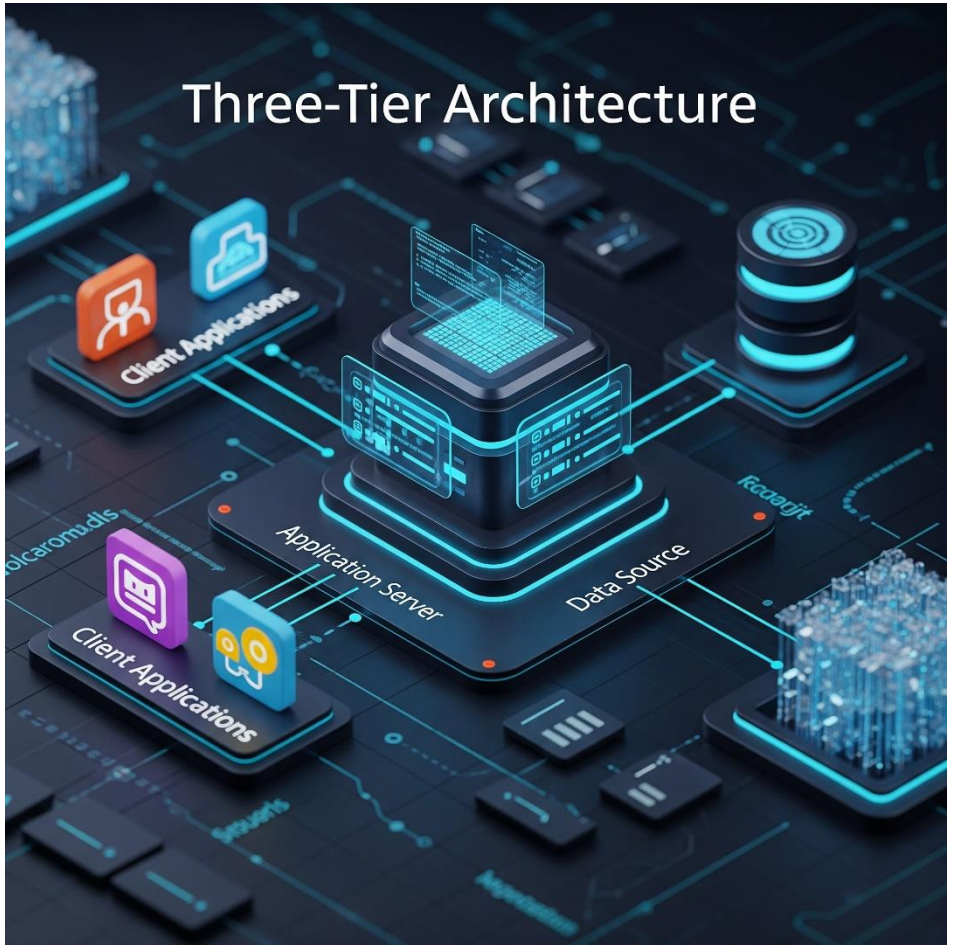
Here information is stored and retrieved from a database or file system. The information is then passed back to the logic tier for processing, and then eventually back to the user.



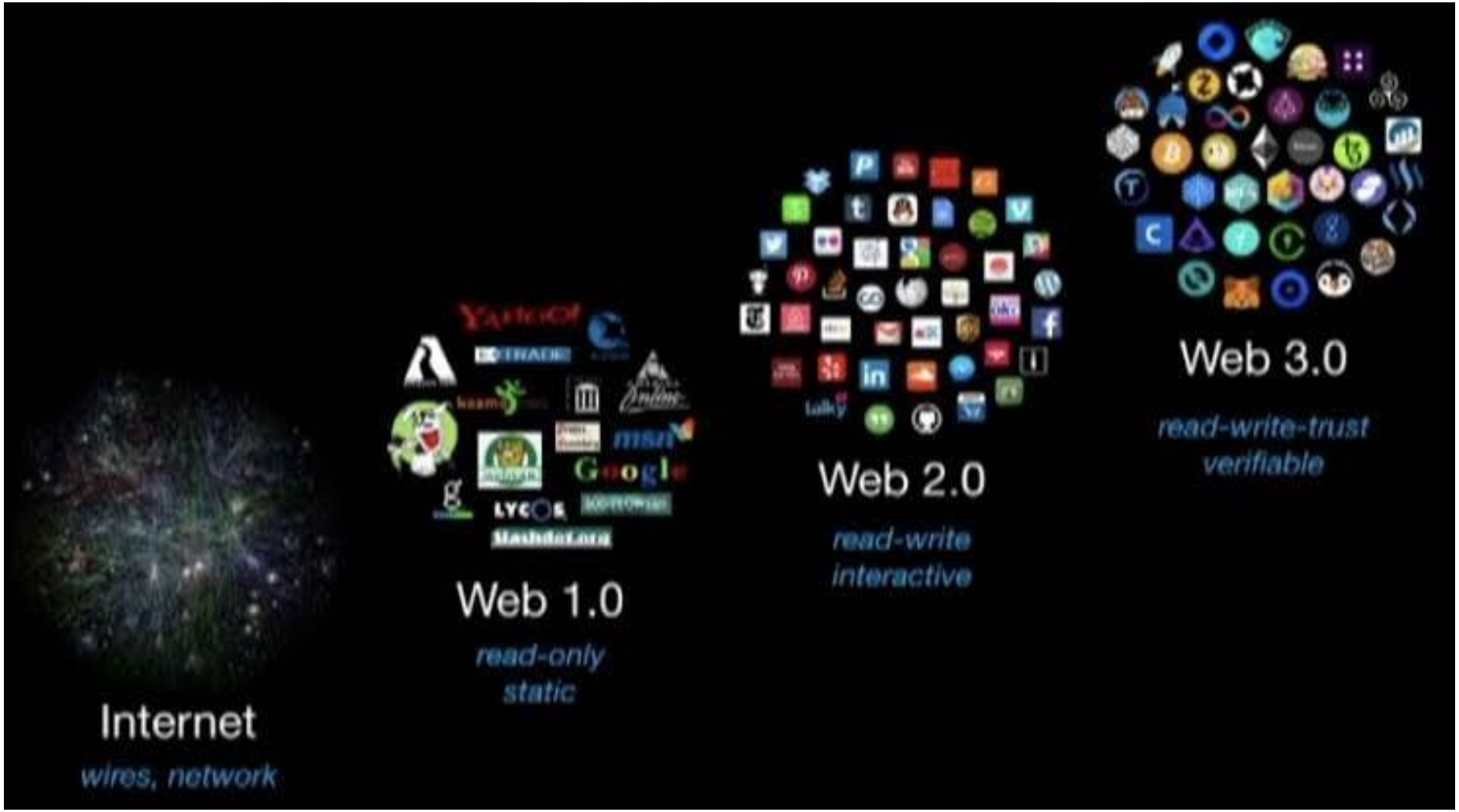

Presentation tier
Application tier
Data tier

Web development & Tree-tier

- **Front-end Web Server (เซิร์ฟเวอร์ส่วนหน้า):** ทำหน้าที่ให้บริการเนื้อหาแบบคงที่ **(Static Content)** เช่น รูปภาพ, **CSS**, **JavaScript** และอาจรวมถึงเนื้อหาแบบไดนามิกบางส่วนที่ถูกแคชไว้ ในแอปพลิเคชันบนเว็บ ส่วนหน้า **(Front-end)** คือเนื้อหาที่แสดงผลโดยเบราว์เซอร์ ซึ่งเนื้อหานี้อาจเป็นแบบคงที่หรือถูกสร้างขึ้นมาแบบไดนามิกก็ได้
- **Middle Dynamic Content Processing and Generation Level Application Server (เซิร์ฟเวอร์แอปพลิเคชันประมวลผลเนื้อหาไดนามิกส่วนกลาง):** เป็นเซิร์ฟเวอร์ที่ทำหน้าที่ประมวลผลและสร้างเนื้อหาแบบไดนามิก ตัวอย่างเช่นเทคโนโลยีที่ใช้ได้แก่ **Symfony**, **Spring**, **ASP.NET**, **Django**, **Rails**, **Node.js**.
- **Back-end Database or Data Store (ฐานข้อมูลส่วนหลัง):** คือฐานข้อมูลหรือแหล่งเก็บข้อมูลที่ประกอบด้วยทั้งชุดข้อมูล **(Data Sets)** และซอฟต์แวร์จัดการฐานข้อมูล **(Database Management System)** ที่ทำหน้าที่จัดการและให้การเข้าถึงข้อมูล




HTML
Front end
HTML
Bront end
Bootstrap
Node.js
node.js
PostgreSQL
Front End
Back End


Internet
wires, network
Web 1.0
read-only
static
Web 2.0
read-write
interactive
Web 3.0
read-write-trust
verifiable

1. **จุดกำเนิดของเว็บ** (1989 - 1990)

•**ผู้ริเริ่ม**: Sir **Tim Berners-Lee** (นักวิทยาศาสตร์จาก CERN, สวิตเซอร์แลนด์)

•**แนวคิด**: สร้างระบบแลกเปลี่ยนข้อมูลระหว่างนักวิจัยทั่วโลก

•**เทคโนโลยี**หลัก

- HTML (HyperText Markup Language)
- HTTP (HyperText Transfer Protocol)
- URL (Uniform Resource Locator)

•**เว็บไซต์แรกของโลก**: http://info.cern.ch

2. **ยุคเว็บ 1.0 (**1991 **-** 2000**) – เว็บแบบอ่านอย่างเดียว**

•เว็บส่วนใหญ่เป็น **แบบสถิต (Static)** ผู้ใช้แค่ **อ่าน** ข้อมูล

•สร้างด้วย HTML ธรรมดา ไม่มีฐานข้อมูล ไม่มีการโต้ตอบ

•ตัวอย่าง: เว็บไซต์บริษัท, หน้าแสดงบทความหรือข้อมูลคงที่

เทคโนโลยีหลัก:

•HTML

•HTTP

•Static Server

3. **ยุคเว็บ 2.0 (**2000 **-** 2010**) – เว็บแบบโต้ตอบ**

•เริ่มมี **เว็บโต้ตอบได้ (Interactive)** เช่น เว็บบอร์ด, โซเชียลมีเดีย

•ผู้ใช้งานสามารถ **เพิ่ม/แบ่งปันเนื้อหาเอง** (User-Generated Content)

•เว็บเริ่มมี **ฐานข้อมูล**, **login, comment, ajax**

ตัวอย่าง Facebook, YouTube, Wikipedia, Blogger

เทคโนโลยีหลัก

•HTML + CSS + JavaScript

•PHP, MySQL, AJAX

•CMS เช่น WordPress, Joomla

4. **ยุคเว็บ** 3.0 (2010 - **ปัจจุบัน) – เว็บอัจฉริยะ**

•เน้น **Web Semantic**: เว็บที่เข้าใจความหมายของข้อมูล

•ใช้ **AI, Machine Learning, Big Data**

•เว็บสามารถปรับเปลี่ยนตามผู้ใช้ (Personalization)

•มีการเชื่อมโยงข้อมูลแบบเปิด (Linked Open Data)

เทคโนโลยีหลัก:

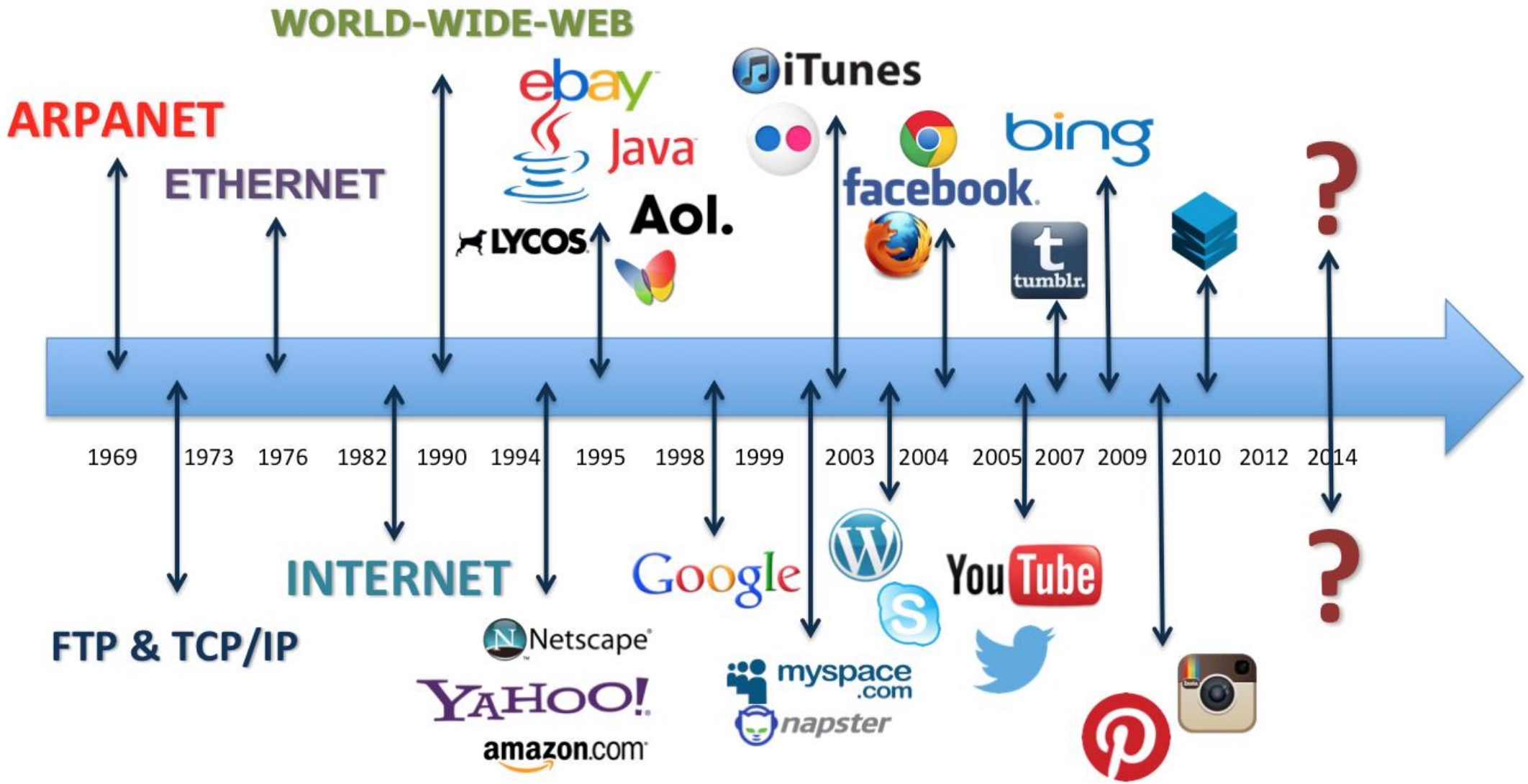
•JSON-LD, RDF, SPARQL

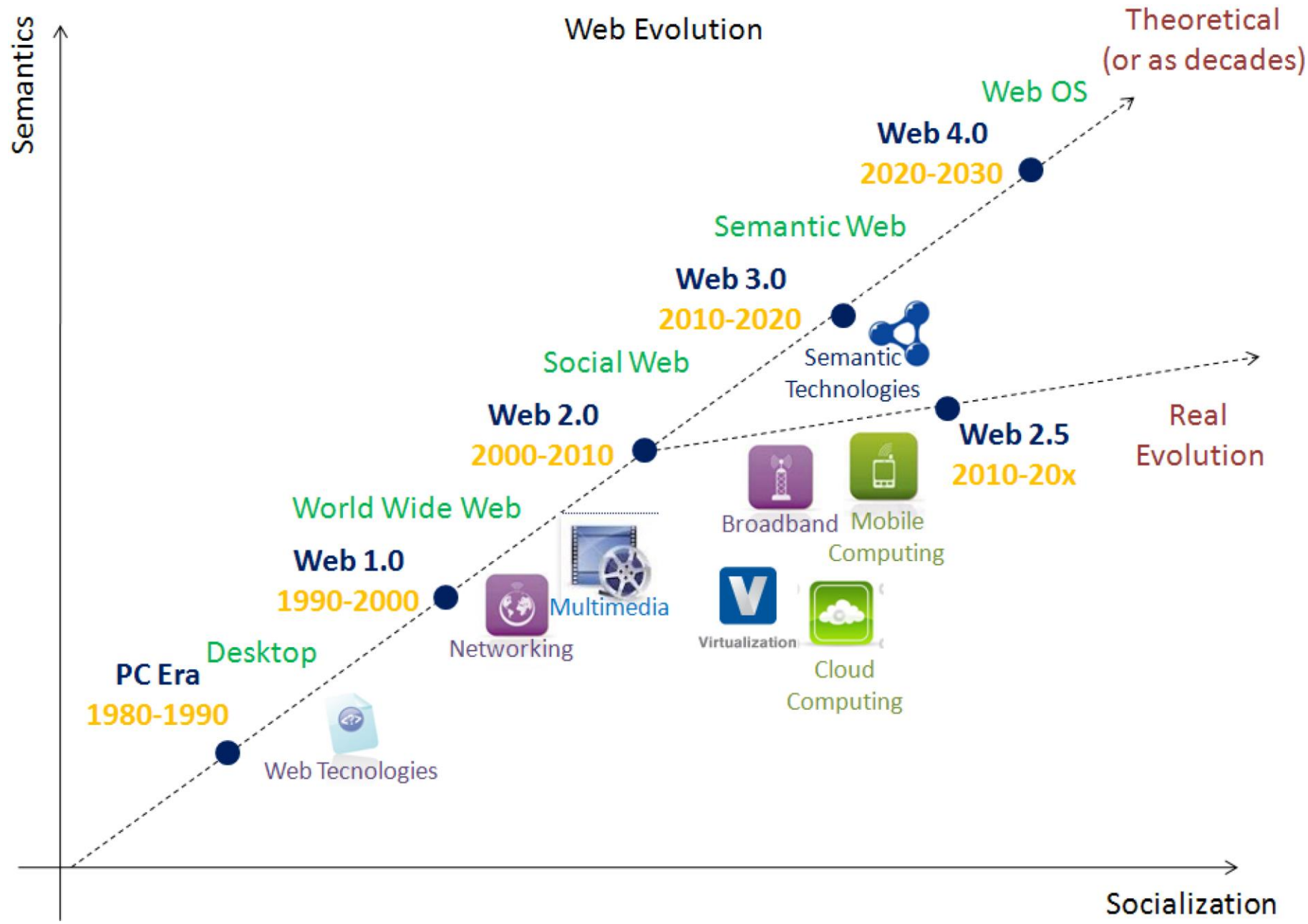
•AI, GraphQL, Progressive Web App (PWA)

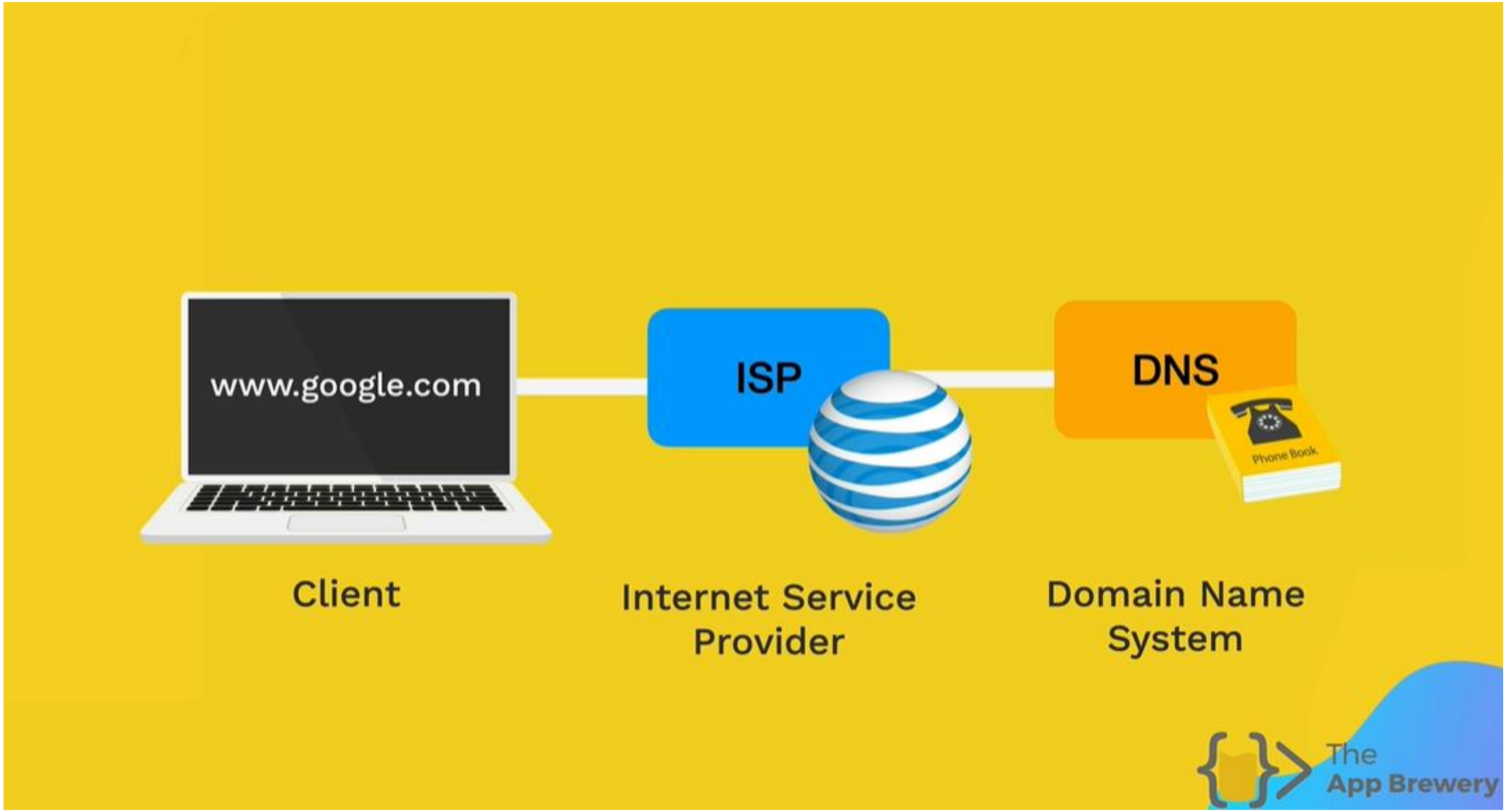
5. **เว็บ** 4.0 **และอนาคต (Web of Everything)**

•กำลังพัฒนา:

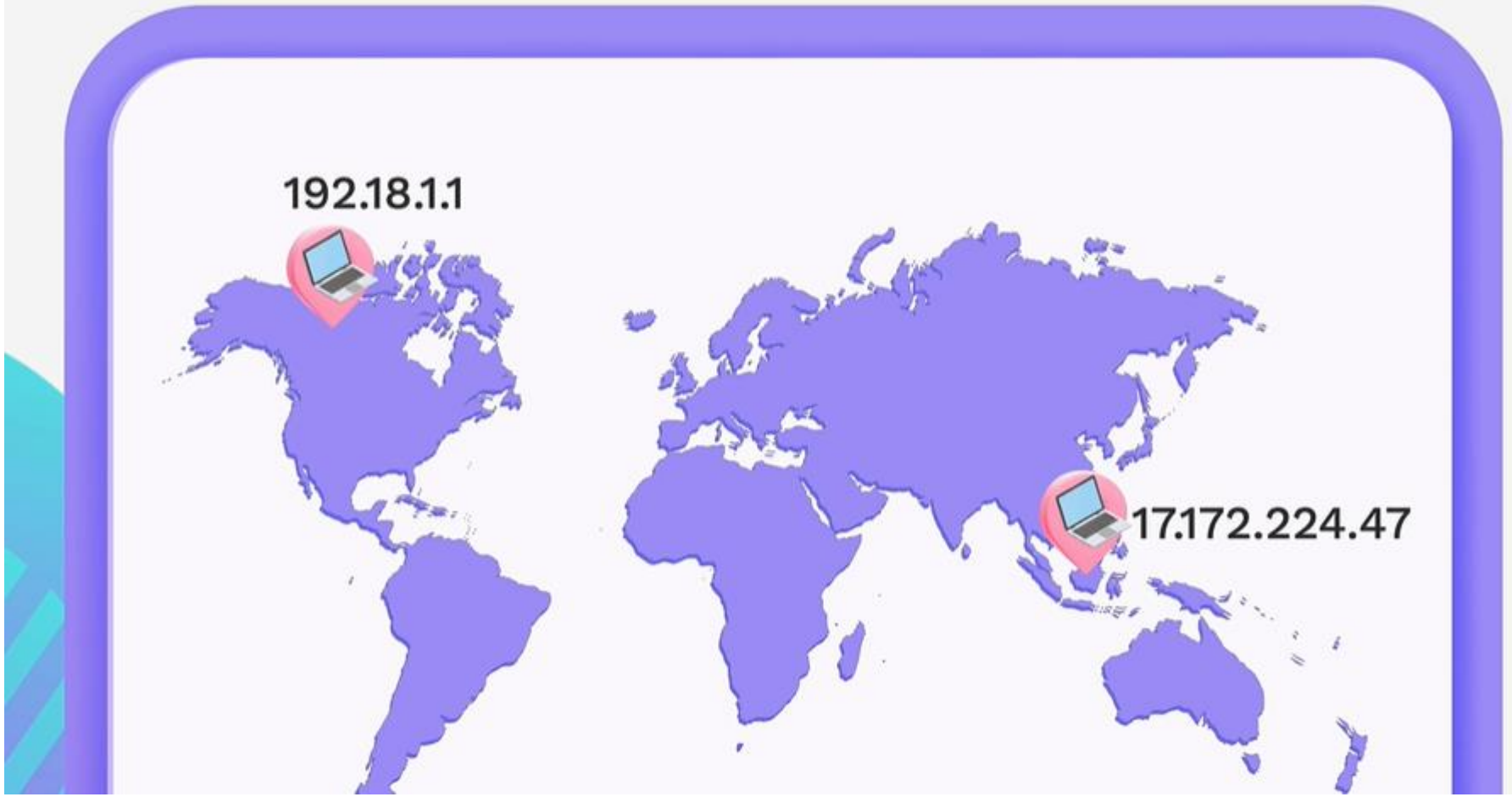
- **Web of Things (WoT)** / Internet of Things (IoT)
- **Metaverse / Virtual Reality Web**
- การผสานข้อมูลข้ามอุปกรณ์ทุกชนิด
- การควบคุมด้วยเสียง สมอง หรืออวตาร

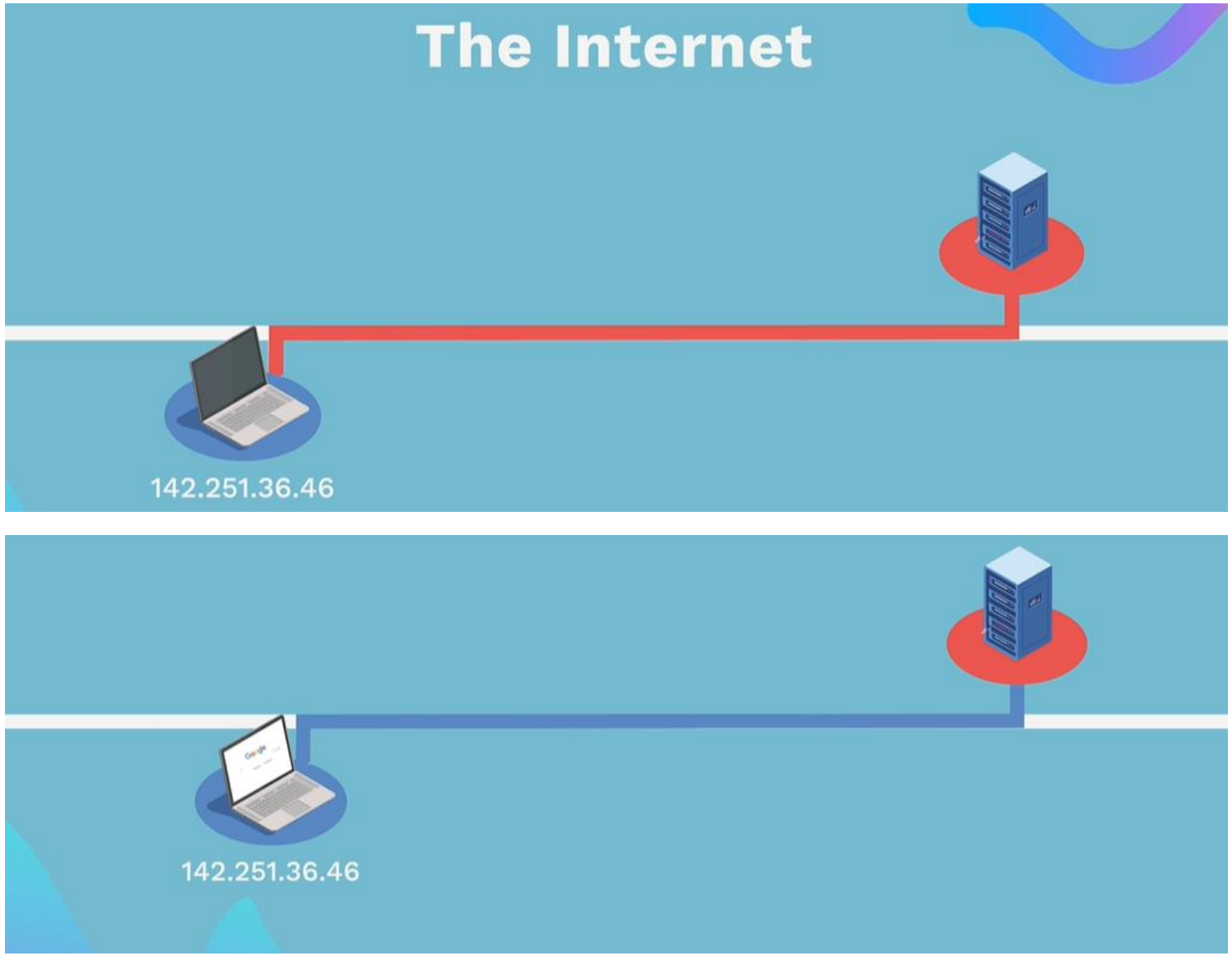

WORLD-WIDE-WEB
ARPANET
ETHERNET
ebay
Java
iTunes
LYCOS
Aol.
bing
facebook
tumblr.
?
1969
1973
1976
1982
1990
1994
1995
1998
1999
2003
2004
2005
2007
2009
2010
2012
2014
INTERNET
Google
YouTube
?
FTP & TCP/IP
Netscape
YAHOO!
amazon.com
myspace
.com
napster

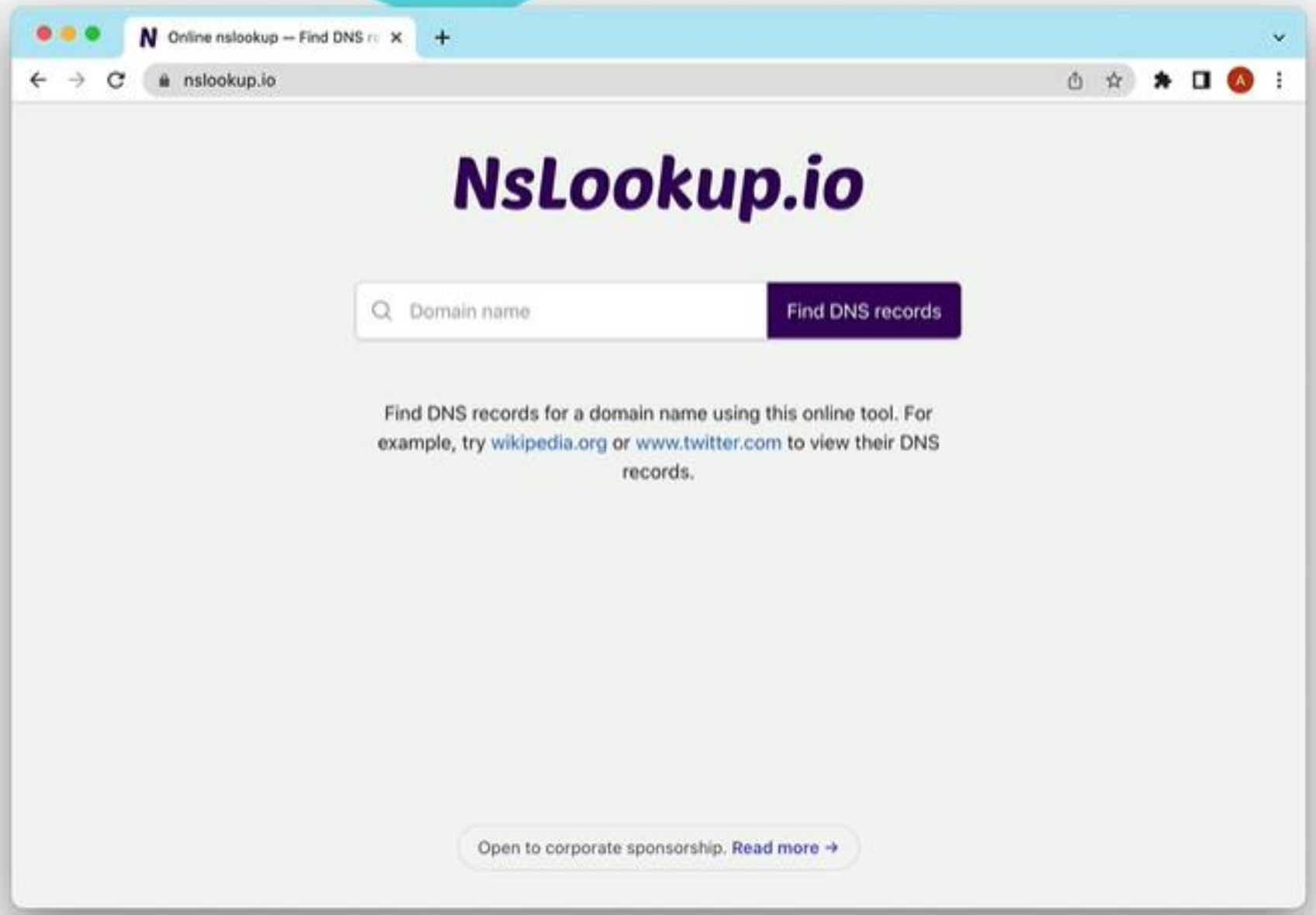

Web Evolution
Semantics
Socialization
Theoretical
(or as decades)
Web OS
Web 4.0
2020-2030
Semantic Web
Web 3.0
2010-2020
Social Web
Semantic
Technologies
Web 2.0
2000-2010
Real
Evolution
Web 2.5
2010-20x
World Wide Web
Broadband
Mobile
Computing
Web 1.0
1990-2000
Multimedia
Networking
Virtualization
Cloud
Computing
Desktop
PC Era
1980-1990
Web Tecnologies


www.google.com
ISP
DNS
Phone Book
Client
Internet Service Provider
Domain Name System
The App Brewery

IP address




The Internet
142.251.36.46
142.251.36.46


Online nslookup — Find DNS
nslookup.io
NsLookup.io
Domain name
Find DNS records
Find DNS records for a domain name using this online tool. For example, try wikipedia.org or www.twitter.com to view their DNS records.
Open to corporate sponsorship. Read more →

DNS records for **google.com**

Cloudflare | Google DNS | OpenDNS | Authoritative | Local DNS

The Cloudflare DNS server responded with these DNS records. Cloudflare will serve these records for as long as the time to live (TTL) has not expired. After this period, Cloudflare will update its cache by querying one of the authoritative name servers.

A records

IPv4 address	Revalidate in
172.217.168.238	2m 52s

AAAA records

IPv6 address	Revalidate in
2a00:1450:400e:80c::200e	49s

CNAME record

No CNAME record found.

TXT records

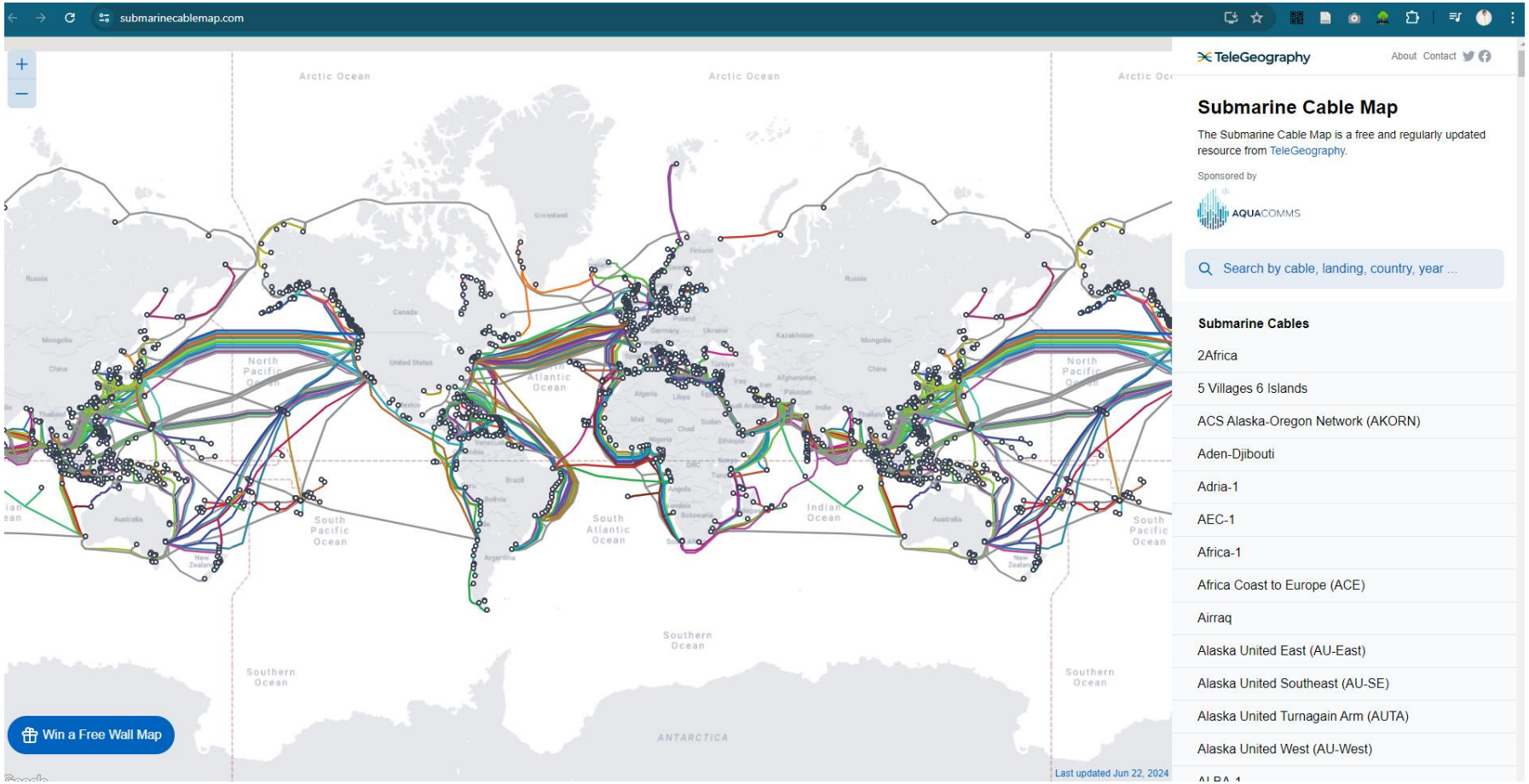
Site ownership verification

Apple | Atlassian | DocuSign | DocuSign | Facebook


www.appbrewery.com

Submarine Cable Map

https://www.submarinecablemap.com/





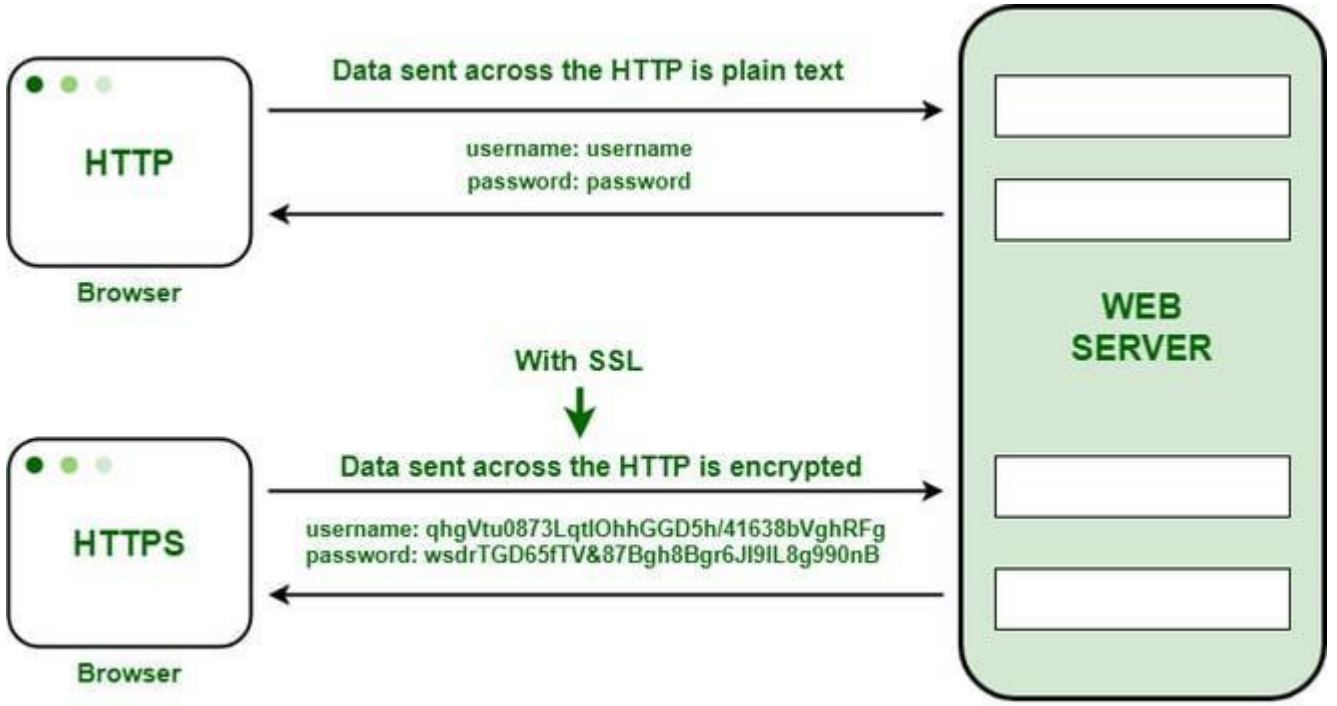
HTTP / HTTPS

HTTPS ใช้ปกป้องอะไรบ้าง?

1.ข้อมูลผู้ใช้ เช่น รหัสผ่าน, เลขบัตรเครดิต

2.ความเป็นส่วนตัว – ไม่ให้แฮกเกอร์รู้ว่าคุณกรอกอะไร

3.ป้องกันการปลอมแปลงหน้าเว็บ (Man-in-the-Middle Attack)



- ทำแบบทดสอบ Online สักครู่

The App Brewery

How Do Websites Work?

HTML, CSS and JS

www.appbrewery.com



Download the Required Software

1. Install Visual Studio Code

This is a free piece of software from Microsoft that you will use to write your code.

https://code.visualstudio.com/

NOTE: You'll see the Atom editor in some of the videos, this software is now being sunsetted by GitHub and in the coming months all the videos will update to use VS Code. But if you hear me say "Open Atom" or "Inside the Atom editor do..." just know that you should now use VS Code.

2. Install Visual Studio Code Extensions

Extensions are like plugins that extend the functionality of VSCode. Here are the ones that I've installed. Click on the link and then click on the "install" button and it should automatically install it in your VSCode application.

Live Preview

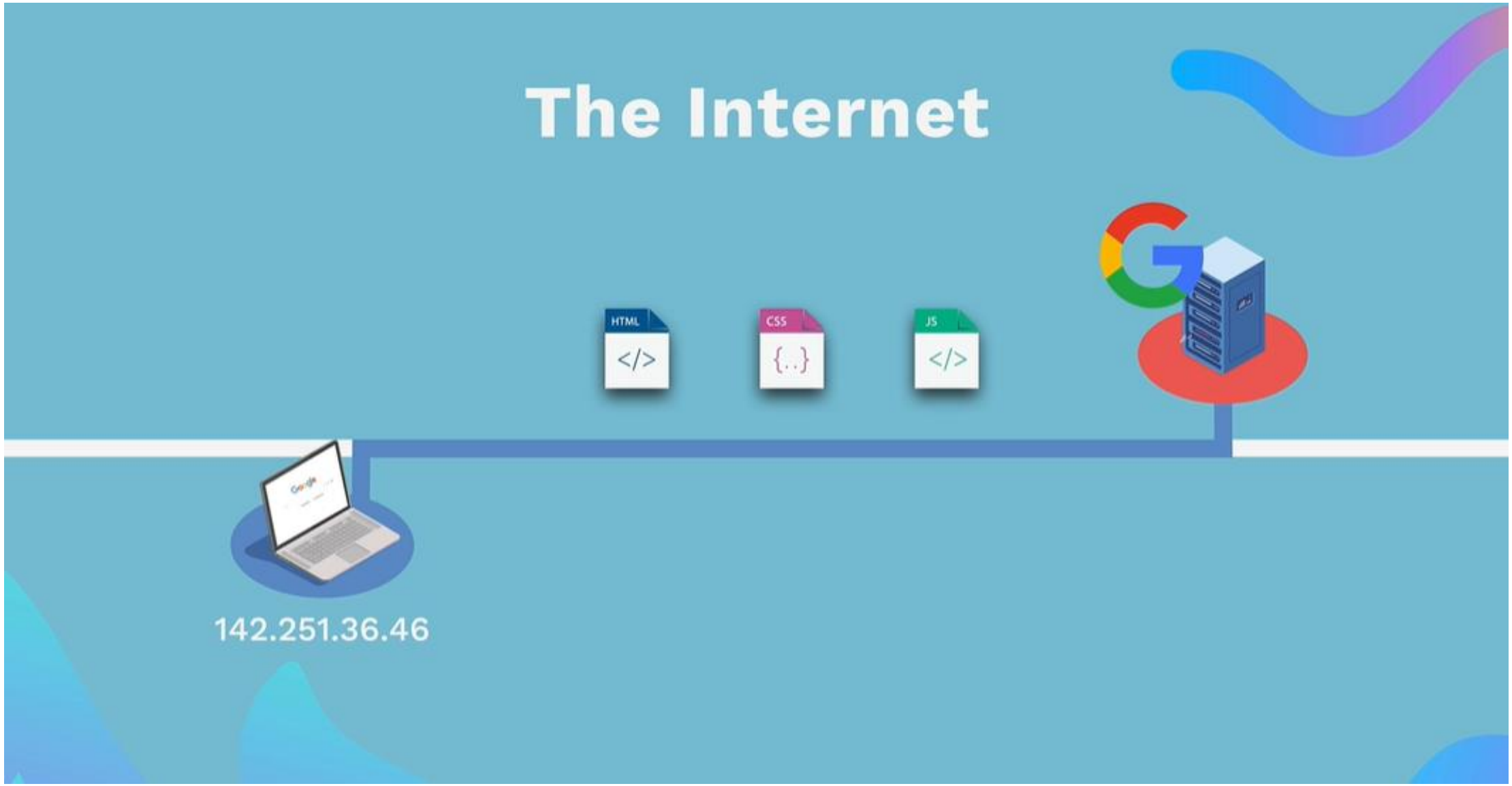
Prettier

vscode-icons

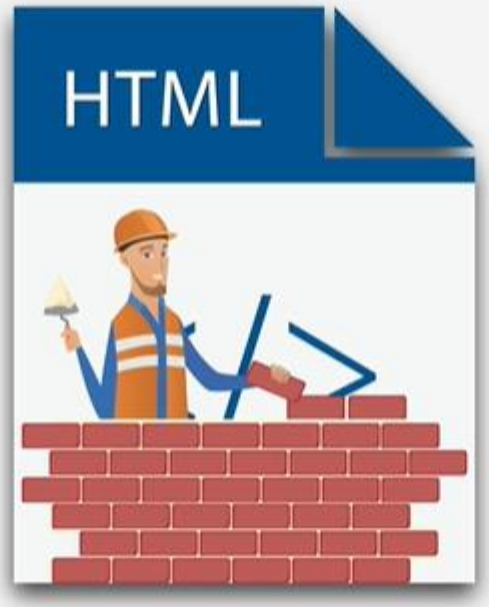
3. Install Google Chrome

This is a free browser from Google that we need for development. Even if you normally use another browser, I recommend downloading Chrome for their developer tool suite which we will use extensively in the course.

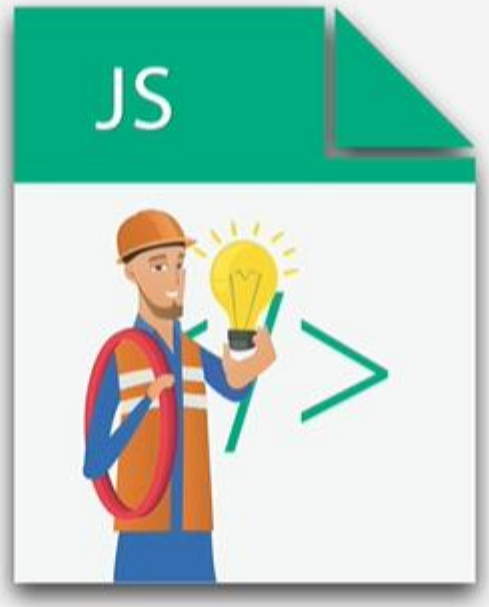
https://www.google.com/intl/en_uk/chrome/


The Internet
HTML
CSS
JS
142.251.36.46

Code Files



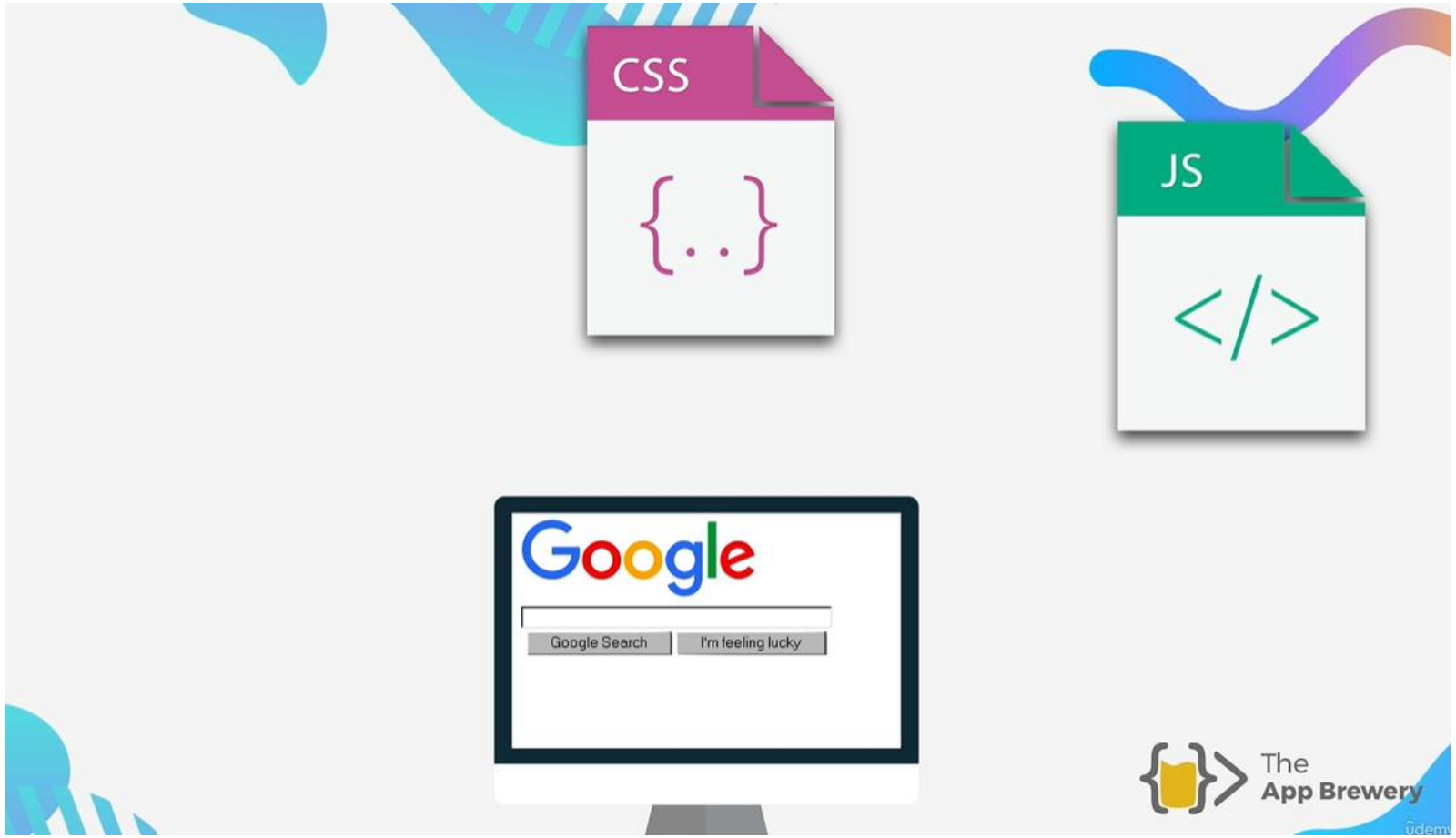


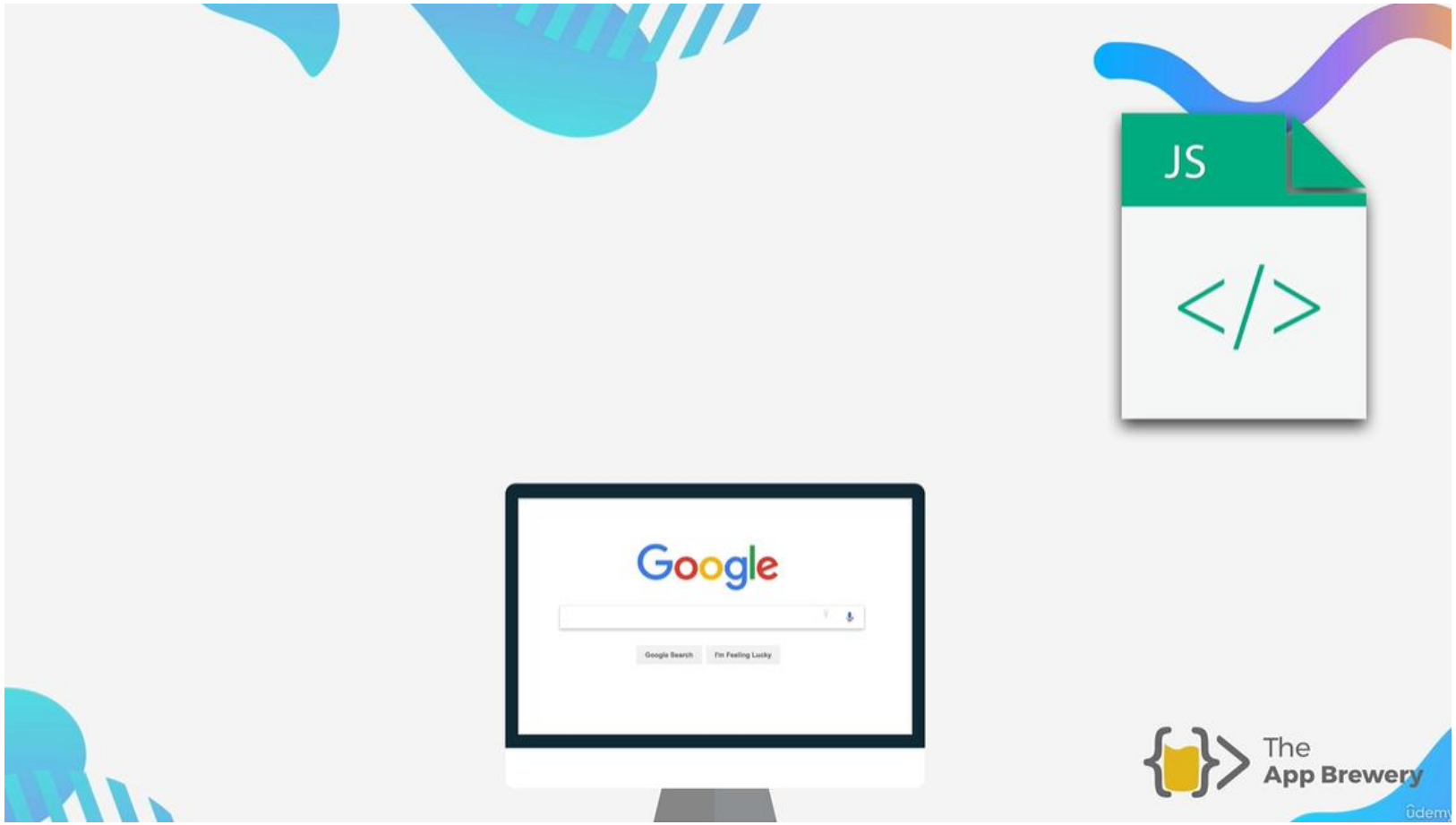


www.appbrewery.com

The App Brewery


HTML
CSS
JS
The
App Brewery

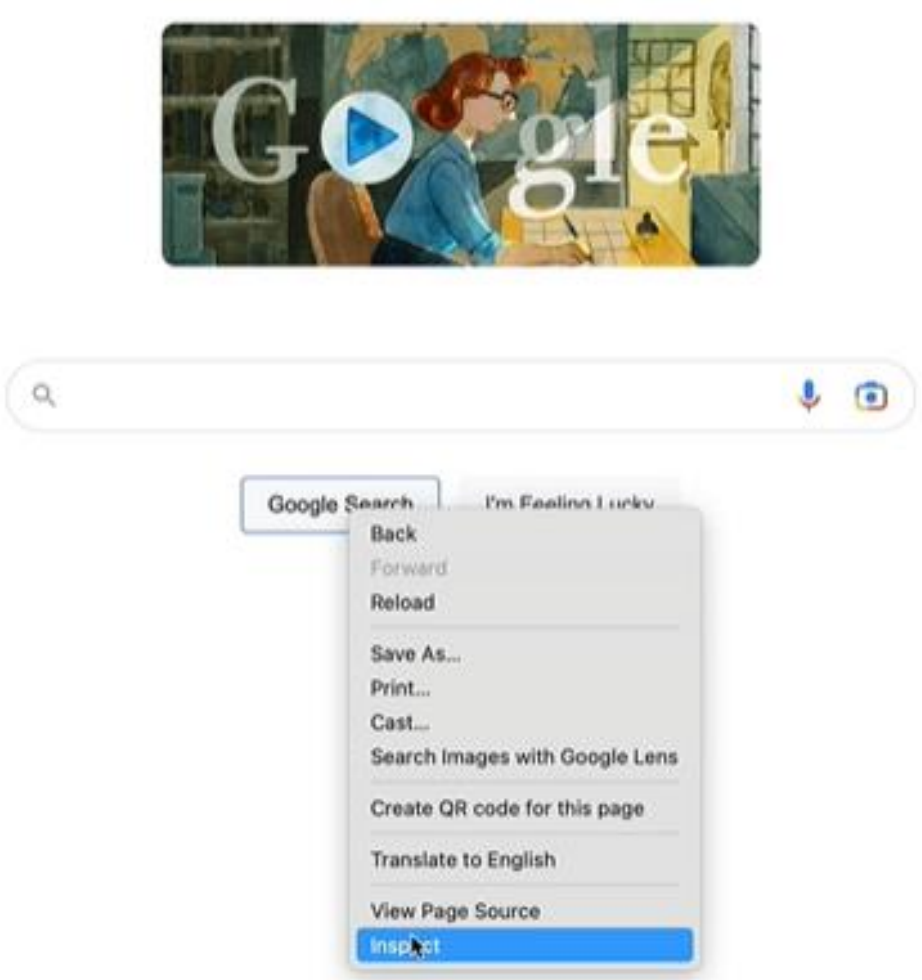

CSS
{..}
JS
</>
Google
Google Search
I'm feeling lucky
The
App Brewery

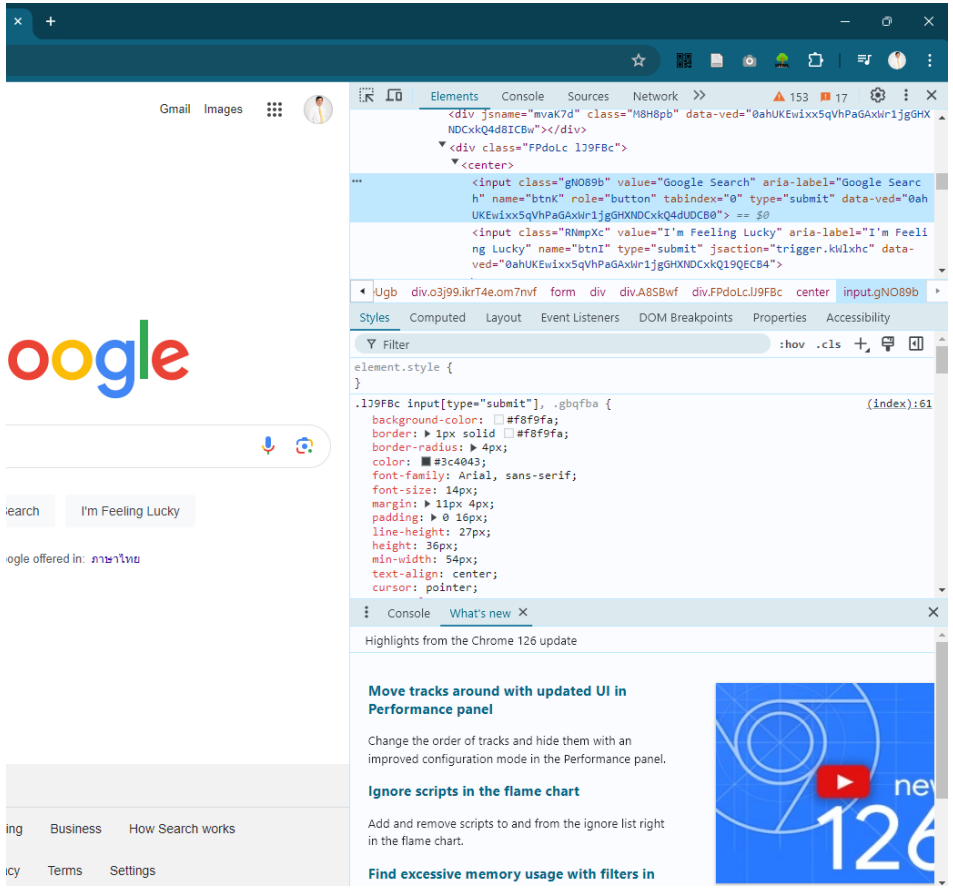

JS
Google
Google Search
I'm Feeling Lucky
The
App Brewery



The App Brewery

www.google.co.th





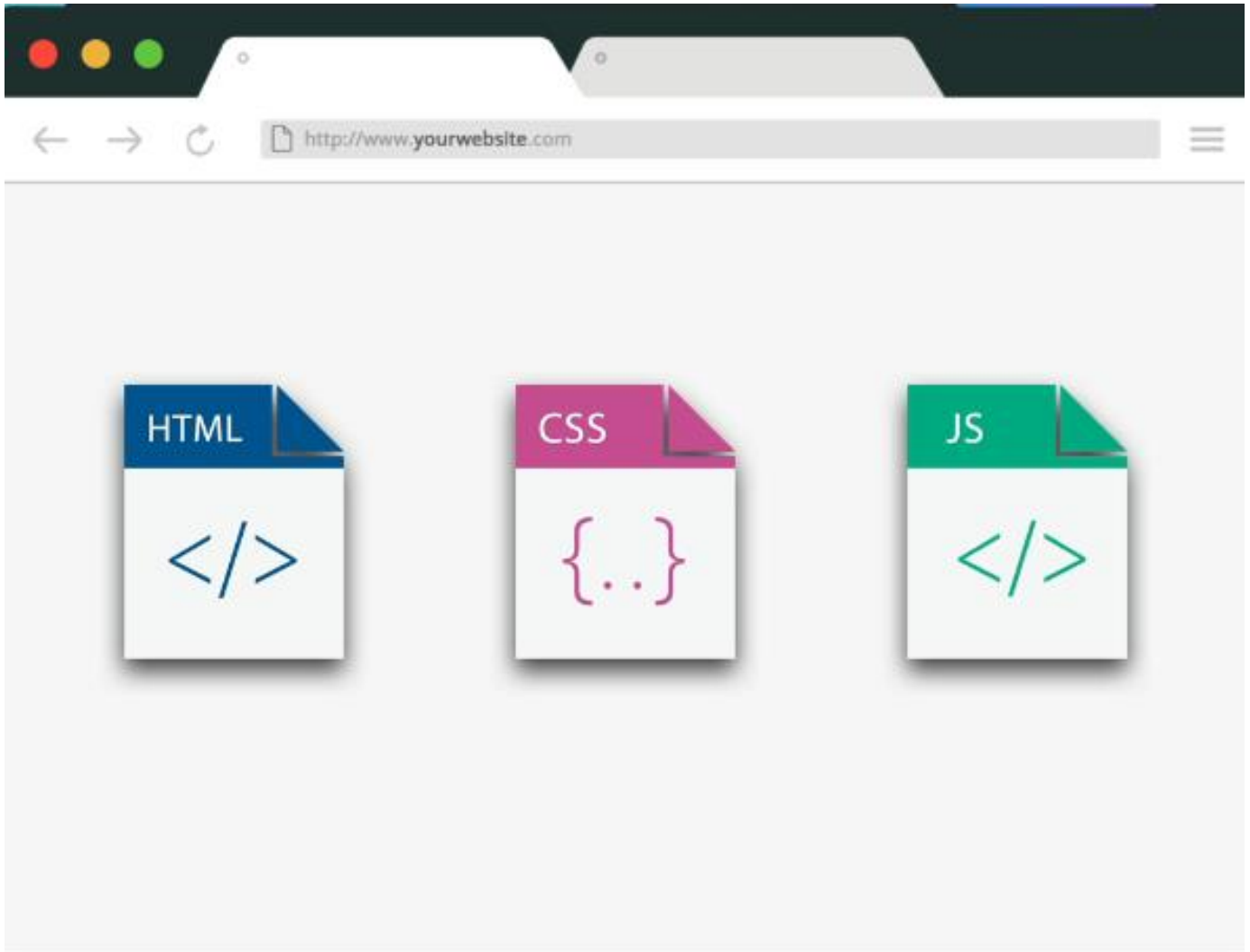
The App Brewery

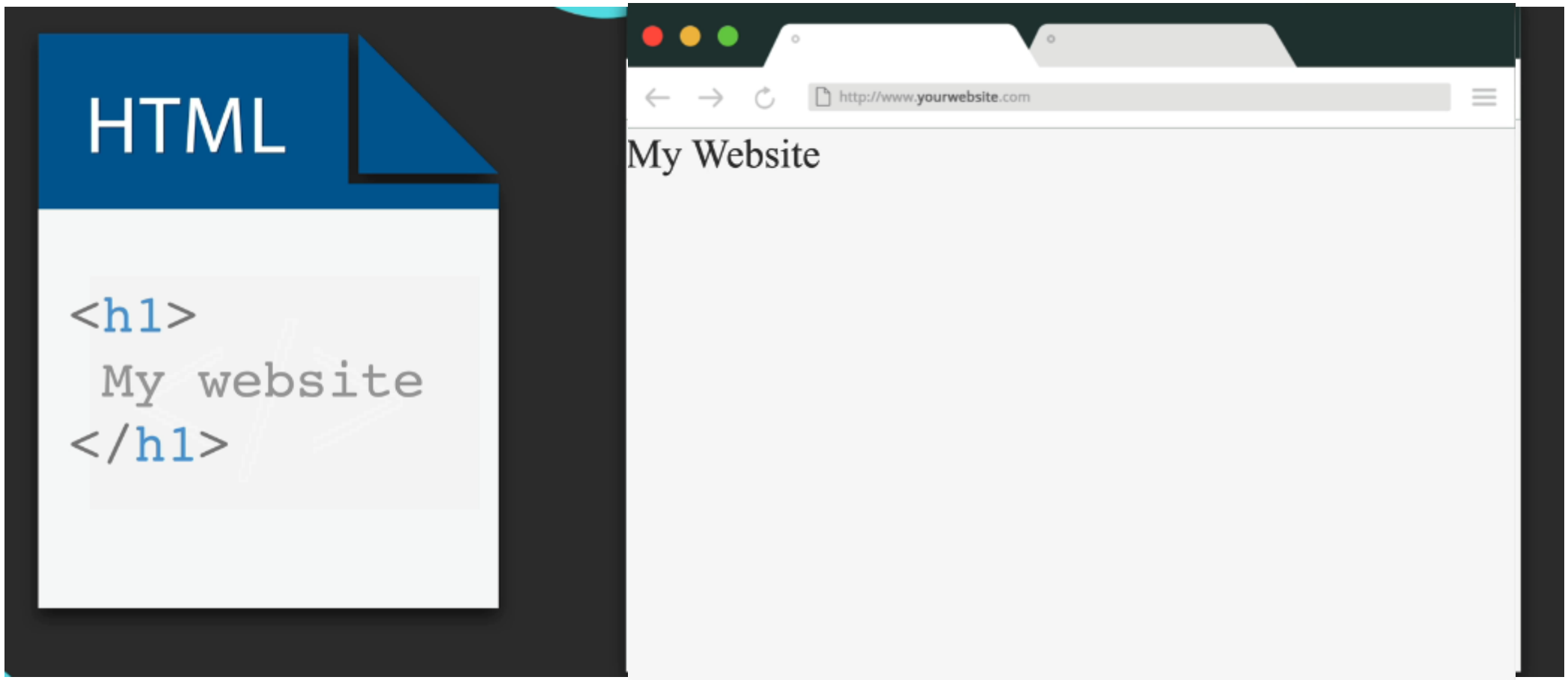
What is HTML?

Introduction to HTML

www.appbrewery.com



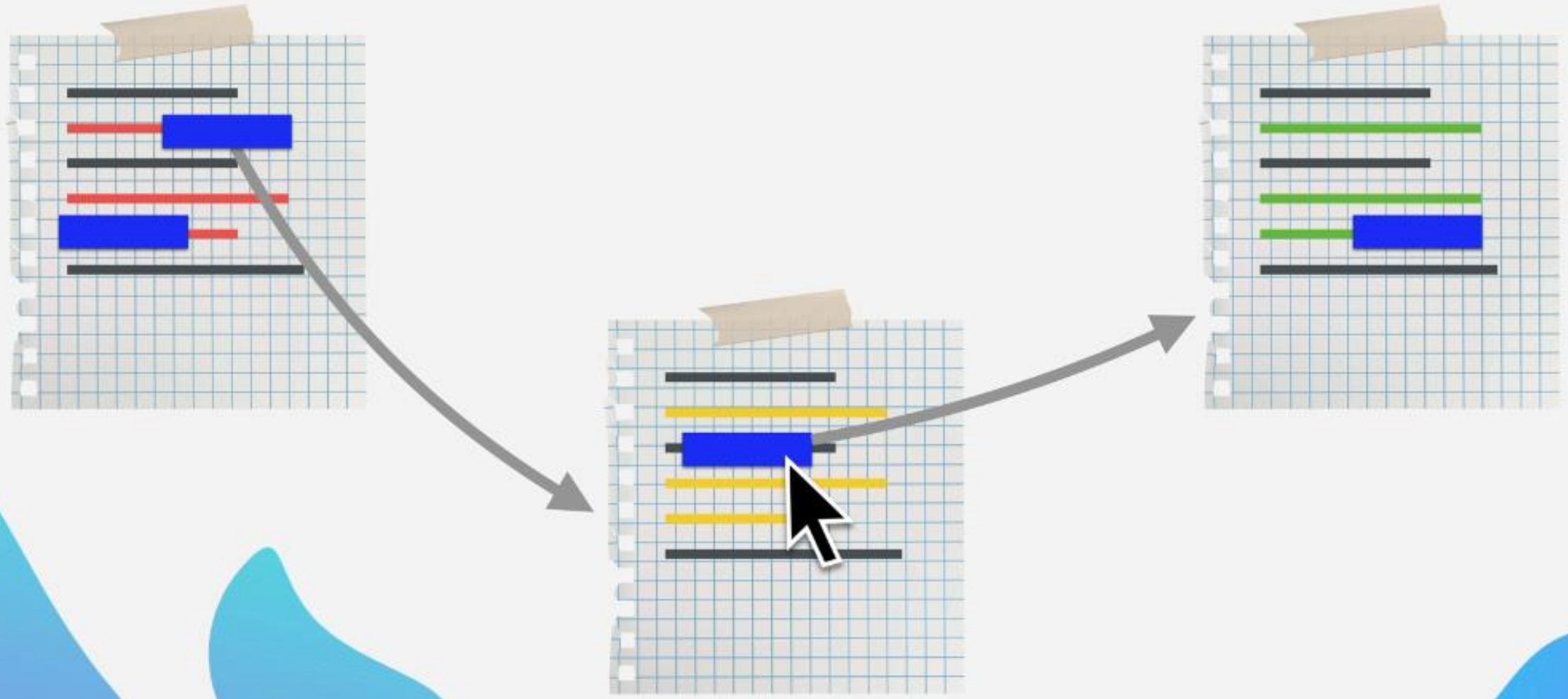

http://www.yourwebsite.com
HTML
</>
CSS
{..}
JS
</>


HTML
<h1>
My website
</h1>
http://www.yourwebsite.com
My Website

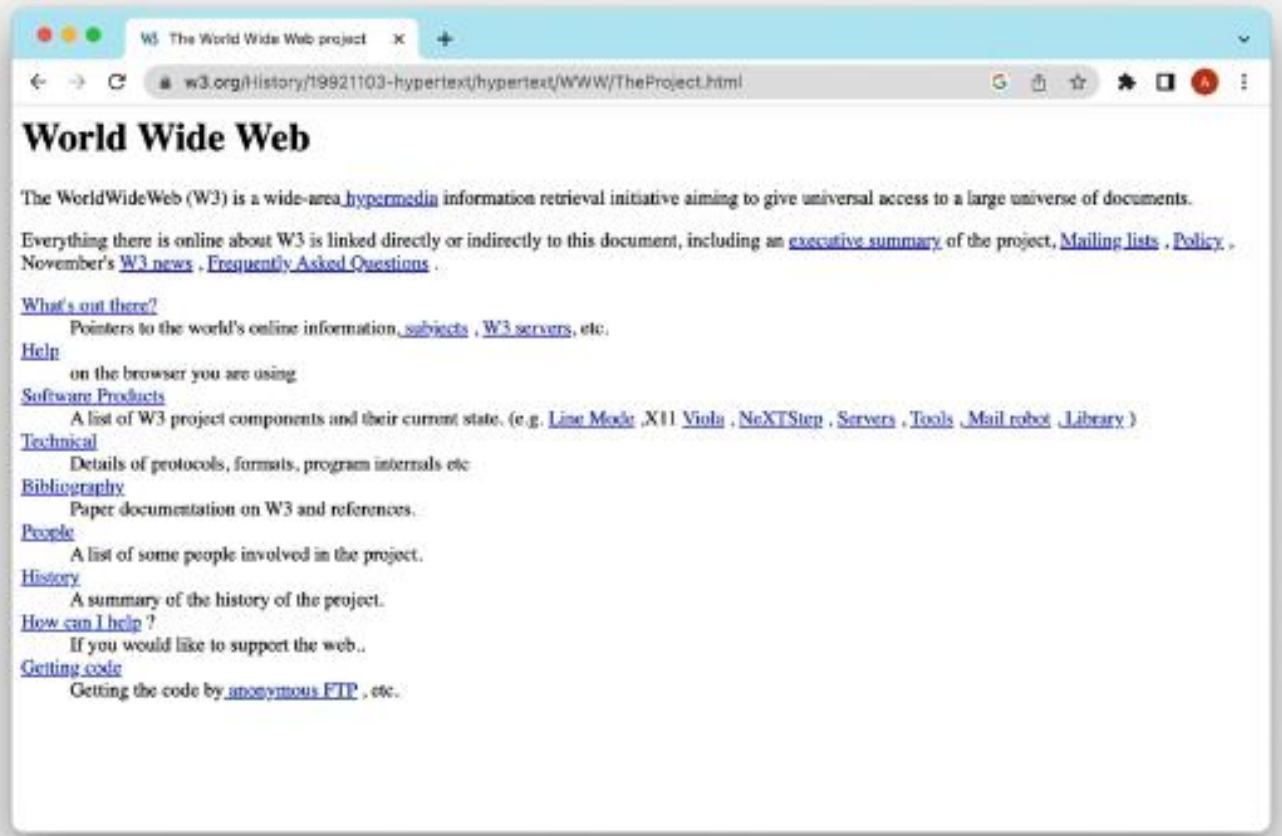
HTML

HyperText Markup Language

HyperText



World's First Website



Markup Language

“This is a quote”

inexplicable growth spurt.

capitalize

when in new york, do as the romans, the australians, or the martians would do.

superscript

E=mc2, even if we don't understand it, should be E=mc².

subscript

Do we breathe $0\hat{2}$ or 0_2?

italics

really
I really want a room of my own.

delete italic or bold or underline

Writers should use *italics*, **boldface** type, and underlining sparingly.

bold

choose one
Please ensure you choose one of the following birthday presents.

flush left

Text has a way of shifting when you're not looking.

flush right

Text can also hang around in the middle, instead of being aligned at the right margin:

J.K. Rowling

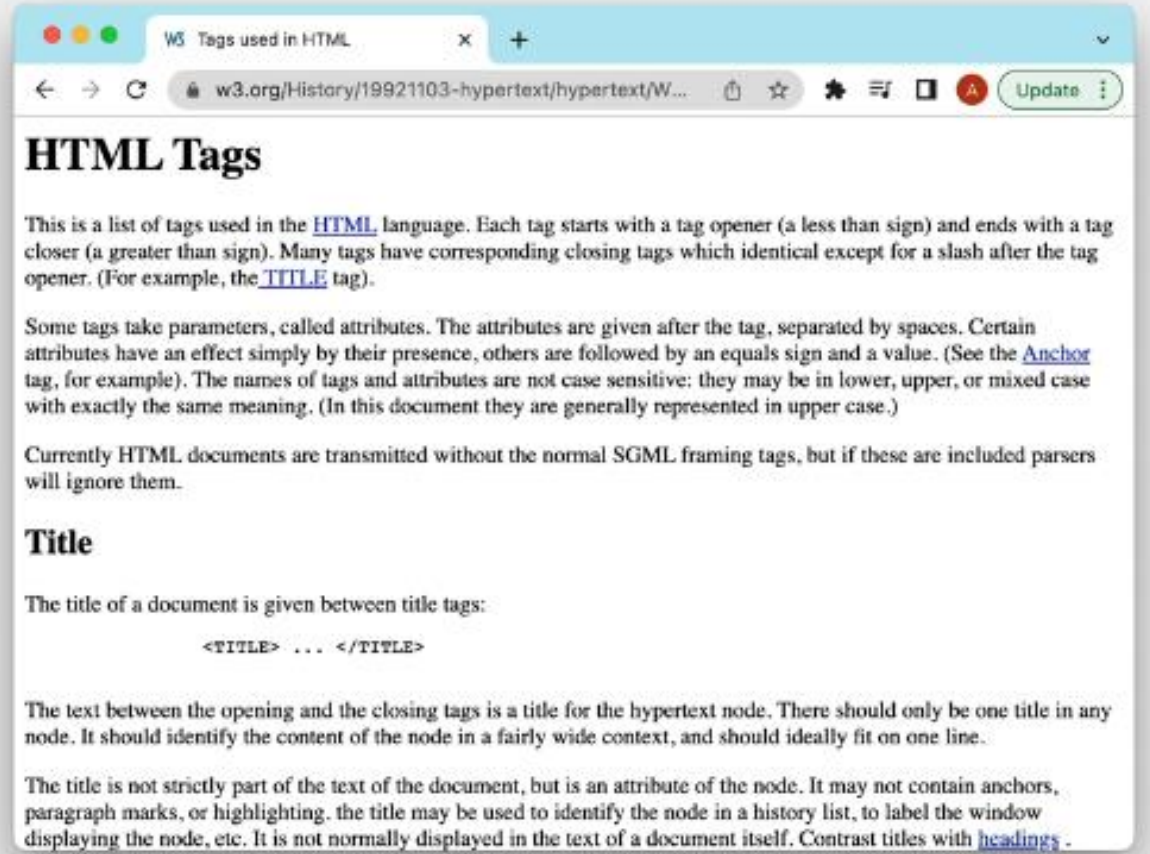
center

This line will be centered.

align

A few great authors:
Norman Rush
David Foster Wallace

How to Markup with HTML?



This tag allows the rest of a file to be read efficiently without parsing. Its presence is an optimisation. There is no closing tag.

Example sections

These styles allow text of fixed-width characters to be embedded absolutely as is into the document. The format is:

```
<LISTING>
        ...
</LISTING>
```

The text between these tags is to be portrayed in a fixed width font, so that any formatting done by character spacing on successive lines will be maintained. Between the opening and closing tags:

- The text may contain any ISO Latin printable characters, including the tag opener, so long as it does not contain the closing tag in full.
- Line boundaries are significant, and are to be interpreted as a move to the start of a new line.
- The ASCII Horizontal Tab (HT) character should be interpreted as the smallest positive nonzero number of spaces which will leave the number of characters so far on the line as a multiple of 8. Its use is not recommended however.

The LISTING tag is portrayed so that at least 132 characters will fit on a line. The XMP tag is portrayed in a font so that at least 80 characters will fit on a line but is otherwise identical to LISTING. The examples of markup are here given using the XMP tag.

Paragraph

This tag indicates a new paragraph. The exact representation of this (indentation, leading, etc) is not defined here, and may be a function of other tags, style sheets etc. The format is simply

```
<P>
```

[In SGML terms, paragraph elements are transmitted in minimised form).

Headings

Several levels (at least six) of heading are supported. Note that a hypertext document tends to need less levels of heading than a normal document whose only structure is given by the nesting of headings. H1 is the highest level of heading, and is recommened for the start of a hypertext node. It is suggested that the first heading be one suitable for a reader who is already browsing in related information, in contrast to the title tag which should identify the node in a wider context.

```
<H1>, <H2>, <H3>, <H4>, <H5>, <H6>
```

These tags are kept as defined in the CERN SGML guide. Their definition is completely historical, deriving from the AAP tag set. A difference is that HTML documents allow headings to be terminated by closing tags:

```
<H2>Second level heading</h2>
```

Address

This tag is for address information, signatures, etc, normally at the top or bottom of a document. typically, it is italic and/or right justified or indented. The format is:

```
<ADDRESS> text ... </ADDRESS>
```

All HTML5 Tags

<a>
<abbr>
<address>
<area>
<article>
<aside>
<audio>
<b>
<base>
<bdi>
<bdo>
<blockquote>
<body>

<button>

<canvas>
<caption>
<cite>
<code>
<col>
<colgroup>
<data>
<datalist>
<dd>
<del>
<details>
<dfn>
<dialog>

<dl>

<dt>

<embed>
<fieldset>
<figcaption>
<figure>
<footer>
<form>
<head>
<header>
<hgroup>
<h1> to <h6>
<hr>
<html>
<i>

<iframe>
<img>
<input>
<ins>
<kbd>
<keygen>
<label>
<legend>

<link>
<main>
<map>
<mark>
<menu>
<menuitem>

<meta>
<meter>
<nav>
<noscript>
<object>

<optgroup>
<option>
<output>
<p>
<param>
<picture>
<pre>
<progress>
<q>

<rp>
<rt>
<ruby>
<s>
<samp>
<script>
<section>
<select>
<small>
<source>

<style>
<sub>
<summary>

<sup>
<svg>
<table>
<tbody>
<td>
<template>
<textarea>
<tfoot>
<th>
<thead>
<title>
<tr>

<var>
<video>
<wbr>

All You Need to Know

<a>
<abbr>
<address>
<area>
<article>
<aside>
<audio>
<b>
<base>
<bdi>
<bdo>
<blockquote>
<body>

<button>
<canvas>
<caption>
<cite>
<code>
<col>
<colgroup>
<data>
<datalist>
<dd>
<del>
<details>
<dfn>
<dialog>

<dl>
<dt>

<embed>
<fieldset>
<figcaption>
<figure>
<footer>
<form>
<head>
<header>
<hgroup>
<h1> to <h6>
<hr>
<html>
<i>
<iframe>
<img>
<input>
<ins>
<kbd>
<keygen>
<label>
<legend>

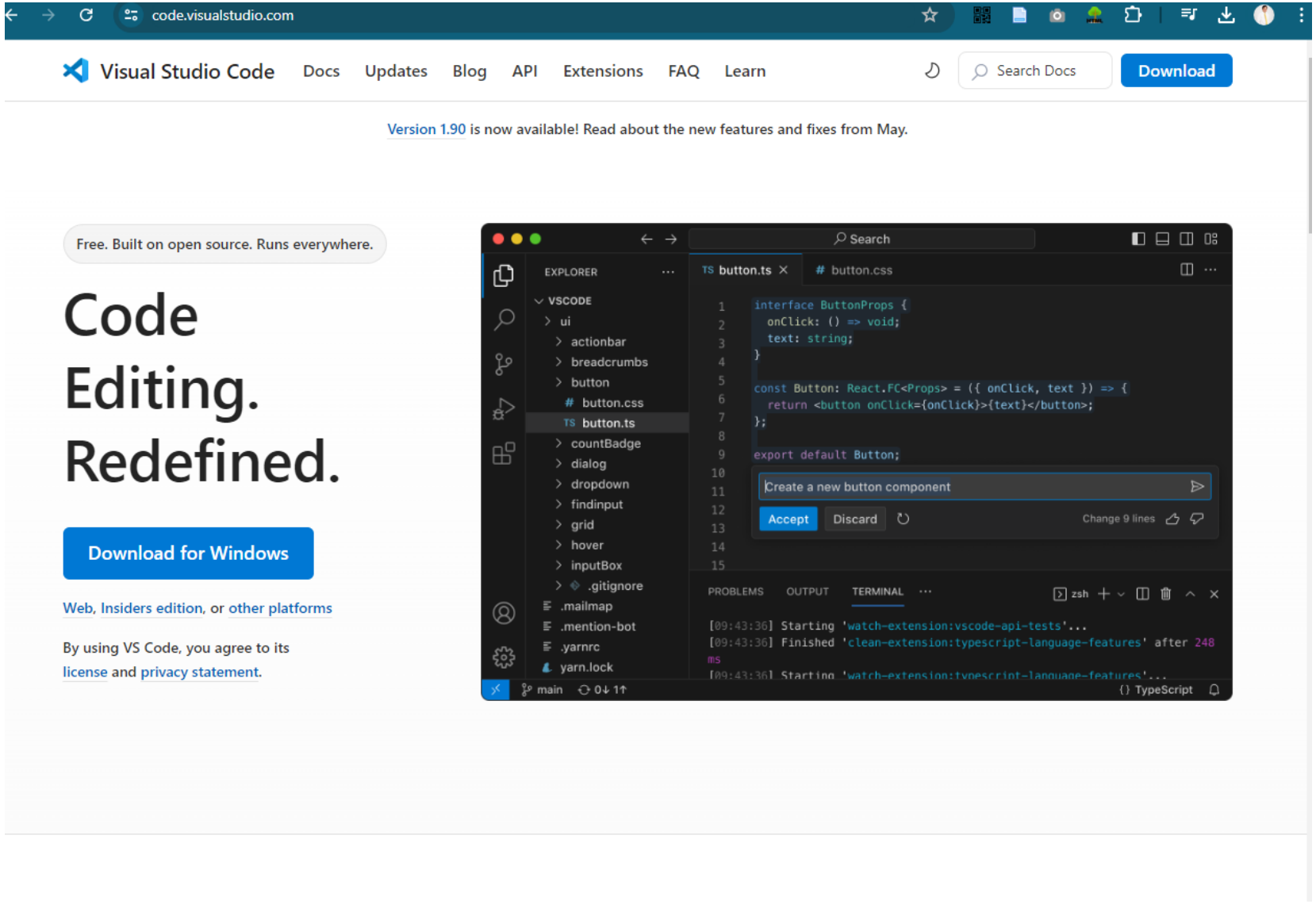
<link>
<main>
<map>
<mark>
<menu>
<menuitem>
<meta>
<meter>
<nav>
<noscript>
<object>

<optgroup>
<option>
<output>
<p>
<param>
<picture>
<pre>
<progress>
<q>
<rp>
<rt>
<ruby>
<s>
<samp>
<script>
<section>
<select>
<small>
<source>

<style>
<sub>
<summary>
<sup>
<svg>
<table>
<tbody>
<td>
<template>
<textarea>
<tfoot>
<th>
<thead>
<title>
<tr>

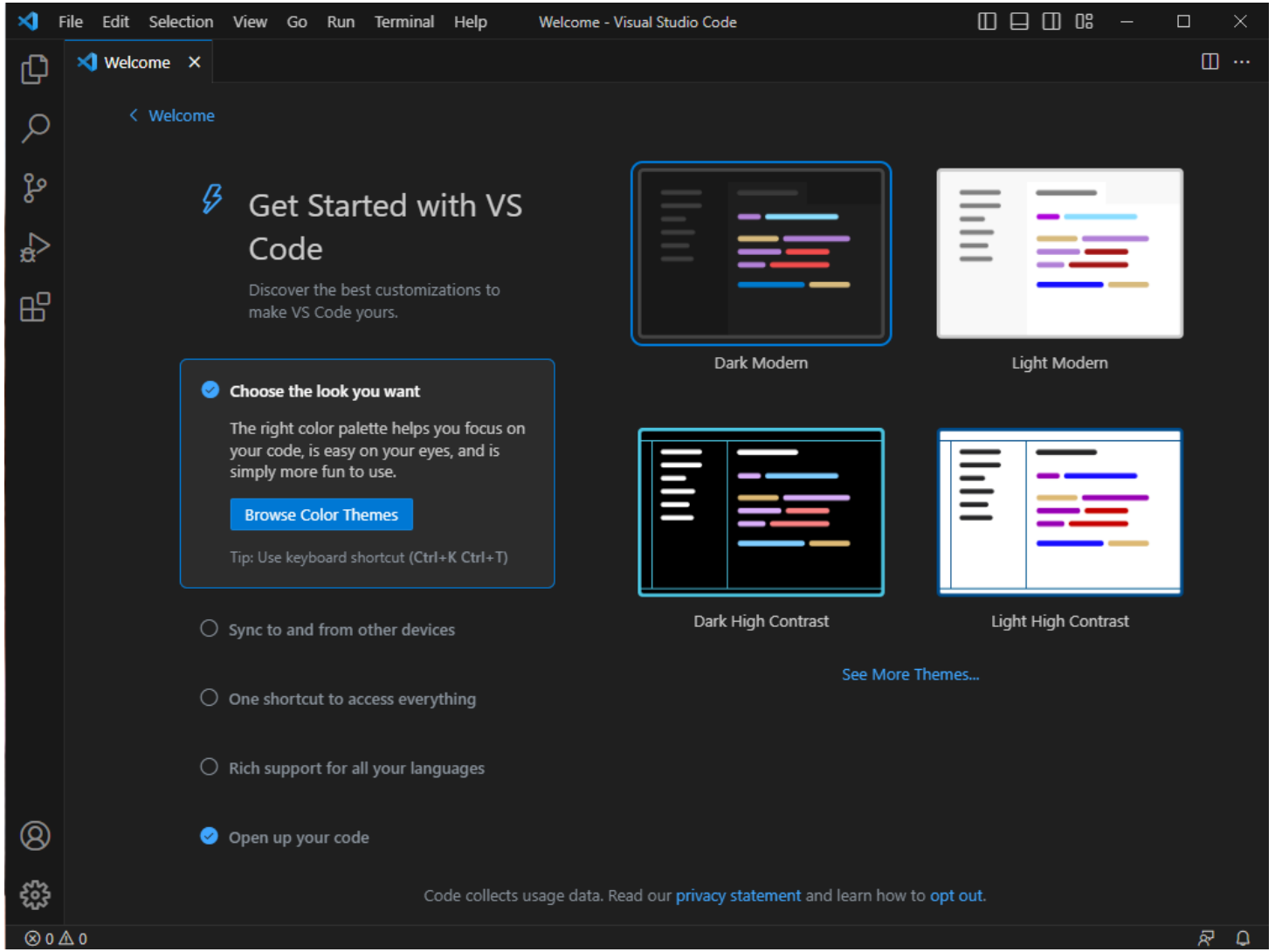
<var>
<video>
<wbr>

Visual Studio Code



Visual Studio Code

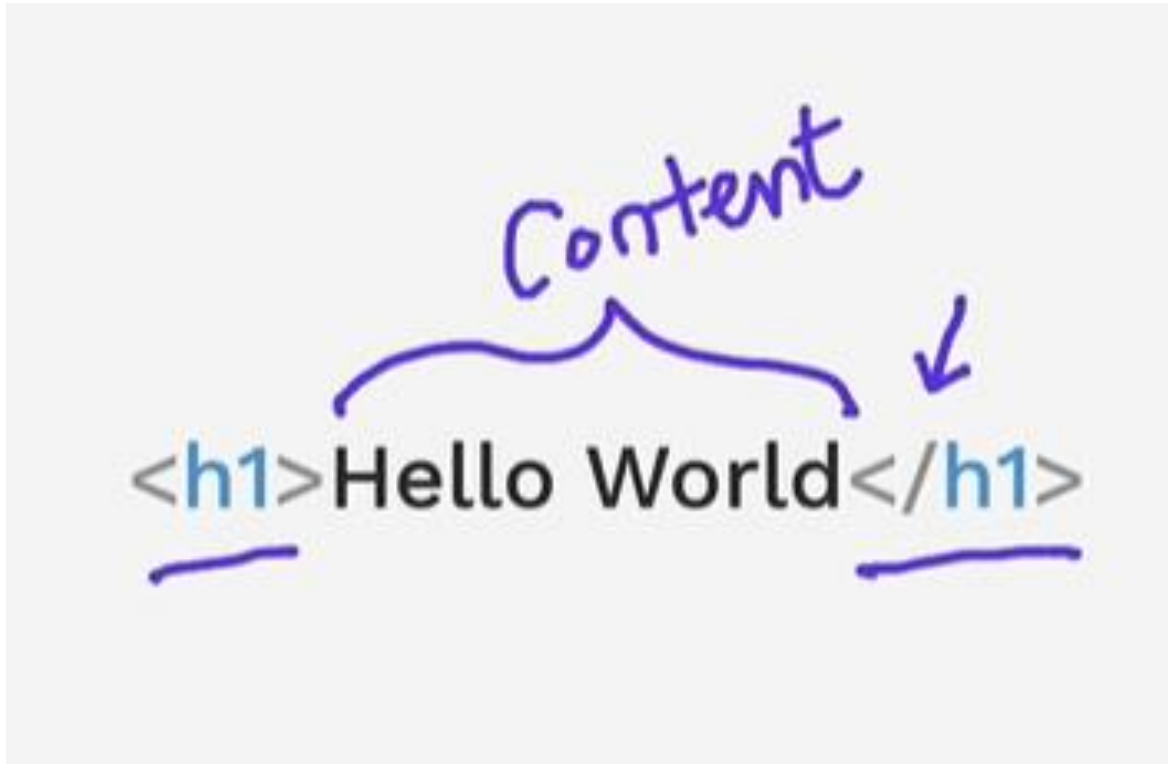


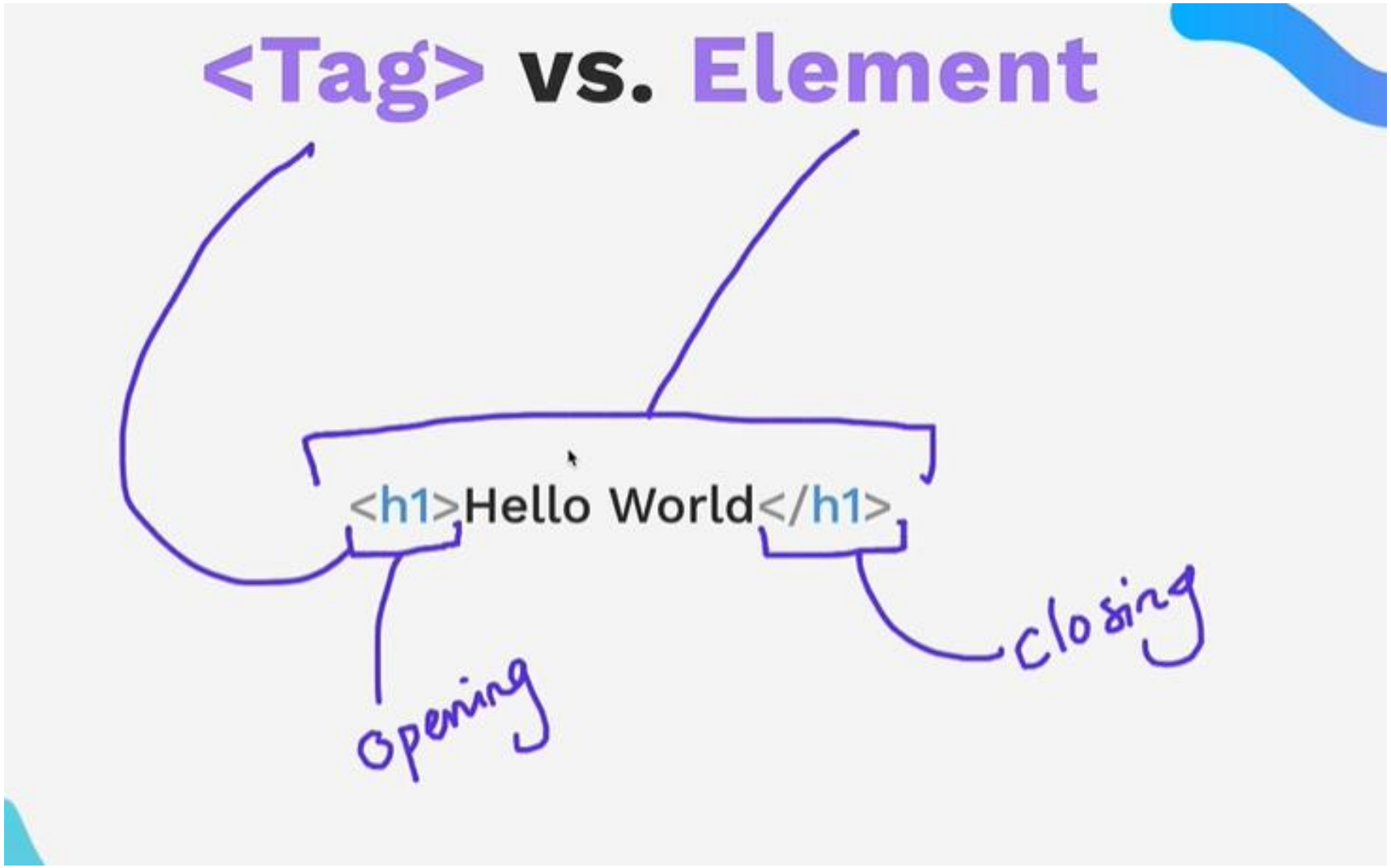


The Heading Element

<h1> to <h6>




Content
<h1>Hello World</h1>


<Tag> vs. Element
<h1>Hello World</h1>
opening
closing

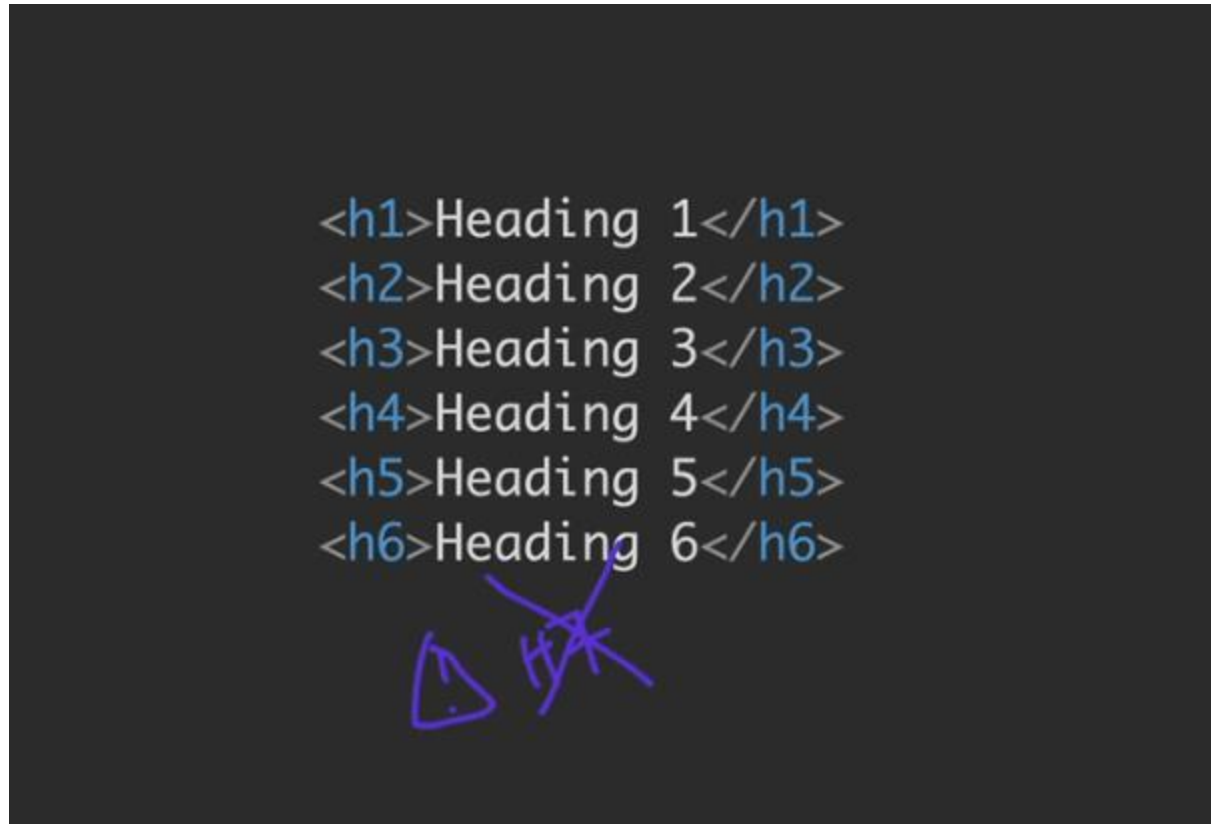
Section Headings

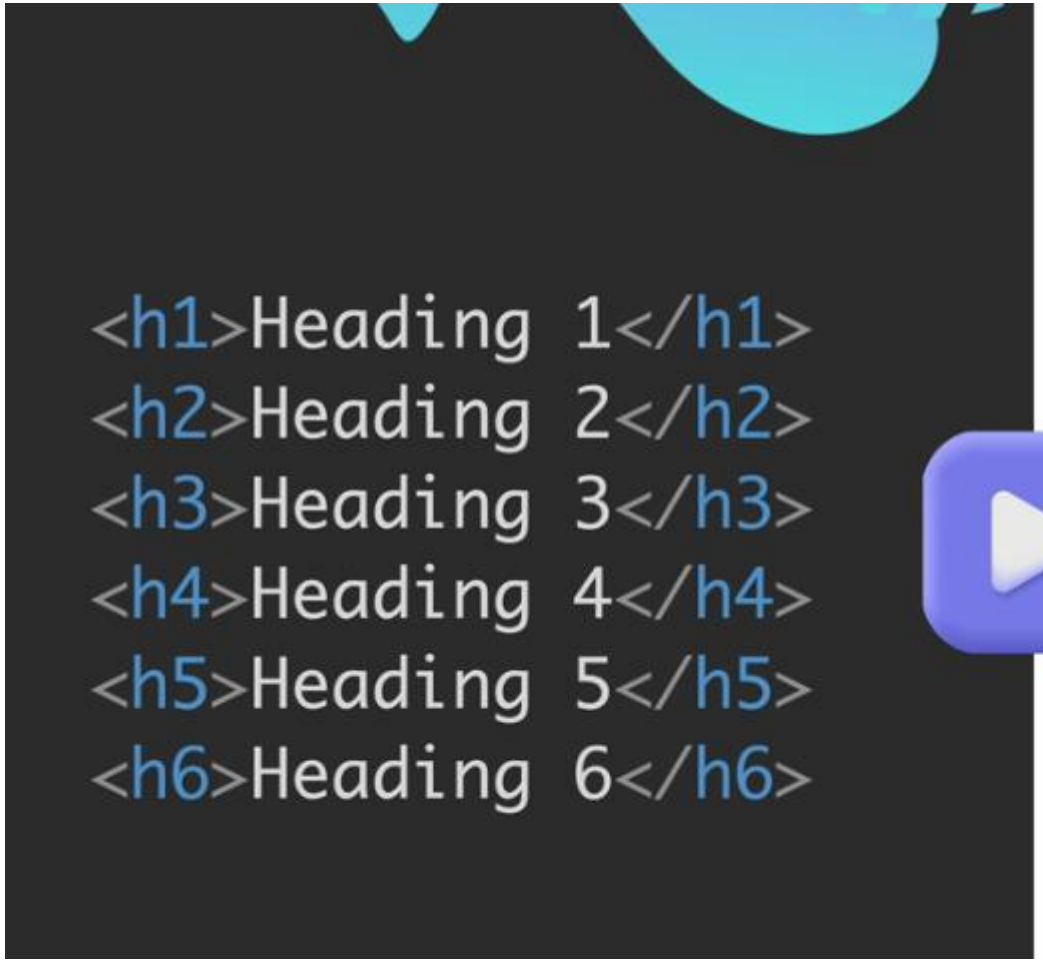
H1

H2

H3?

TABLE OF CONTENTS 1
REVISION HISTORY 2
APPROVALS 2
1. PURPOSE 3
2. POLICY 3
3. SCOPE 3
4. REFERENCES 3
5. DEFINITIONS 3
6. ACRONYMS 4
7. RESPONSIBILITIES 4
8. PROCEDURE 4
8.1 Record Type and Retention 4
9. QUALITY SYSTEM RECORDS 5
10. DEVICE MASTER RECORDS (DMR) 5
11. DEVICE HISTORY FILE (DHR'S) 6
12. COMPLIANT AND COMPLAINT INVESTIGATION FILES 6
12.1 Obsolete Records 6
12.2 Quality Records 7
12.3 Quality Manager (QM) 7
12.4 Non-exhaustive List of Quality Records 7
13. RECORDS 7


<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>



Heading 1

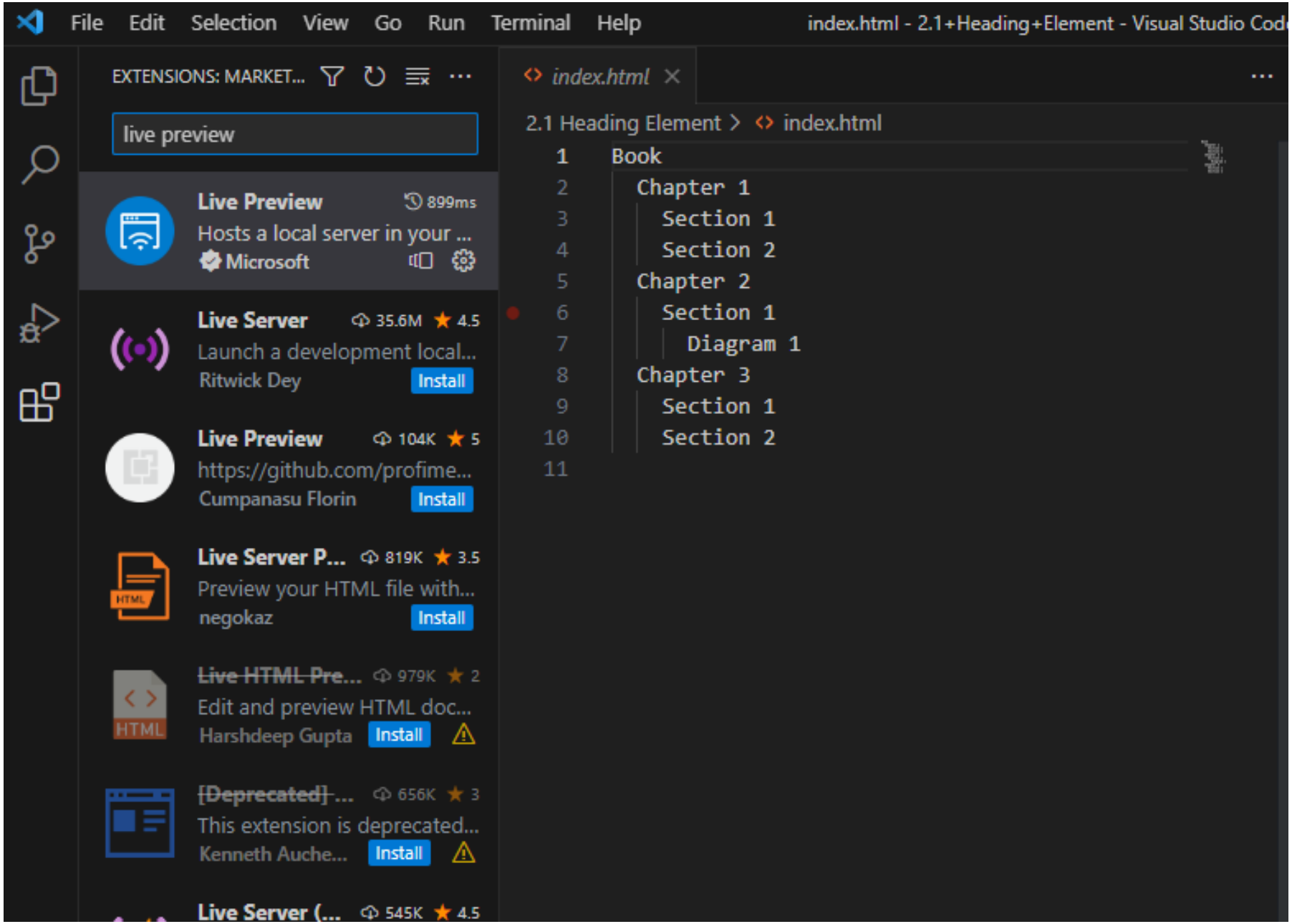
Heading 2

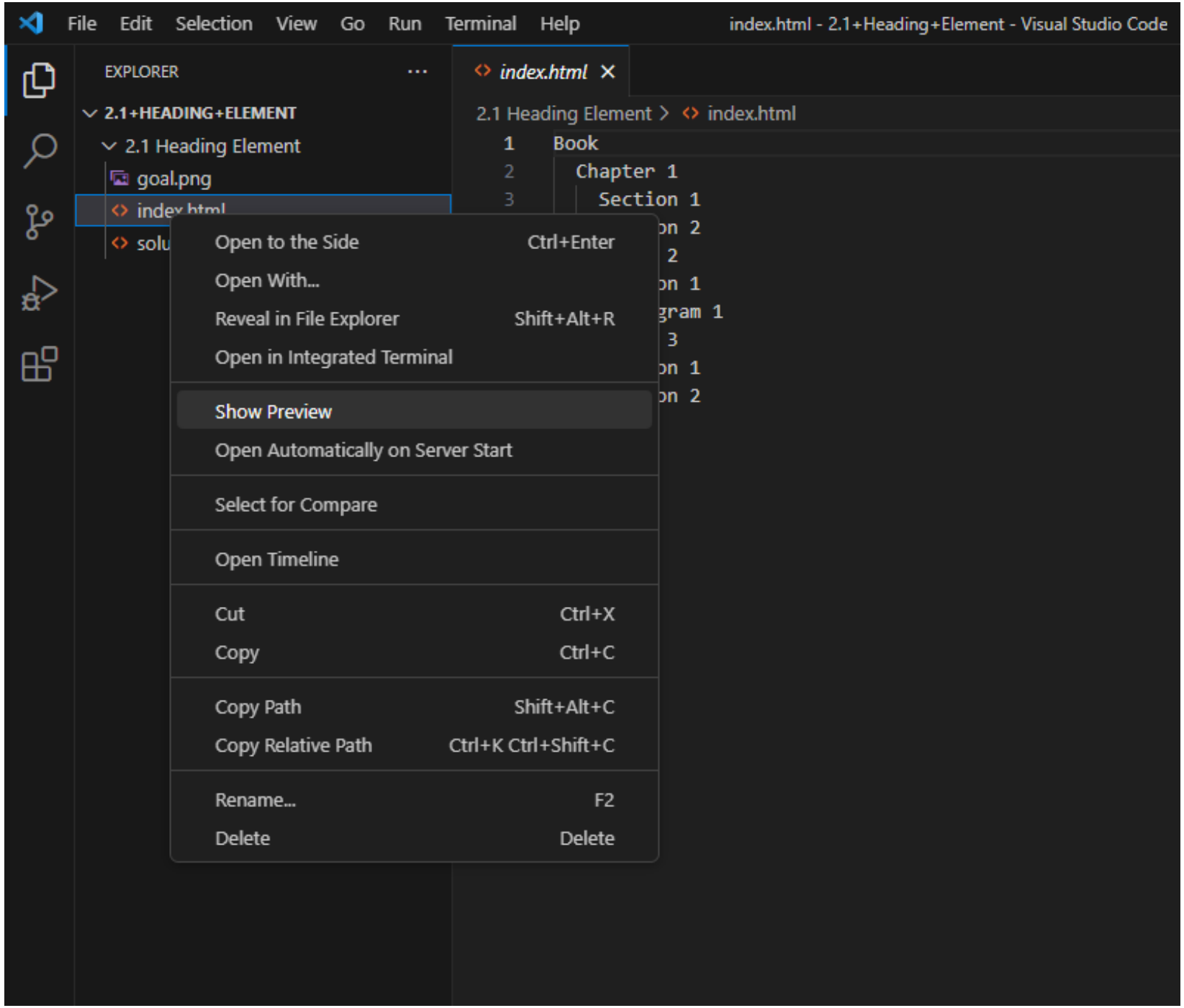
Heading 3

Heading 4

Heading 5

Heading 6

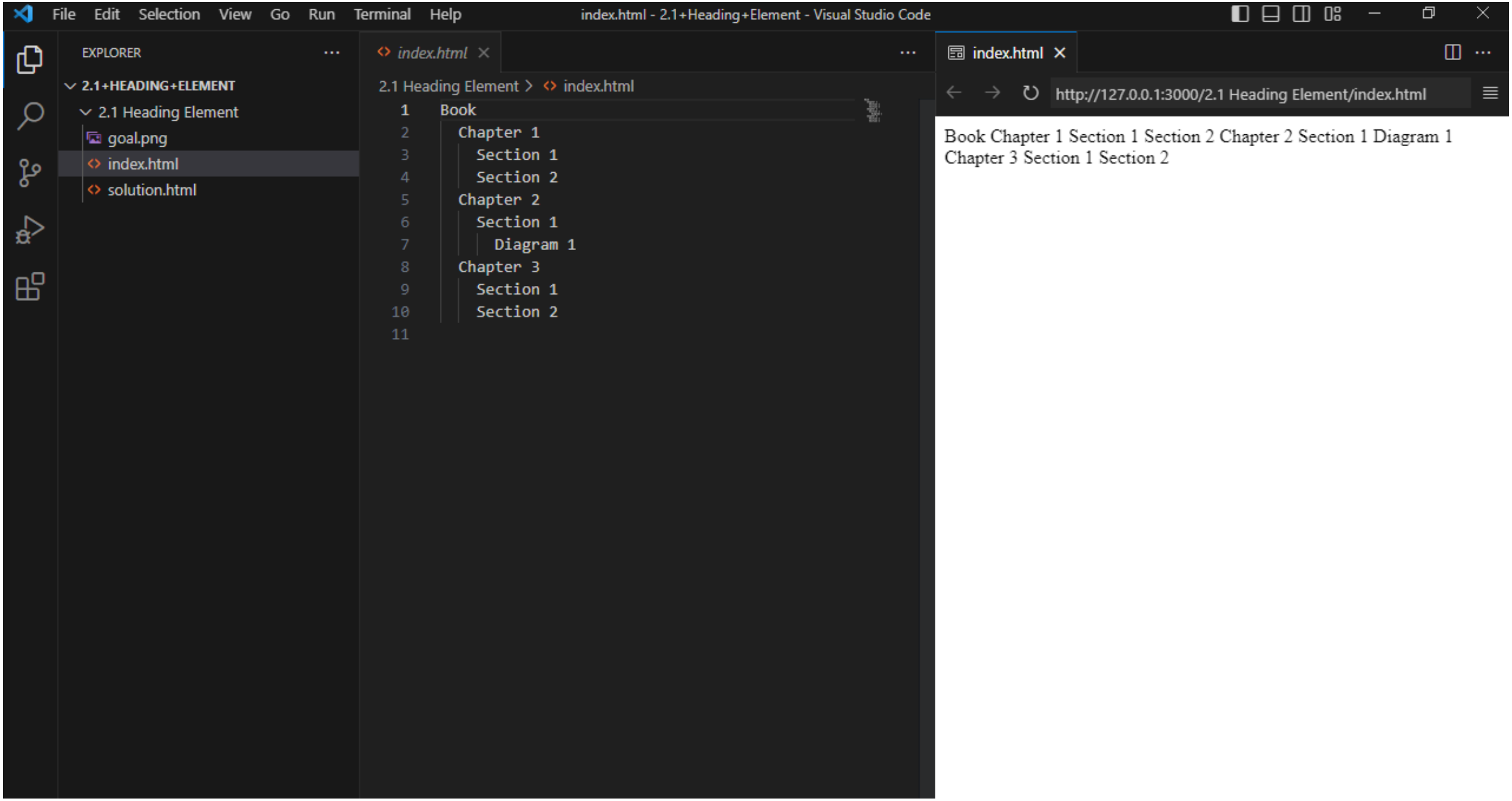

File Edit Selection View Go Run Terminal Help
index.html - 2.1+Heading+Element - Visual Studio Cod
EXTENSIONS: MARKET...
live preview
Live Preview 899ms
Hosts a local server in your ...
Microsoft
Live Server 35.6M 4.5
Launch a development local...
Ritwick Dey
Install
Live Preview 104K 5
https://github.com/profime...
Cumpanasu Florin
Install
Live Server P... 819K 3.5
Preview your HTML file with...
negokaz
Install
Live HTML Pre... 979K 2
Edit and preview HTML doc...
Harshdeep Gupta
Install
[Deprecated] ... 656K 3
This extension is deprecated...
Kenneth Auche...
Install
Live Server (... 545K 4.5
index.html
2.1 Heading Element > index.html
1 Book
2 Chapter 1
3 Section 1
4 Section 2
5 Chapter 2
6 Section 1
7 Diagram 1
8 Chapter 3
9 Section 1
10 Section 2
11

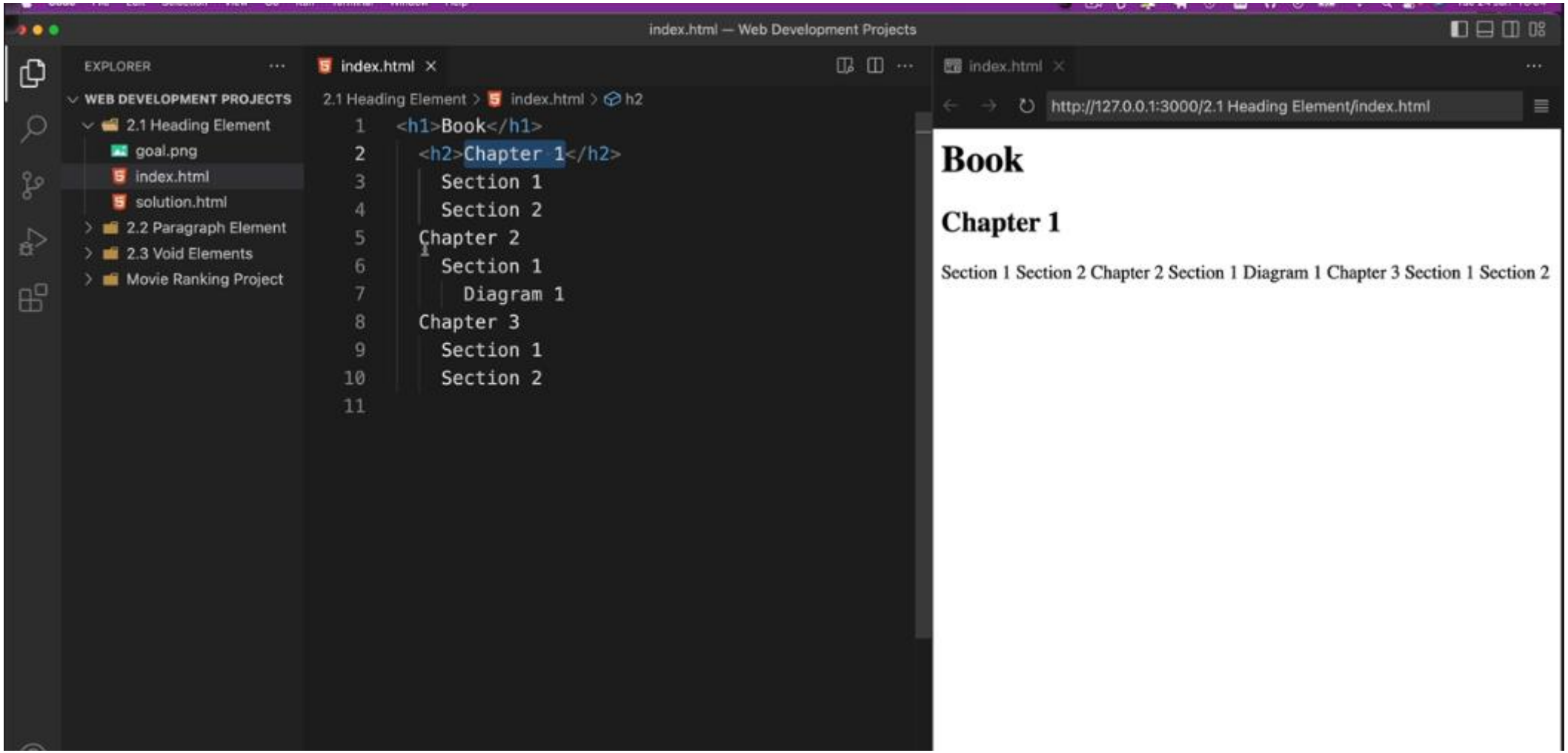

File Edit Selection View Go Run Terminal Help
index.html - 2.1+Heading+Element - Visual Studio Code
EXPLORER
2.1+HEADING+ELEMENT
2.1 Heading Element
goal.png
index.html
solu
index.html
2.1 Heading Element > index.html
1 Book
2 Chapter 1
3 Section 1
Open to the Side Ctrl+Enter
Open With...
Reveal in File Explorer Shift+Alt+R
Open in Integrated Terminal
Show Preview
Open Automatically on Server Start
Select for Compare
Open Timeline
Cut Ctrl+X
Copy Ctrl+C
Copy Path Shift+Alt+C
Copy Relative Path Ctrl+K Ctrl+Shift+C
Rename... F2
Delete Delete

Try an Exercise



2.1 Heading
Element.zip


File Edit Selection View Go Run Terminal Help
index.html - 2.1+Heading+Element - Visual Studio Code
EXPLORER
2.1+HEADING+ELEMENT
2.1 Heading Element
goal.png
index.html
solution.html
index.html
2.1 Heading Element > index.html
1 Book
2 Chapter 1
3 Section 1
4 Section 2
5 Chapter 2
6 Section 1
7 Diagram 1
8 Chapter 3
9 Section 1
10 Section 2
11
index.html
http://127.0.0.1:3000/2.1 Heading Element/index.html
Book Chapter 1 Section 1 Section 2 Chapter 2 Section 1 Diagram 1
Chapter 3 Section 1 Section 2

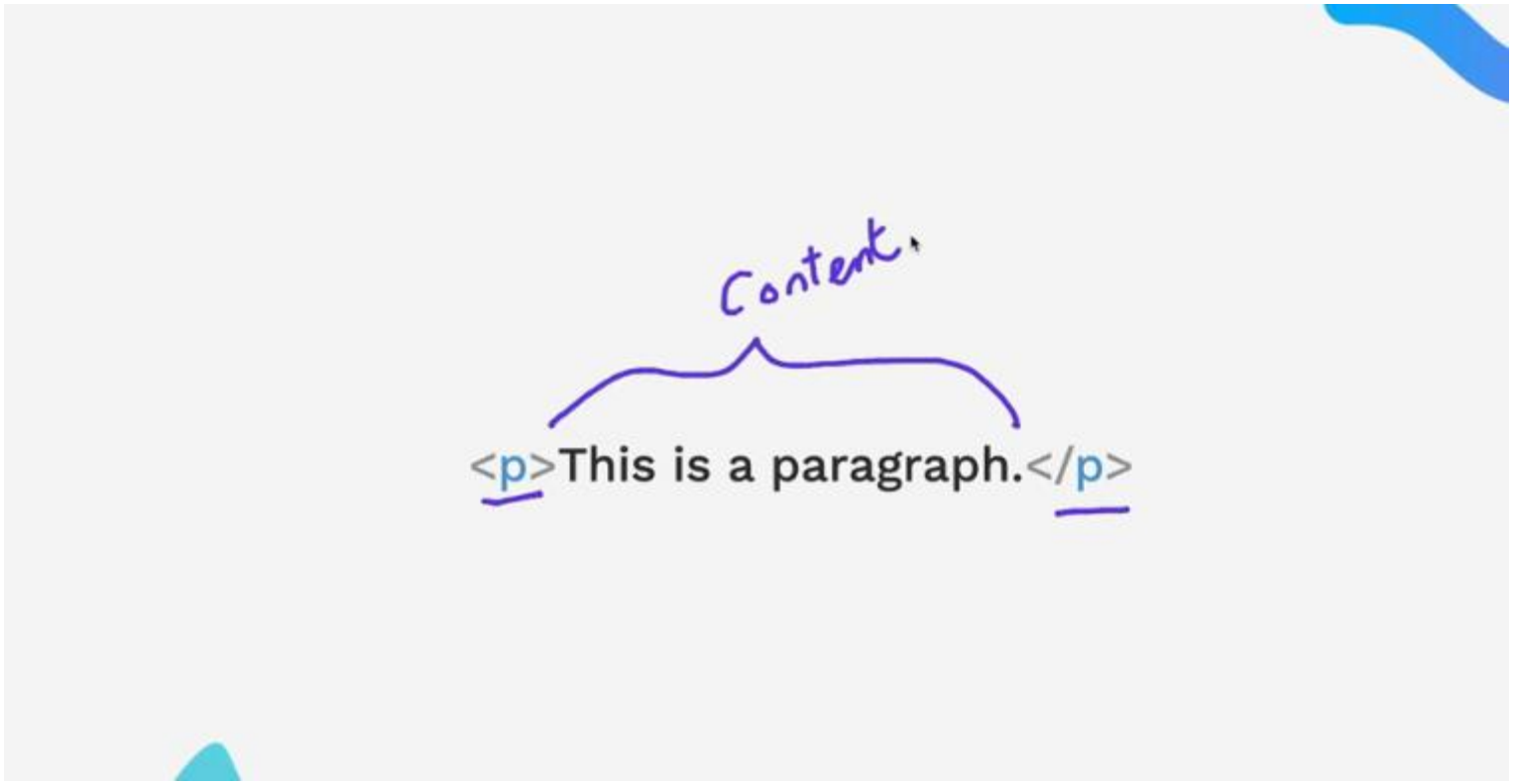

index.html — Web Development Projects
EXPLORER
WEB DEVELOPMENT PROJECTS
2.1 Heading Element
goal.png
index.html
solution.html
2.2 Paragraph Element
2.3 Void Elements
Movie Ranking Project
index.html
2.1 Heading Element > index.html > h2
<h1>Book</h1>
<h2>Chapter 1</h2>
Section 1
Section 2
Chapter 2
Section 1
Diagram 1
Chapter 3
Section 1
Section 2
index.html
http://127.0.0.1:3000/2.1 Heading Element/index.html
Book
Chapter 1
Section 1 Section 2 Chapter 2 Section 1 Diagram 1 Chapter 3 Section 1 Section 2

- https://developer.mozilla.org/en-US/docs/Web/HTML/Element/Heading_Elements

The Paragraph Element

<p>




Content
This is a paragraph.

This is a paragraph
This is a paragraph



This is a paragraph This is a paragraph

This is a paragraph
This is a paragraph



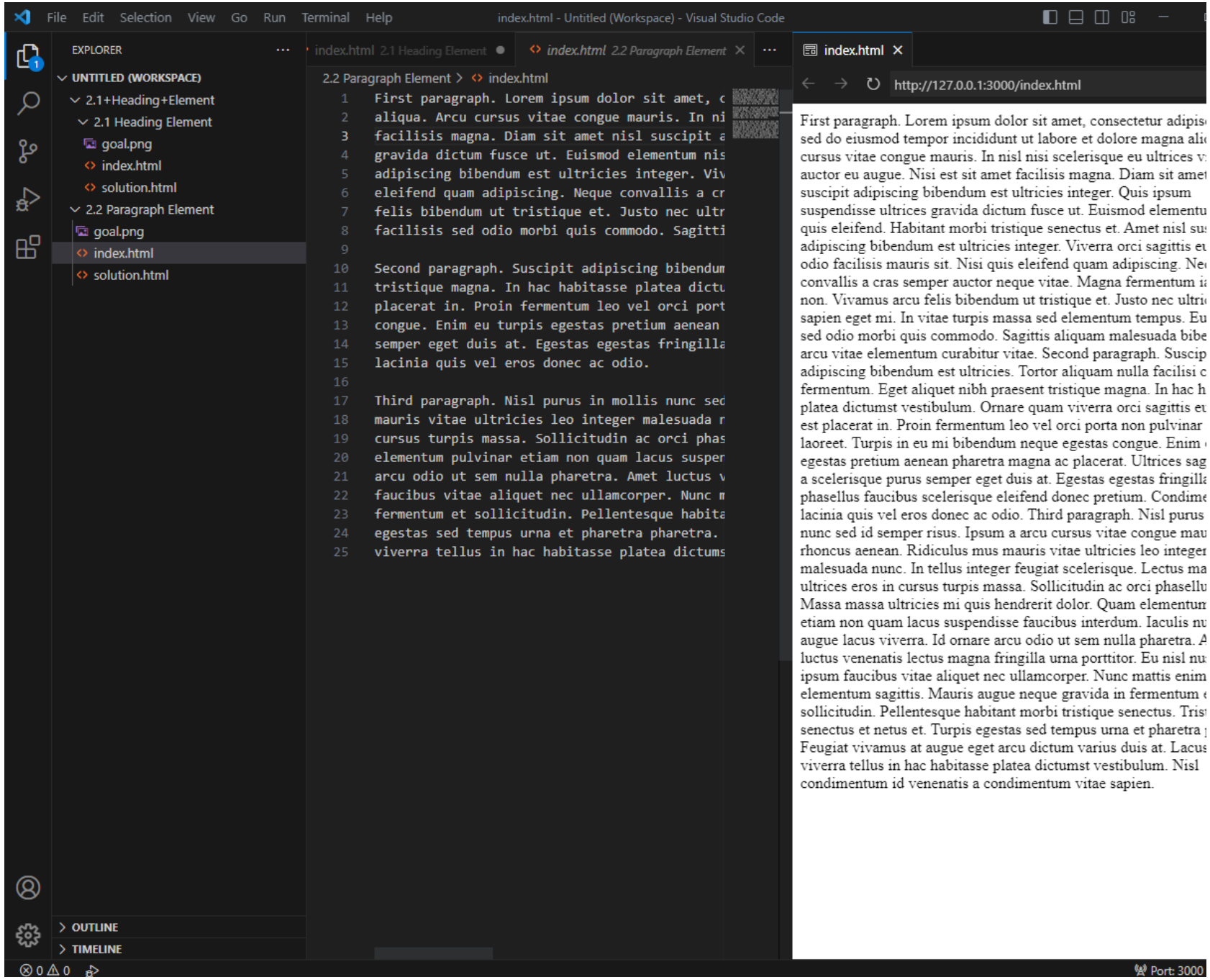
This is a paragraph

This is a paragraph

Try an Exercise



2.2 Paragraph Element.zip



https://www.lipsum.com/

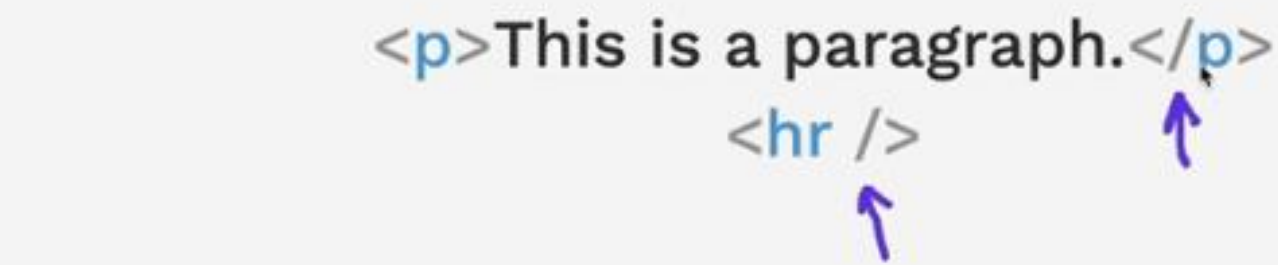
Lorem Ipsum คือข้อความตัวอย่างที่ใช้กันอย่างแพร่หลายในงานออกแบบสิ่งพิมพ์ การจัดหน้าหนังสือ เว็บไซต์ หรือแอปพลิเคชัน เพื่อ **แทนที่เนื้อหาจริง** ชั่วคราวระหว่างขั้นตอนการออกแบบ

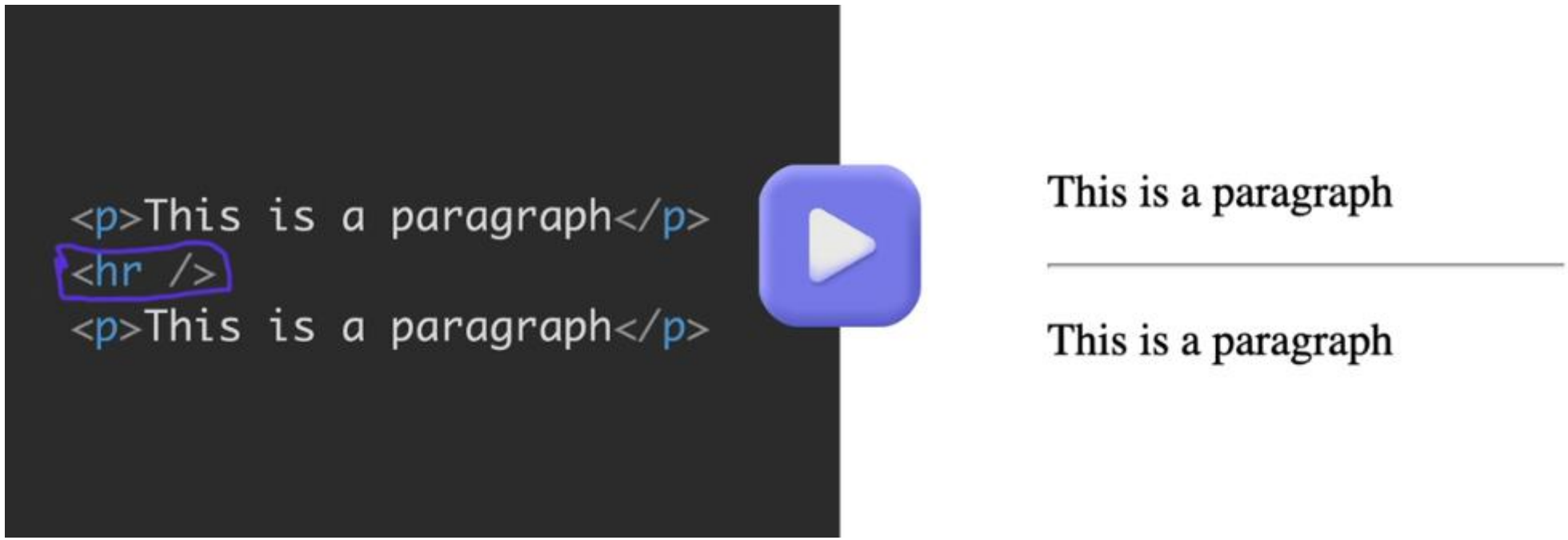
Void Elements

<hr /> and



Horizontal Rule Element





This is a paragraph

This is a paragraph

```
<p>
To see a World in a Grain of Sand
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand
And Eternity in an hour.
<p>
```



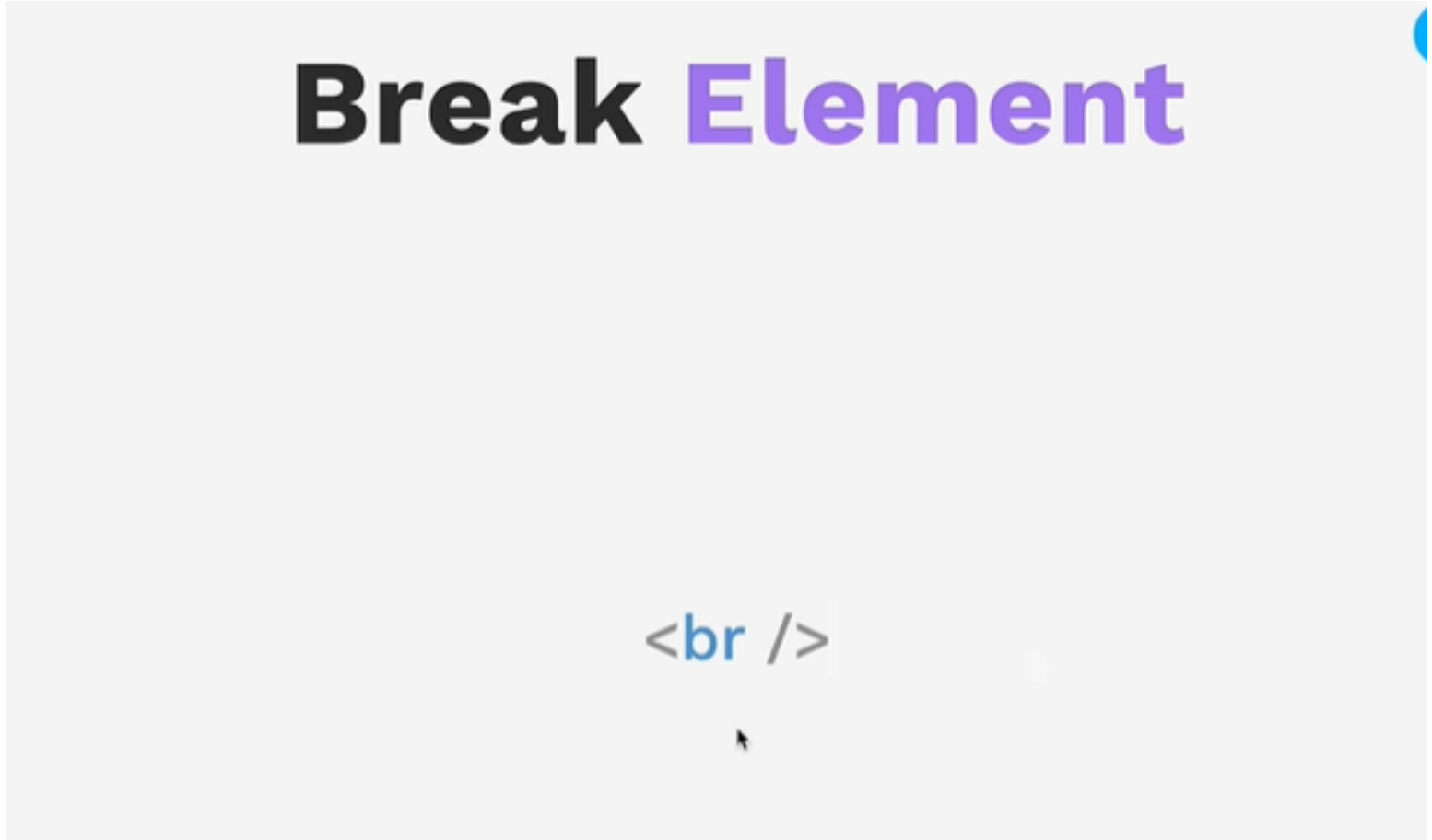
To see a World in a Grain of Sand And a Heaven in a Wild Flower, Hold Infinity in

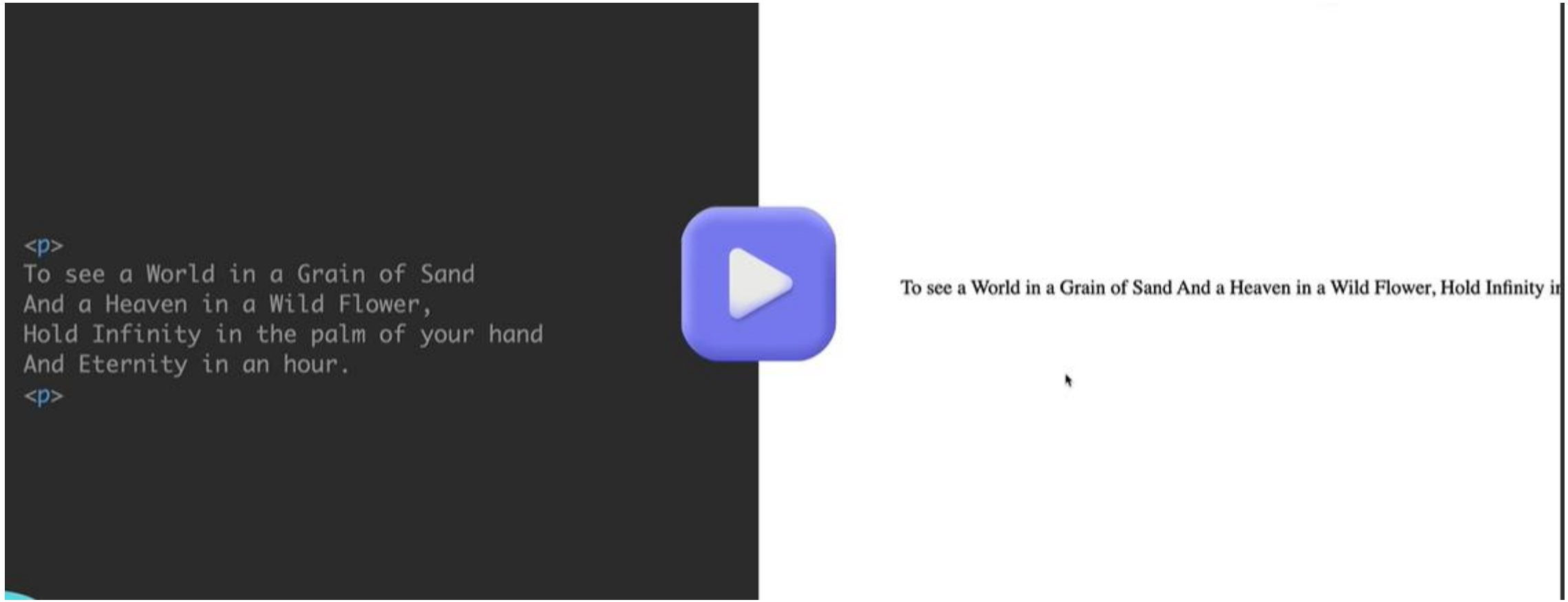
```
<p>
To see a World in a Grain of Sand<br />
And a Heaven in a Wild Flower,<br />
Hold Infinity in the palm of your hand<br />
And Eternity in an hour.<br />
<p>
```

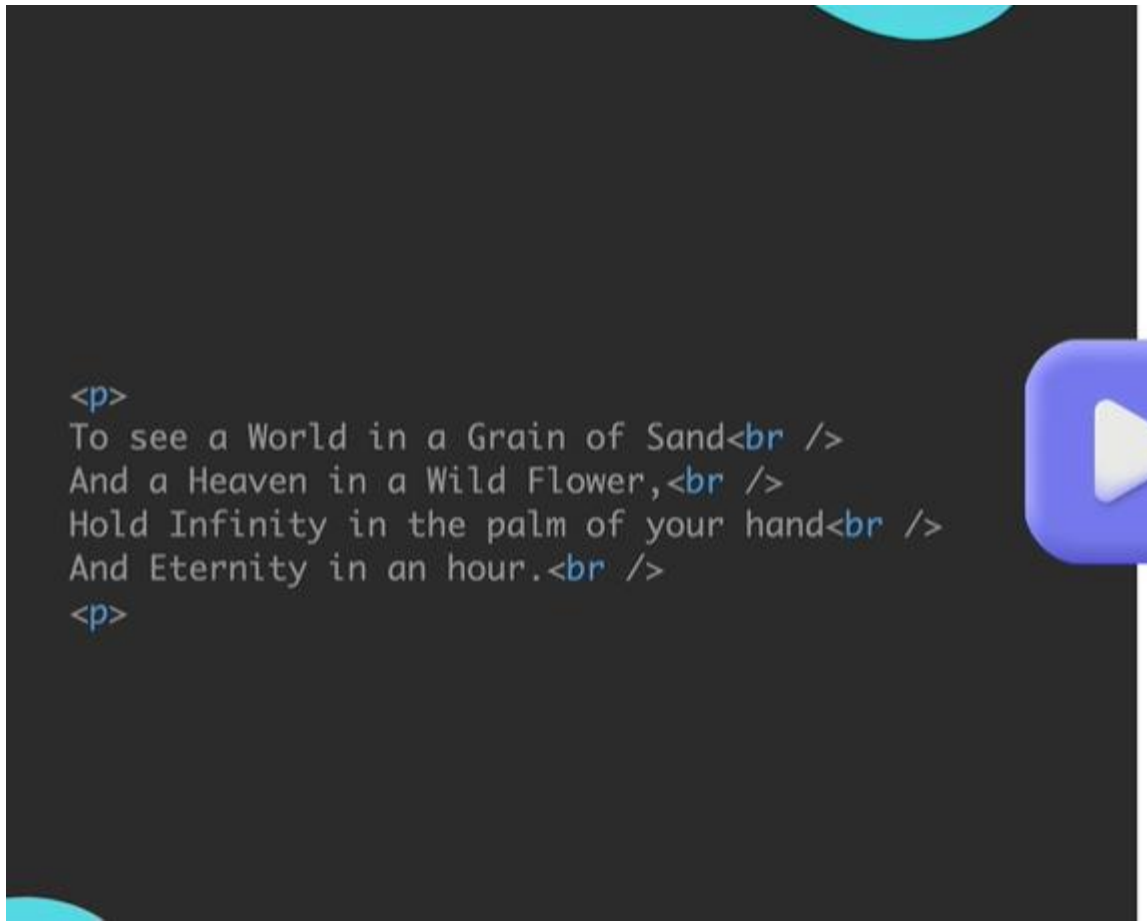


To see a World in a Grain of Sand
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand
And Eternity in an hour.

Break Element




<p>
To see a World in a Grain of Sand
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand
And Eternity in an hour.
<p>
To see a World in a Grain of Sand And a Heaven in a Wild Flower, Hold Infinity in



To see a World in a Grain of Sand
And a Heaven in a Wild Flower,
Hold Infinity in the palm of your hand
And Eternity in an hour.

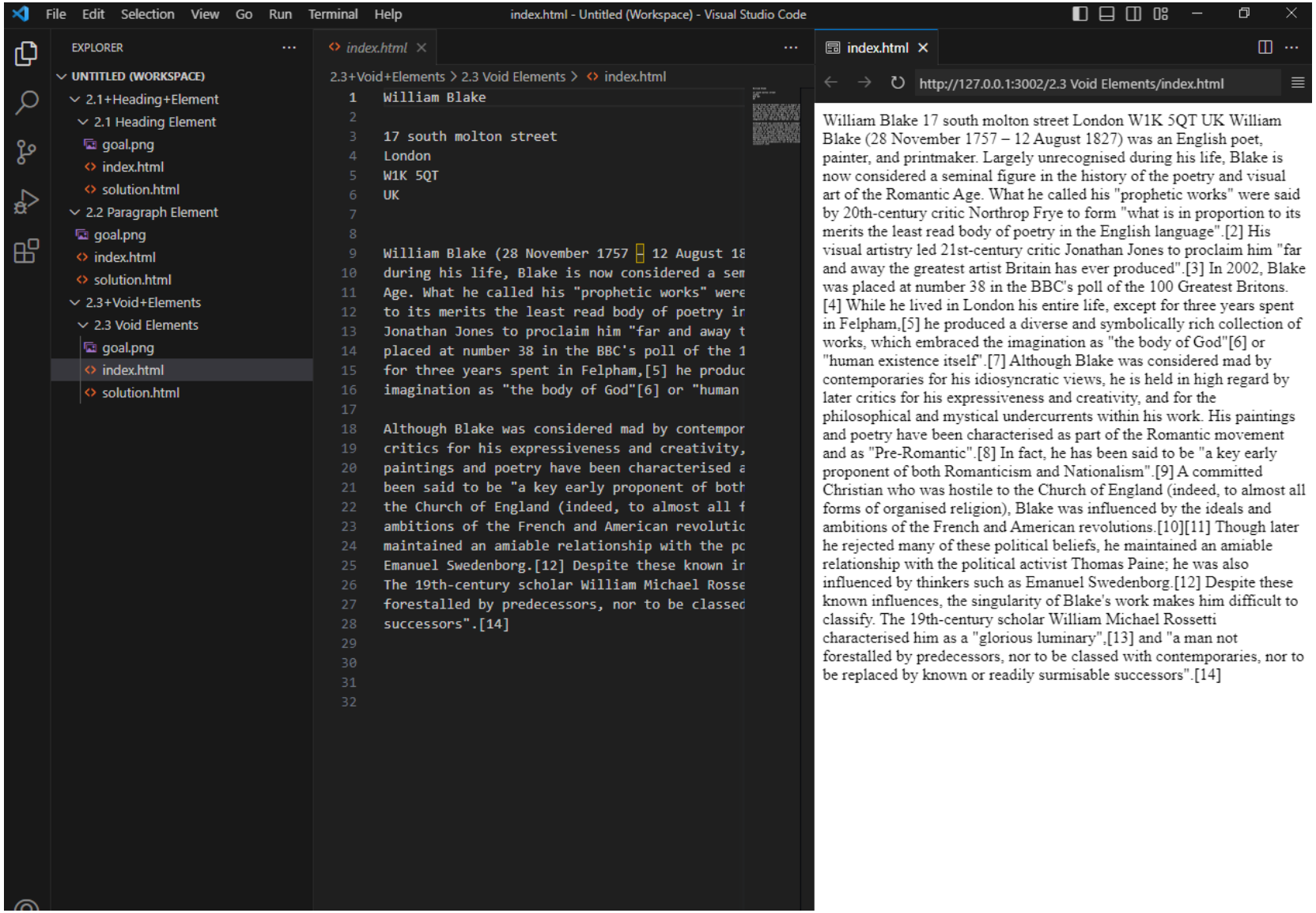
ตัวอย่างของ Void Elements ที่พบบ่อย

-
 ขึ้นบรรทัดใหม่ (line break)
- <img> แสดงภาพ
- <hr> เส้นคั่นแนวนอน
- <input> ช่องกรอกข้อมูล
- <meta> ข้อมูลเมตา เช่น charset, viewport
- <link> เชื่อมโยงกับไฟล์ภายนอก เช่น CSS
- <source> ระบุแหล่งไฟล์มีเดีย เช่น <video> หรือ <audio>
- <area> พื้นที่ในแผนที่ภาพ (image map)
- <base> กำหนด URL พื้นฐานของหน้า
- <col> ระบุลักษณะของคอลัมน์ใน <colgroup>
- <embed> ฝังไฟล์มีเดียเช่น PDF, Flash
- <track> คำบรรยายหรือคำอธิบายสำหรับ <video>

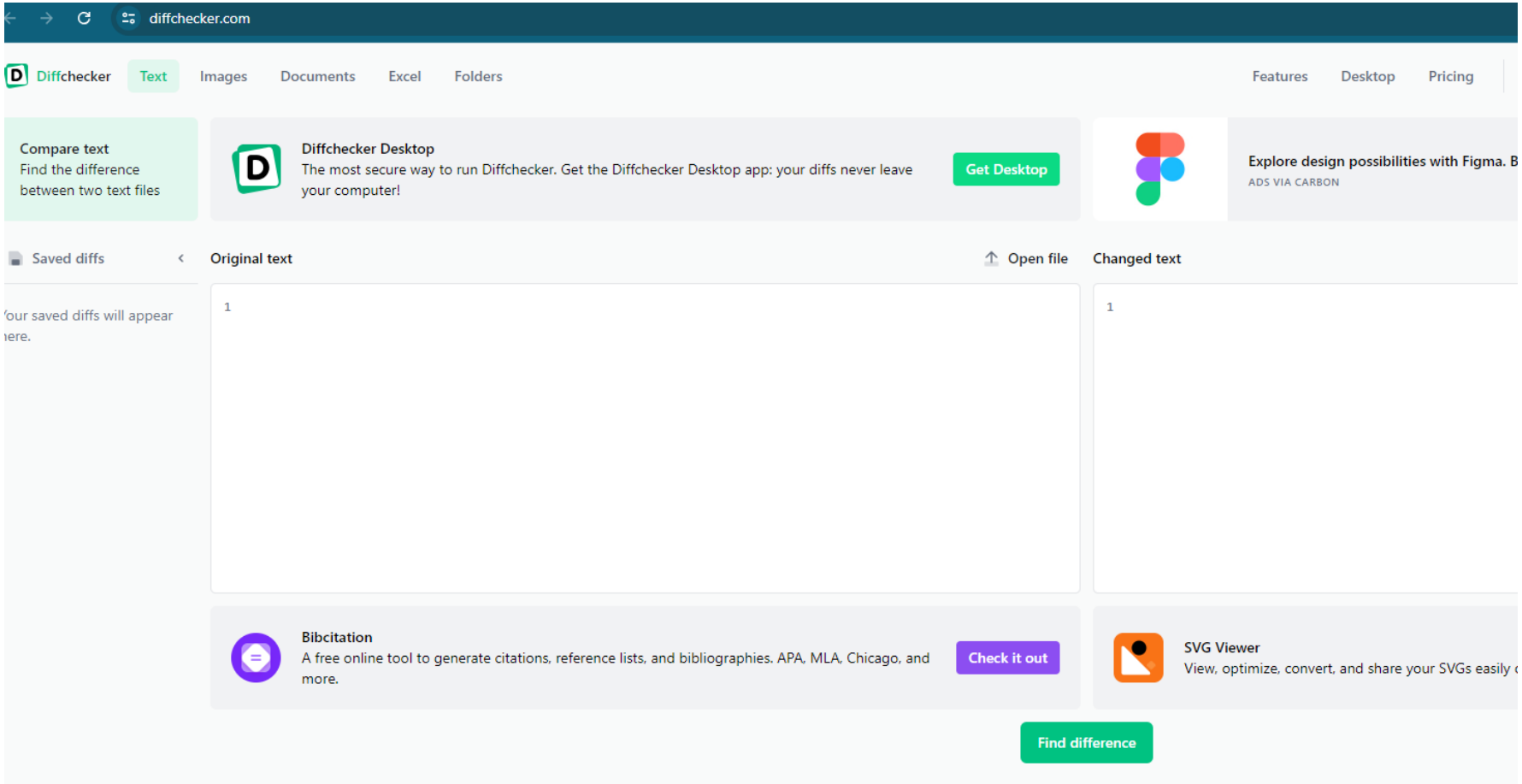
Try an Exercise



2.3 Void Elements.zip


File Edit Selection View Go Run Terminal Help
index.html - Untitled (Workspace) - Visual Studio Code
EXPLORER
UNTITLED (WORKSPACE)
2.1+Heading+Element
2.1 Heading Element
goal.png
index.html
solution.html
2.2 Paragraph Element
goal.png
index.html
solution.html
2.3+Void+Elements
2.3 Void Elements
goal.png
index.html
solution.html
index.html
2.3+Void+Elements > 2.3 Void Elements > index.html
William Blake
17 south molton street
London
W1K 5QT
UK
William Blake (28 November 1757 – 12 August 18
during his life, Blake is now considered a sem
Age. What he called his "prophetic works" were
to its merits the least read body of poetry in
Jonathan Jones to proclaim him "far and away t
placed at number 38 in the BBC's poll of the 1
for three years spent in Felpham,[5] he produc
imagination as "the body of God"[6] or "human
Although Blake was considered mad by contempor
critics for his expressiveness and creativity,
paintings and poetry have been characterised a
been said to be "a key early proponent of both
the Church of England (indeed, to almost all f
ambitions of the French and American revolutic
maintained an amiable relationship with the pc
Emanuel Swedenborg.[12] Despite these known in
The 19th-century scholar William Michael Rosse
forestalled by predecessors, nor to be classed
successors".[14]
index.html
http://127.0.0.1:3002/2.3 Void Elements/index.html
William Blake 17 south molton street London W1K 5QT UK William Blake (28 November 1757 – 12 August 1827) was an English poet, painter, and printmaker. Largely unrecognised during his life, Blake is now considered a seminal figure in the history of the poetry and visual art of the Romantic Age. What he called his "prophetic works" were said by 20th-century critic Northrop Frye to form "what is in proportion to its merits the least read body of poetry in the English language".[2] His visual artistry led 21st-century critic Jonathan Jones to proclaim him "far and away the greatest artist Britain has ever produced".[3] In 2002, Blake was placed at number 38 in the BBC's poll of the 100 Greatest Britons.[4] While he lived in London his entire life, except for three years spent in Felpham,[5] he produced a diverse and symbolically rich collection of works, which embraced the imagination as "the body of God"[6] or "human existence itself".[7] Although Blake was considered mad by contemporaries for his idiosyncratic views, he is held in high regard by later critics for his expressiveness and creativity, and for the philosophical and mystical undercurrents within his work. His paintings and poetry have been characterised as part of the Romantic movement and as "Pre-Romantic".[8] In fact, he has been said to be "a key early proponent of both Romanticism and Nationalism".[9] A committed Christian who was hostile to the Church of England (indeed, to almost all forms of organised religion), Blake was influenced by the ideals and ambitions of the French and American revolutions.[10][11] Though later he rejected many of these political beliefs, he maintained an amiable relationship with the political activist Thomas Paine; he was also influenced by thinkers such as Emanuel Swedenborg.[12] Despite these known influences, the singularity of Blake's work makes him difficult to classify. The 19th-century scholar William Michael Rossetti characterised him as a "glorious luminary",[13] and "a man not forestalled by predecessors, nor to be classed with contemporaries, nor to be replaced by known or readily surmisable successors".[14]

https://www.diffchecker.com/



Don't

```
<p>
Paragraph 1<br />
Paragraph 2<br />
</p>
```

Do

```
<p>Paragraph 1</p>
<p>Paragraph 2</p>
```

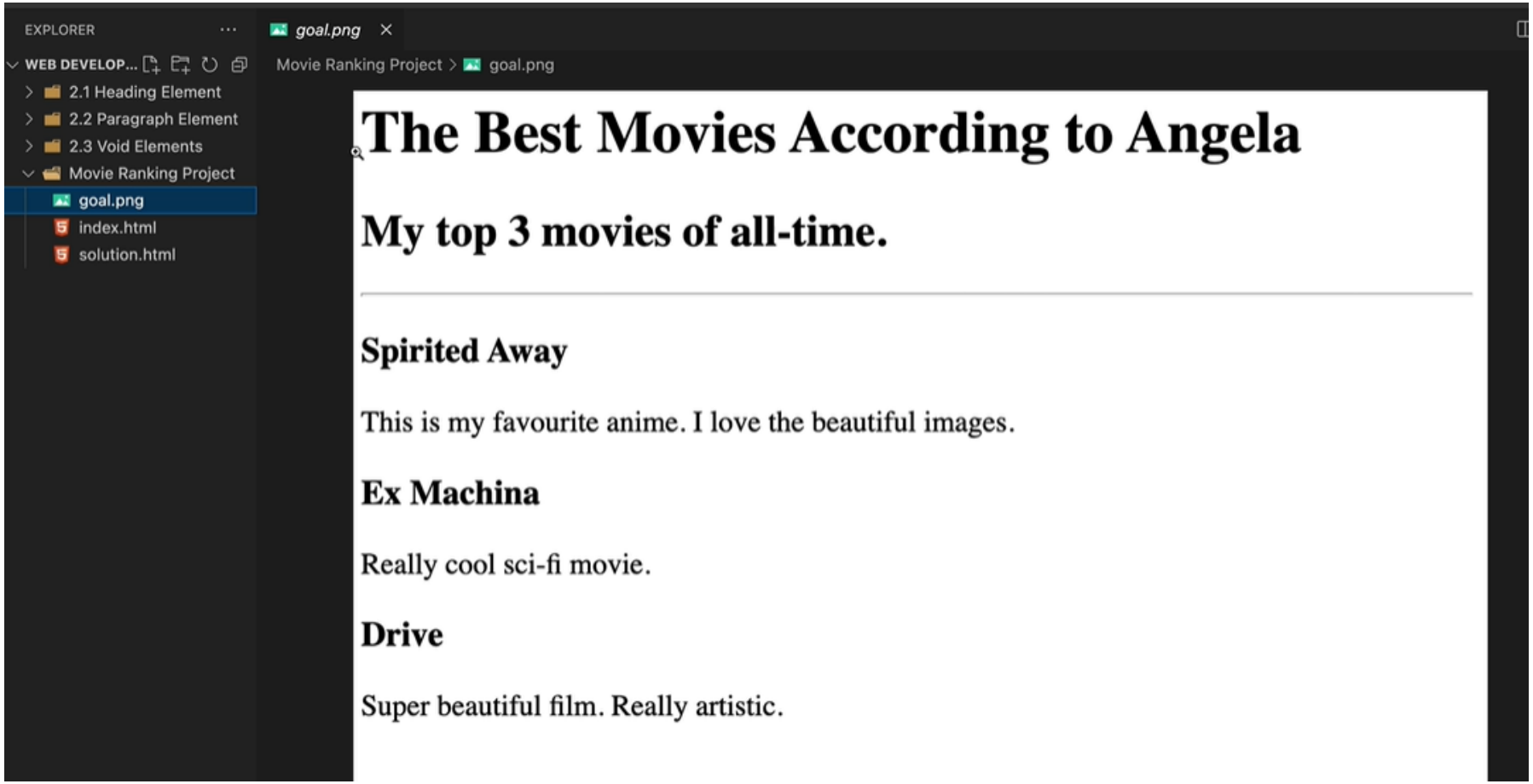
The App Brewery

Favourite Movie Project

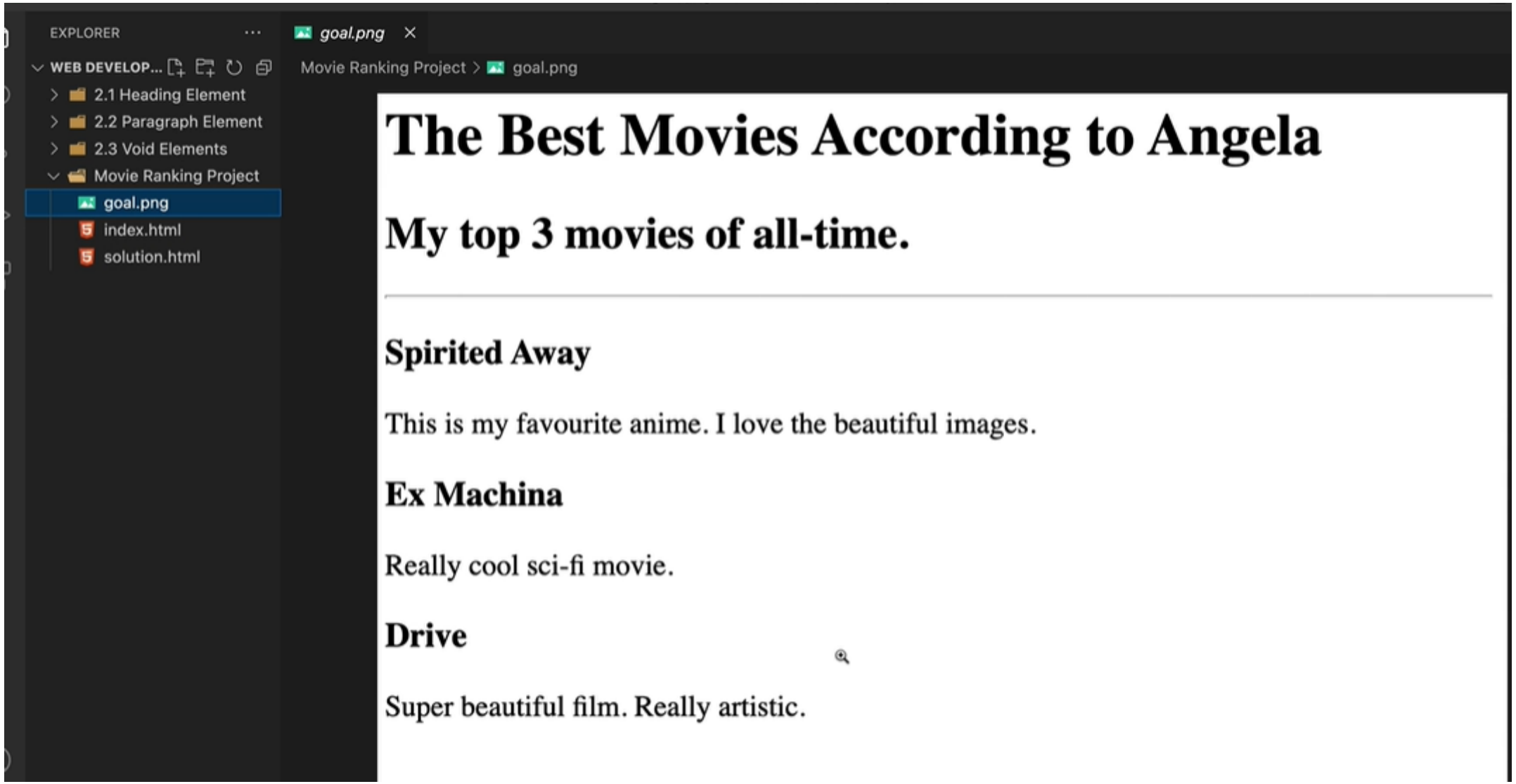
A website for your favourites.



www.appbrewery.com


EXPLORER
goal.png
WEB DEVELOP...
Movie Ranking Project > goal.png
2.1 Heading Element
2.2 Paragraph Element
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Movie Ranking Project
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solution.html
The Best Movies According to Angela
My top 3 movies of all-time.
Spirited Away
This is my favourite anime. I love the beautiful images.
Ex Machina
Really cool sci-fi movie.
Drive
Super beautiful film. Really artistic.

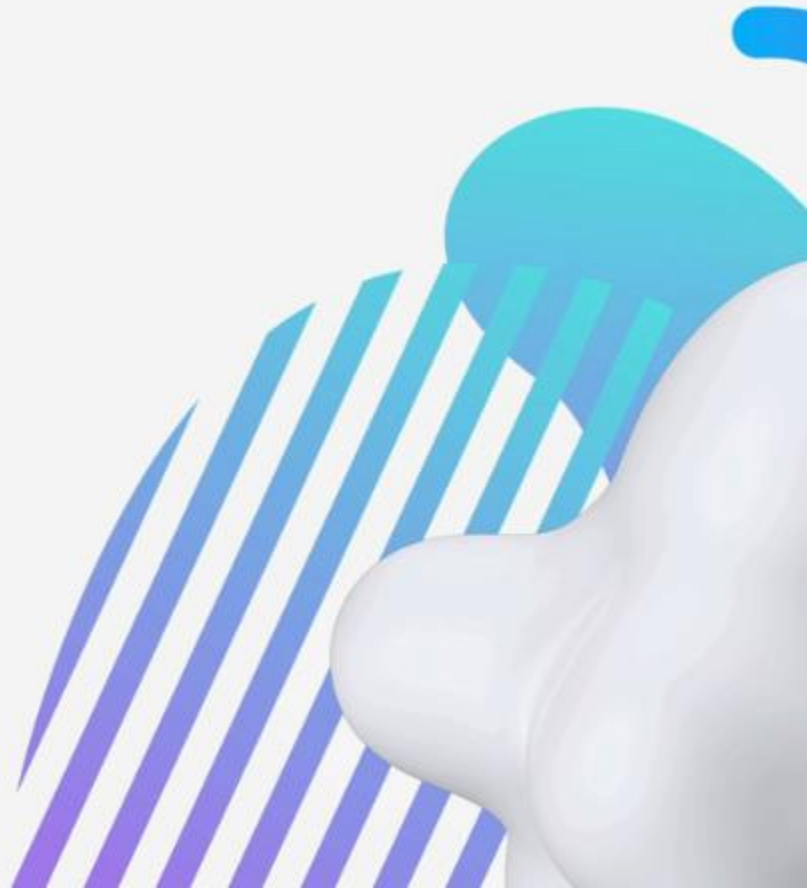

oject
ZIP
Movie Ranking
Project.zip


EXPLORER
goal.png
WEB DEVELOP...
Movie Ranking Project > goal.png
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The Best Movies According to Angela
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Really cool sci-fi movie.
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Super beautiful film. Really artistic.

The App Brewery

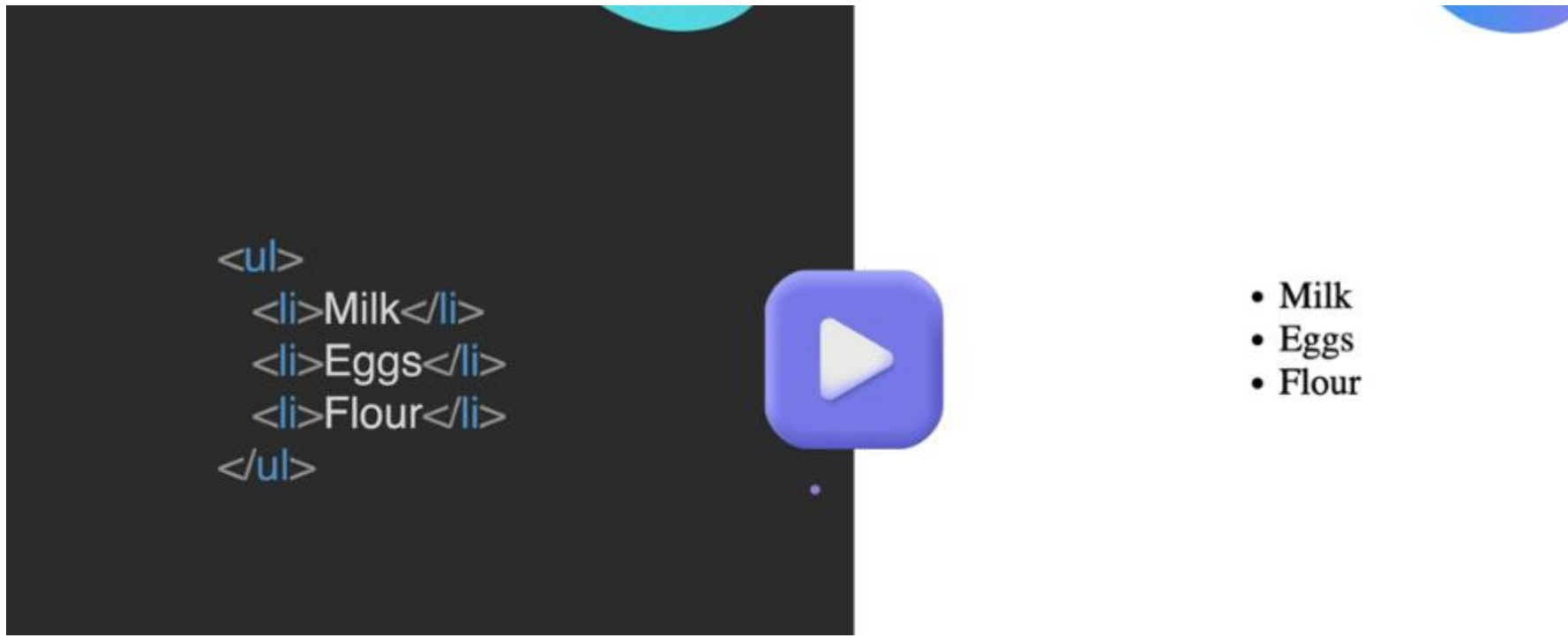
The List Elements

Ordered and Unordered Lists



Unordered List

```
<ul>
  <li>Milk</li>
  <li>Eggs</li>
  <li>Flour</li>
</ul>
```



Milk
Eggs
Flour

• Milk
• Eggs
• Flour

Ordered List + List Items

```
<ol>
  <li>Milk</li>
  <li>Eggs</li>
  <li>Flour</li>
</ol>
```



1. Milk
2. Eggs
3. Flour

Ten Most Wanted Fugitives — | 27 Things That Must Have Be

fbi.gov/wanted/topten

Ten Most Wanted Fugitives

Notice: The official FBI Ten Most Wanted Fugitives list is maintained on the FBI website. This information may be copied and distributed, however, any unauthorized alteration of any portion of the FBI's Ten Most Wanted Fugitives posters is a violation of federal law (18 U.S.C., Section 709). Persons who make or reproduce these alterations are subject to prosecution and, if convicted, shall be fined or imprisoned for not more than one year, or both.

Filter by: Filter by title, description, category... Filter by Year Filter

Sort by: Newest

Results: 10 Items



ARNOLDO JIMENEZ



OMAR ALEXANDER CARDENAS



ALEXIS FLORES



YULAN ADONAY ARCHAGA CARIAS

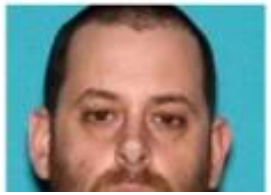


BHADRESHKUMAR CHETANBHAI PATEL











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Filter by:

Filter by title, description, category...

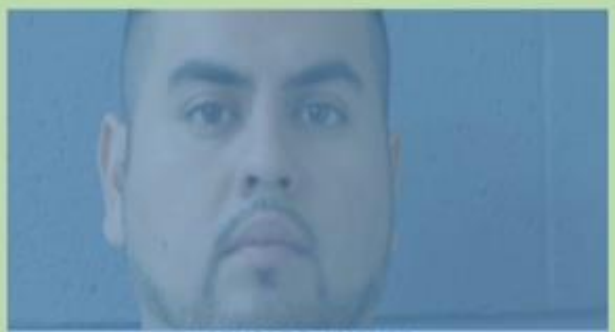
Filter by

Year

Filter

Sort by:

li.portal-type-person.castle-grid-block-item 460 × 286.94



ARNOLDO JIMENEZ



OMAR ALEXANDER CARDENAS



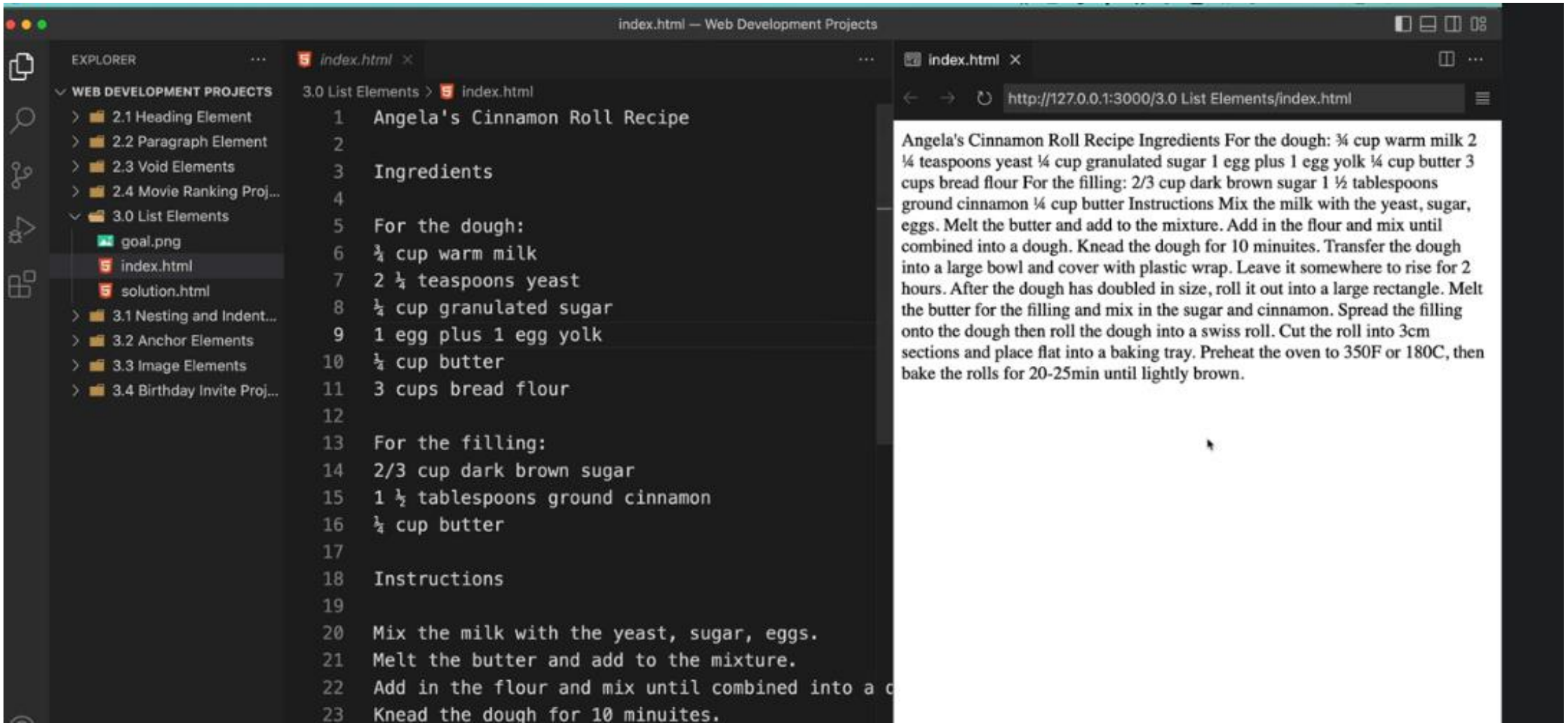


```
<div class="query-results pat-pager" id="query-results-0f737222c5054a
1a120bce207b0446a">
  <div class="row top-total">…</div>
  <ul class="full-grid wanted-grid-natural infinity
            castle-grid-block-xs-2 castle-grid-block-sm-2castle-grid-
    block-md-3 castle-grid-block-lg-5 dt-grid" style>
      ::before
    <li class="portal-type-person castle-grid-block-item">…</li> == $0
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
    <li class="portal-type-person castle-grid-block-item">…</li>
      ::after
  </ul>
  <p class="read-more text-center bottom-total visualClear"> 1 - 10 of
  10 Results </p>
 </div>
</div>
<div class="clearfix">…</div>
</div>
</div>
</div>
</div>
::after
</div>
<div class="mosaic-grid-row row">…</div>
<div class="mosaic-grid-row row">…</div>
</div>
</section>
::after
</div>
</div>
<div class="main-footer" id="main-footer">…</div>
</div>
<div id="mobile-navigation-wrapper">…</div>
<section class="meta-tile-container" id="meta-tile-meta-sticky-footer" aria-labels="meta-s
```

… lock-sm-2castle-grid-block-md-3.castle-grid-block-lg-5.dt-grid li.portal-type-person.castle-grid-block-item

Console What's New ×

Highlights from the Chrome 109 update


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3.1 Nesting and Indent...
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3.3 Image Elements
3.4 Birthday Invite Proj...
index.html
3.0 List Elements > index.html
1 Angela's Cinnamon Roll Recipe
2
3 Ingredients
4
5 For the dough:
6 ¾ cup warm milk
7 2 ¼ teaspoons yeast
8 ¼ cup granulated sugar
9 1 egg plus 1 egg yolk
10 ¼ cup butter
11 3 cups bread flour
12
13 For the filling:
14 2/3 cup dark brown sugar
15 1 ½ tablespoons ground cinnamon
16 ¼ cup butter
17
18 Instructions
19
20 Mix the milk with the yeast, sugar, eggs.
21 Melt the butter and add to the mixture.
22 Add in the flour and mix until combined into a
23 Knead the dough for 10 minuites.
index.html
http://127.0.0.1:3000/3.0 List Elements/index.html
Angela's Cinnamon Roll Recipe Ingredients For the dough: ¾ cup warm milk 2 ¼ teaspoons yeast ¼ cup granulated sugar 1 egg plus 1 egg yolk ¼ cup butter 3 cups bread flour For the filling: 2/3 cup dark brown sugar 1 ½ tablespoons ground cinnamon ¼ cup butter Instructions Mix the milk with the yeast, sugar, eggs. Melt the butter and add to the mixture. Add in the flour and mix until combined into a dough. Knead the dough for 10 minuites. Transfer the dough into a large bowl and cover with plastic wrap. Leave it somewhere to rise for 2 hours. After the dough has doubled in size, roll it out into a large rectangle. Melt the butter for the filling and mix in the sugar and cinnamon. Spread the filling onto the dough then roll the dough into a swiss roll. Cut the roll into 3cm sections and place flat into a baking tray. Preheat the oven to 350F or 180C, then bake the rolls for 20-25min until lightly brown.

Nesting and Indentation

How to write good code



```
<ul>
  <li>Wake up and brush teeth</li>
  <li>Drink 500ml warm water</li>
  <li>Do yoga for an hour</li>
  <li>Make omelette
    <ul>
      <li>Whisk eggs with milk</li>
      <li>Add butter to pan</li>
      <li>Add in eggs and stir</li>
      <li>When solid add salt</li>
    </ul>
  </li>
  <li>Start work</li>
</ul>
```



- Wake up and brush teeth
- Drink 500ml warm water
- Do yoga for an hour
- Make omelette
  - Whisk eggs with milk
  - Add butter to pan
  - Add in eggs and stir
  - When solid add salt
- Start work



A
B

Ba
Bb

Bb1
Bb2
Bb3

Bc

C

• A
• B
◦ Ba
◦ Bb
1. Bb1
2. Bb2
3. Bb3
◦ Bc
• C

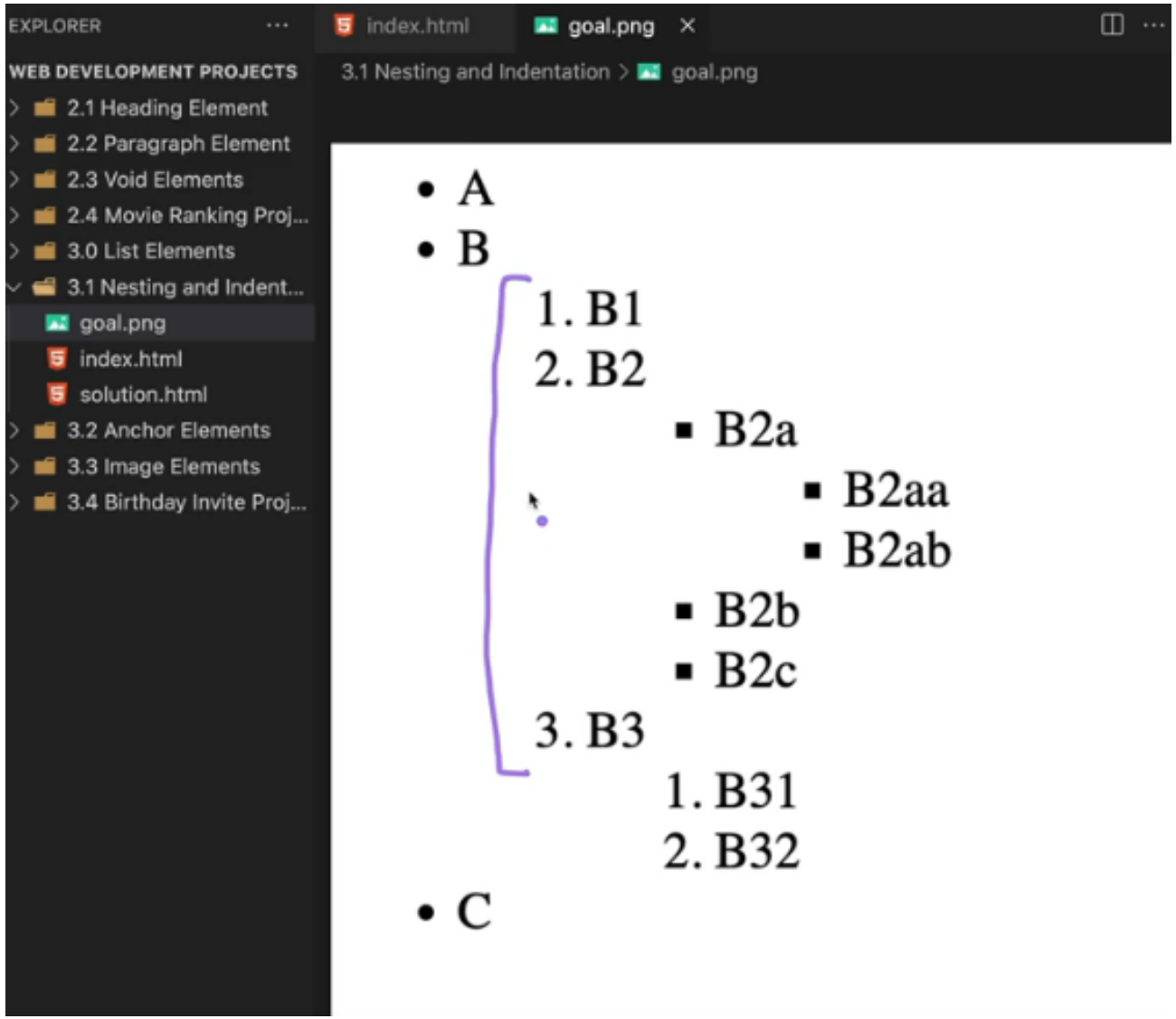
A
B

Ba
Bb

Bb1
Bb2
Bb3

Bc

C


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3.3 Image Elements
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goal.png
3.1 Nesting and Indentation > goal.png
A
B
1. B1
2. B2
B2a
B2aa
B2ab
B2b
B2c
3. B3
1. B31
2. B32
C

The Anchor Element

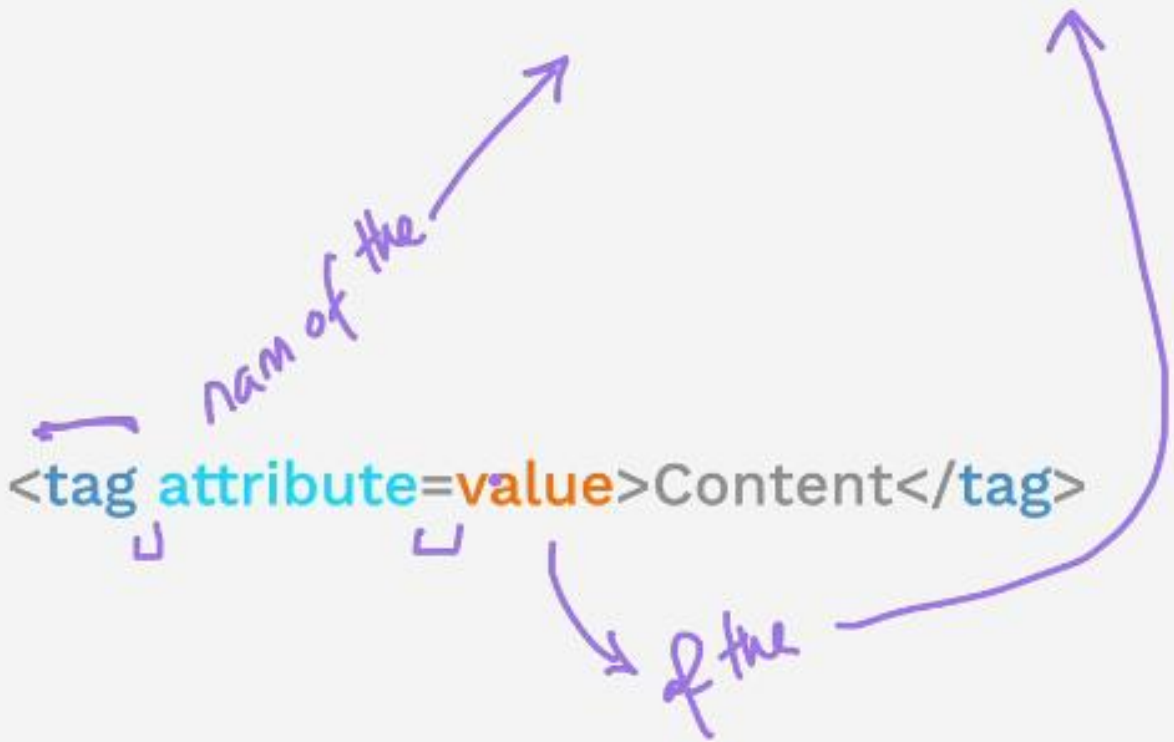
Understanding HTML Attributes



```
<a>This is a link</a>
```

```
<a href="http://www.google.com">This is a link</a>
```

HTML Attributes



HTML Attributes



```
<tag attribute=value anotherattribute=value>Content</tag>
```

```
<a>This is a link to Google</a>
```



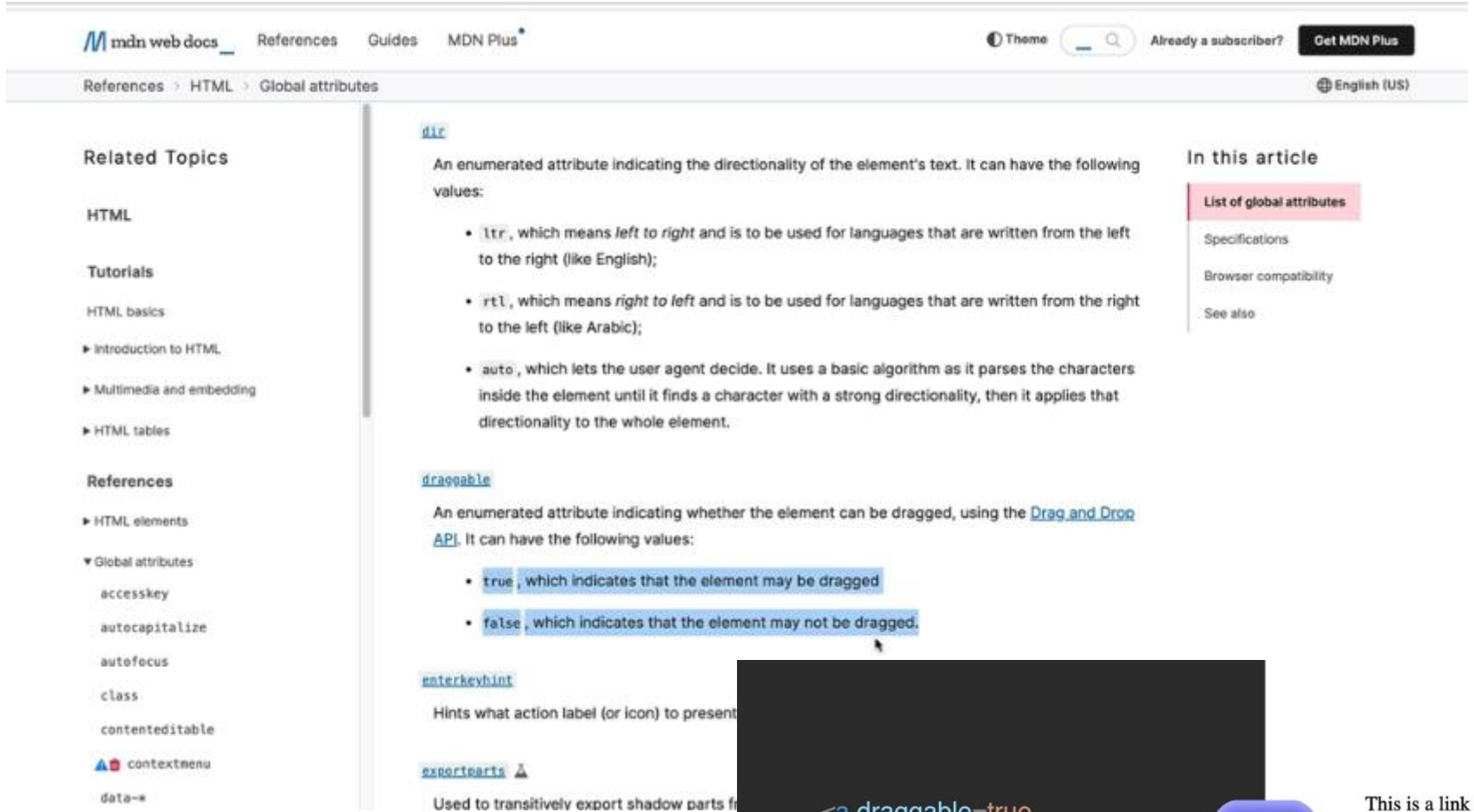
This is a link to Google

```
<a href="https://www.google.com">
This is a link to Google
</a>
```

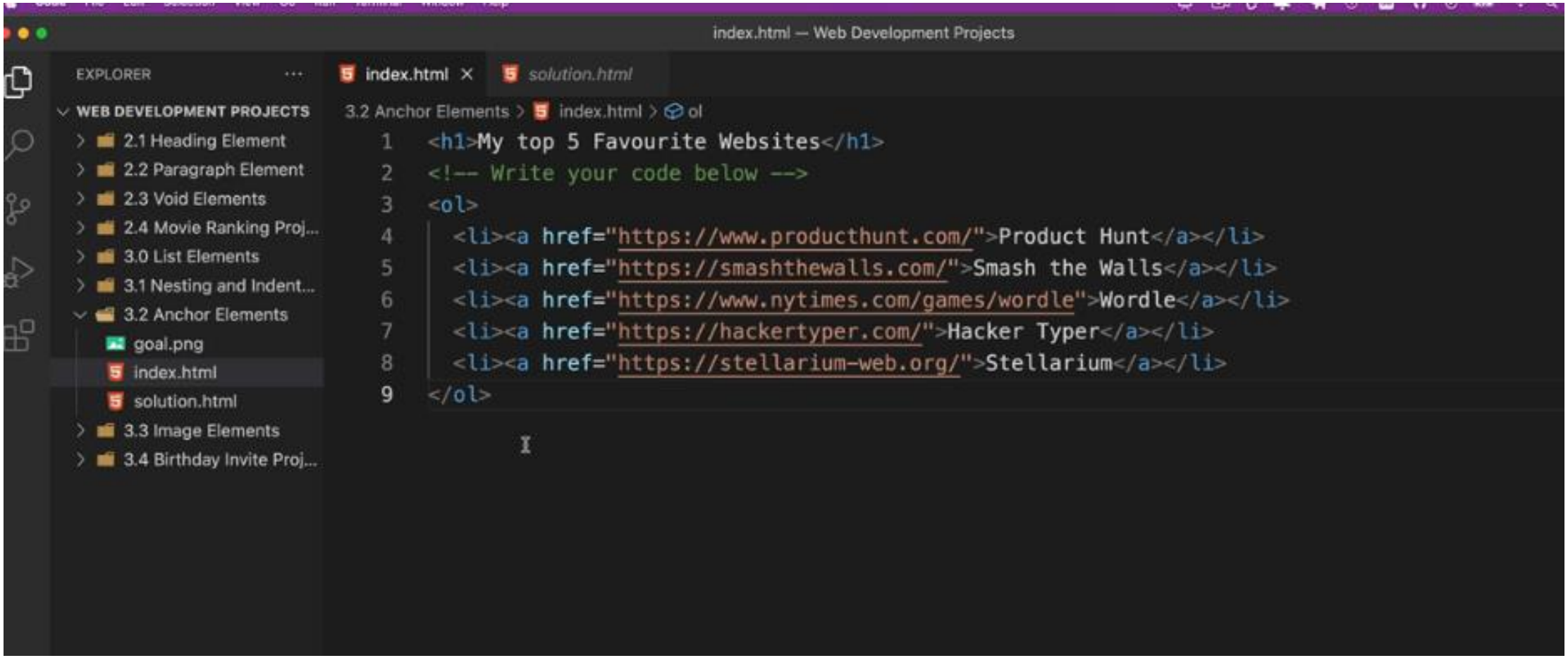
target



This is a link to Google


mdn web docs _
References
Guides
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Already a subscriber?
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References › HTML › Global attributes
English (US)
Related Topics
HTML
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HTML basics
Introduction to HTML
Multimedia and embedding
HTML tables
References
HTML elements
Global attributes
accesskey
autocapitalize
autofocus
class
contenteditable
contextmenu
data-*
dir
An enumerated attribute indicating the directionality of the element's text. It can have the following values:
ltr, which means left to right and is to be used for languages that are written from the left to the right (like English);
rtl, which means right to left and is to be used for languages that are written from the right to the left (like Arabic);
auto, which lets the user agent decide. It uses a basic algorithm as it parses the characters inside the element until it finds a character with a strong directionality, then it applies that directionality to the whole element.
draggable
An enumerated attribute indicating whether the element can be dragged, using the Drag and Drop API. It can have the following values:
true, which indicates that the element may be dragged
false, which indicates that the element may not be dragged.
enterkeyhint
Hints what action label (or icon) to present
exportparts
Used to transitively export shadow parts f
In this article
List of global attributes
Specifications
Browser compatibility
See also
<a draggable=true
This is a link to Google

This is a link to Google
This is a link to Google


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solution.html
3.3 Image Elements
3.4 Birthday Invite Proj...
index.html
solution.html
3.2 Anchor Elements > index.html > ol
1 <h1>My top 5 Favourite Websites</h1>
2 <!-- Write your code below -->
3
4 Product Hunt
5 Smash the Walls
6 Wordle
7 Hacker Typer
8 Stellarium
9

The Image Element

Adding images to our websites



location of image

```
<img src="url" />
```

Source of the image

```
<img src="https://picsum.photos/200" />
```





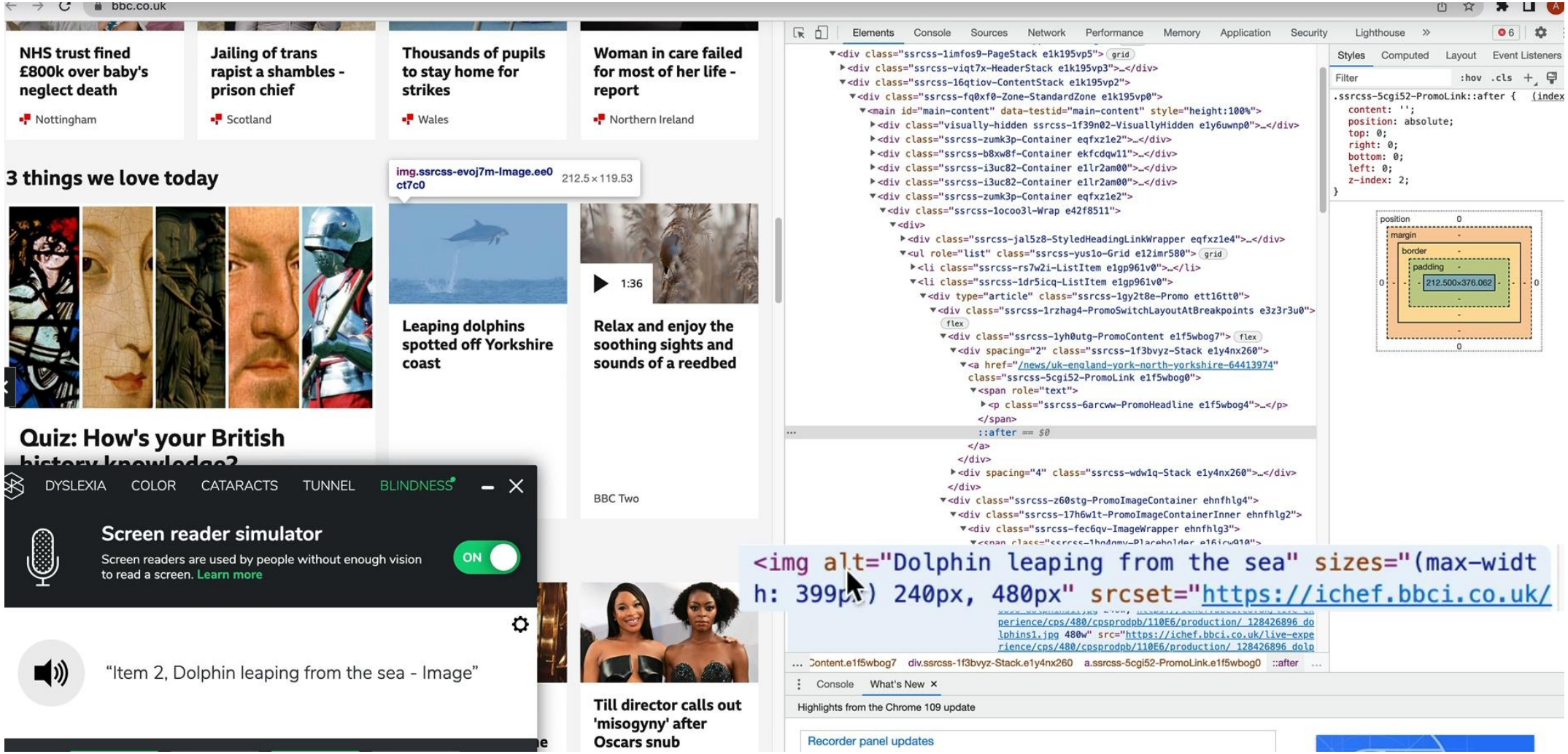
The Alt Attribute

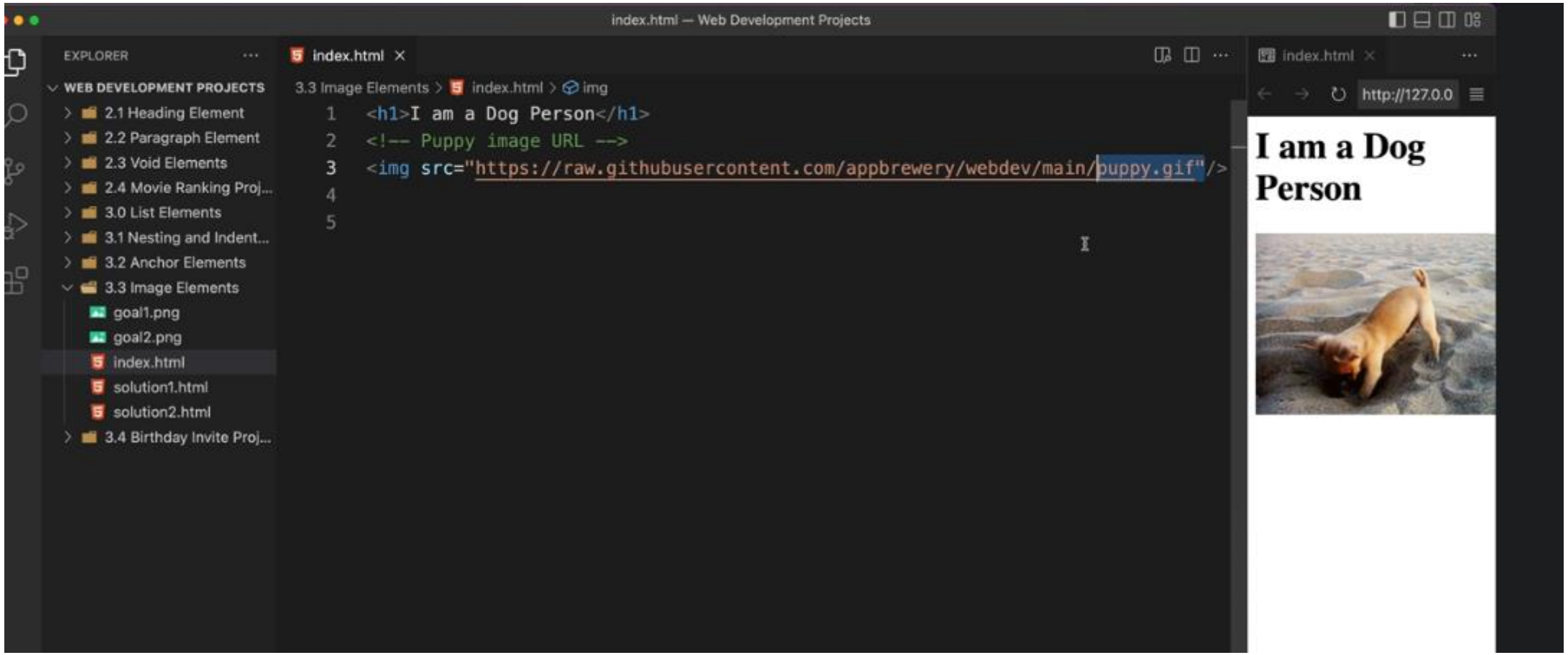
```
<img src="https://picsum.photos/200" alt="forest at sunset"/>
```

alternative text description

alt ย่อมาจาก **"alternative text"** หรือ **ข้อความทางเลือก**
มันคือ **ข้อความที่อธิบายภาพ** สำหรับกรณีที่:

- ภาพโหลดไม่ขึ้น
- ผู้ใช้งานใช้โปรแกรมอ่านหน้าจอ (screen reader) เช่น คนตาบอด
- บอทของ Google ต้องเข้าใจเนื้อหาของภาพเพื่อจัดอันดับ

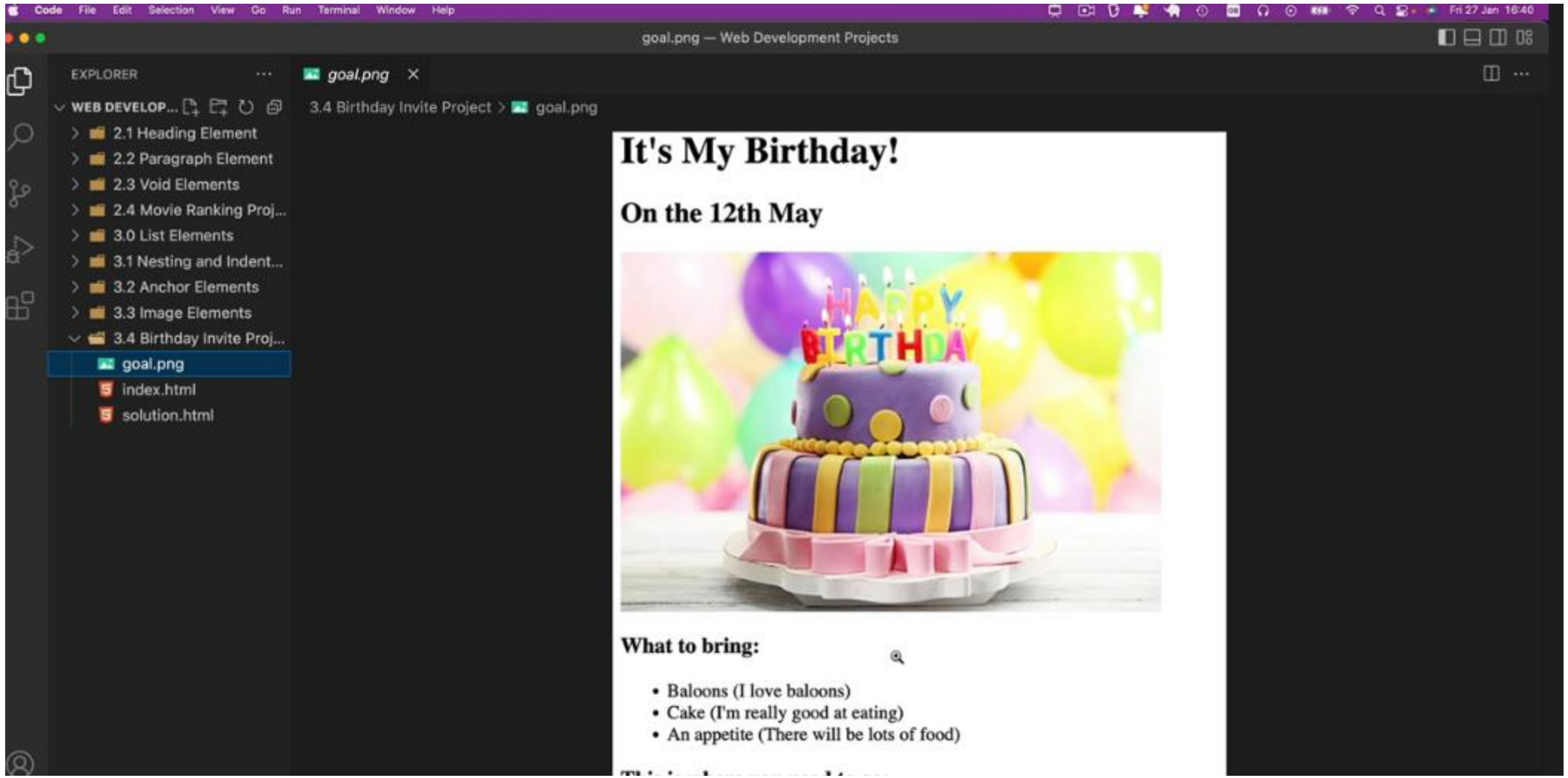

bbc.co.uk
NHS trust fined £800k over baby's neglect death
Nottingham
Jailing of trans rapist a shambles - prison chief
Scotland
Thousands of pupils to stay home for strikes
Wales
Woman in care failed for most of her life - report
Northern Ireland
3 things we love today
img.ssrcss-evoj7m-Image.ee0ct7c0
212.5×119.53
Leaping dolphins spotted off Yorkshire coast
1:36
Relax and enjoy the soothing sights and sounds of a reedbed
BBC Two
Quiz: How's your British
DYSLEXIA COLOR CATARACTS TUNNEL BLINDNESS
Screen reader simulator
Screen readers are used by people without enough vision to read a screen. Learn more
ON
"Item 2, Dolphin leaping from the sea - Image"
Till director calls out 'misogyny' after Oscars snub
Elements Console Sources Network Performance Memory Application Security Lighthouse
Styles Computed Layout Event Listeners
.ssrcss-5cgi52-PromoLink::after {
content: '';
position: absolute;
top: 0;
right: 0;
bottom: 0;
left: 0;
z-index: 2;
}
position 0
margin
border
padding
212.500×376.062
<img alt="Dolphin leaping from the sea" sizes="(max-width: 399px) 240px, 480px" srcset="https://ichef.bbci.co.uk/
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Recorder panel updates


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goal2.png
index.html
solution1.html
solution2.html
3.4 Birthday Invite Proj...
index.html
3.3 Image Elements > index.html > img
1 <h1>I am a Dog Person</h1>
2 <!-- Puppy image URL -->
3
4
5
index.html
http://127.0.0
I am a Dog Person

Birthday Invite Project

A website for your next party




Code File Edit Selection View Go Run Terminal Window Help
Fri 27 Jan 16:40
goal.png — Web Development Projects
EXPLORER
goal.png
WEB DEVELOP...
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index.html
solution.html
3.4 Birthday Invite Project > goal.png
It's My Birthday!
On the 12th May
HAPPY BIRTHDAY
What to bring:
Baloons (I love baloons)
Cake (I'm really good at eating)
An appetite (There will be lots of food)